Getting Started

with Solid Edge
Version 18

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Chapter

1 Welcome

Solid Edge is the CAD component of the new UGS Velocity Series, the industry's first comprehensive, preconfigured portfolio of digital product design, analysis and data management software for the PLM mid-market.

Thousands of companies have taken advantage of innovative Solid Edge technologies, including Rapid Blue for shape creation, Insight for design data management, practical tools for hybrid 2D/3D design and an unmatched four-step program for moving from 2D to 3D. With version 18, Solid Edge once again delivers hundreds of customer-requested enhancements, along with new tools to further help reduce costs and speed product delivery. Solid Edge customers continue to enjoy an exceptional return on investment for a low total cost of ownership.

With its groundbreaking Insight technology, Solid Edge became the only mainstream mechanical system to merge design management capabilities with the CAD tools that designers use every day. Version 18 builds on the success of Insight to set a new standard in CAD/PDM Integration, letting customers choose from a range of easily scalable cPDM solutions. Solid Edge Insight continues to provide proven management capabilities for single-site, departmental teams with basic workflow needs. A new Solid Edge integration provides seamless and transparent connectivity between Solid Edge and the powerful Teamcenter 2005 platform, including the newly announced Teamcenter Express solution. This new level of CAD and cPDM connectivity ensures that all essential Solid Edge commands are encapsulated and makes sophisticated data management functions transparent to the Solid Edge user. Solid Edge-related data is easily captured for reuse in future projects without placing an additional burden on the CAD user, while full scalability means customers can grow their cPDM solution to meet growing business demands without starting from scratch. Now customers with varying needs can take advantage of single-vendor support for both CAD and PDM.

Solid Edge Version 18 also introduces two complementary methods for designs to be analyzed and validated much earlier in the design cycle. The first – Femap Express – is built into Solid Edge and provides preconfigured, best practice, process guidance for fast and accurate analysis of solid and sheet metal parts. Second, Solid Edge integrates further with Femap, the leading CAD-independent finite element analysis (FEA) tool for advanced analysis to ensure quality and to optimize materials within complete assemblies. Both solutions utilize well-proven Nastran solver technology for accurate and repeatable results.

A new wire harness design package utilizes electrical circuit diagrams from popular eCAD systems to automate wire harness designs in Solid Edge. Electrical and mechanical design teams can now collaborate more closely and create a complete digital mock-up that includes cables and wires. Dedicated tools deliver streamlined methods to create wires, cables, and bundles. Correct cut-length information is easily accessible for manufacturing.

Solid Edge diagramming in Version 18 lets users develop electrical, P&ID, and other diagrams by leveraging industry-standard libraries of 2D Blocks (also referred to as symbols). These innovative 2D capabilities further strengthen Solid Edge's ability to create

and maintain 2D drawings and remove the need for companies to maintain separate CAD systems for 2D and 3D requirements.

Solid Edge Mold Tooling established a powerful step-by-step process workflow for removing much of the repetition prevalent in the design of plastic injection molds. Version 18 extends this dramatic time-saving potential beyond tooling design and into manufacturing with a new Electrode Design application. Users are guided through a logical step-by-step approach to develop single or compound electrodes that remain associative to the original mold design.

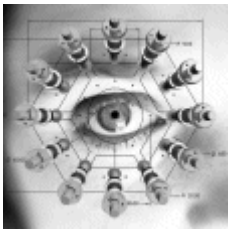
A new upgrade to the existing integration of Solid Edge and NX Machining now provides single-button *associative transfer* of Solid Edge Part, Sheet Metal, or Assembly documents to NX Machining that are ready for manufacturing. Updates to Solid Edge files result in geometry and tool path updates in NX Machining.

Solid Edge Version 18 delivers enhanced capabilities for weldments, including improved BOM creation, more assembly features that allow more joint perpetration types, and the new flexible 'groove weld' command that allows more weld-bead types and the ability to flood fill a joint or span a gap between materials. Documentation is improved by new reporting and weld-symbol representation in drawings, including a new weld texture for a realistic appearance.

Solid Edge version 18 continues to deliver new and innovative technologies that maintain Solid Edge as a market leader. In addition to major projects, the Solid Edge development team maintains a focus on customer-driven enhancements to existing technologies. With every release, Solid Edge makes it easier and more practical to work with massive assemblies and their related drawings. Version 18 also offers improved tools and techniques to help with the design of complex parts, common in consumer products, while the drafting environment adds new drawing view capabilities and even faster performance. New translation capabilities allow users to share even more file types such as the industry-standard JT file format.

Once again the new release of Solid Edge is packed with new features and productivity tools to allow you to streamline your processes and improve your designs. With version 18, the Solid Edge community will once again be the first to benefit from the CAD industry's most functional innovations, first to complete their designs, and first to market with an error-free product.

Solid Edge Insight Technology



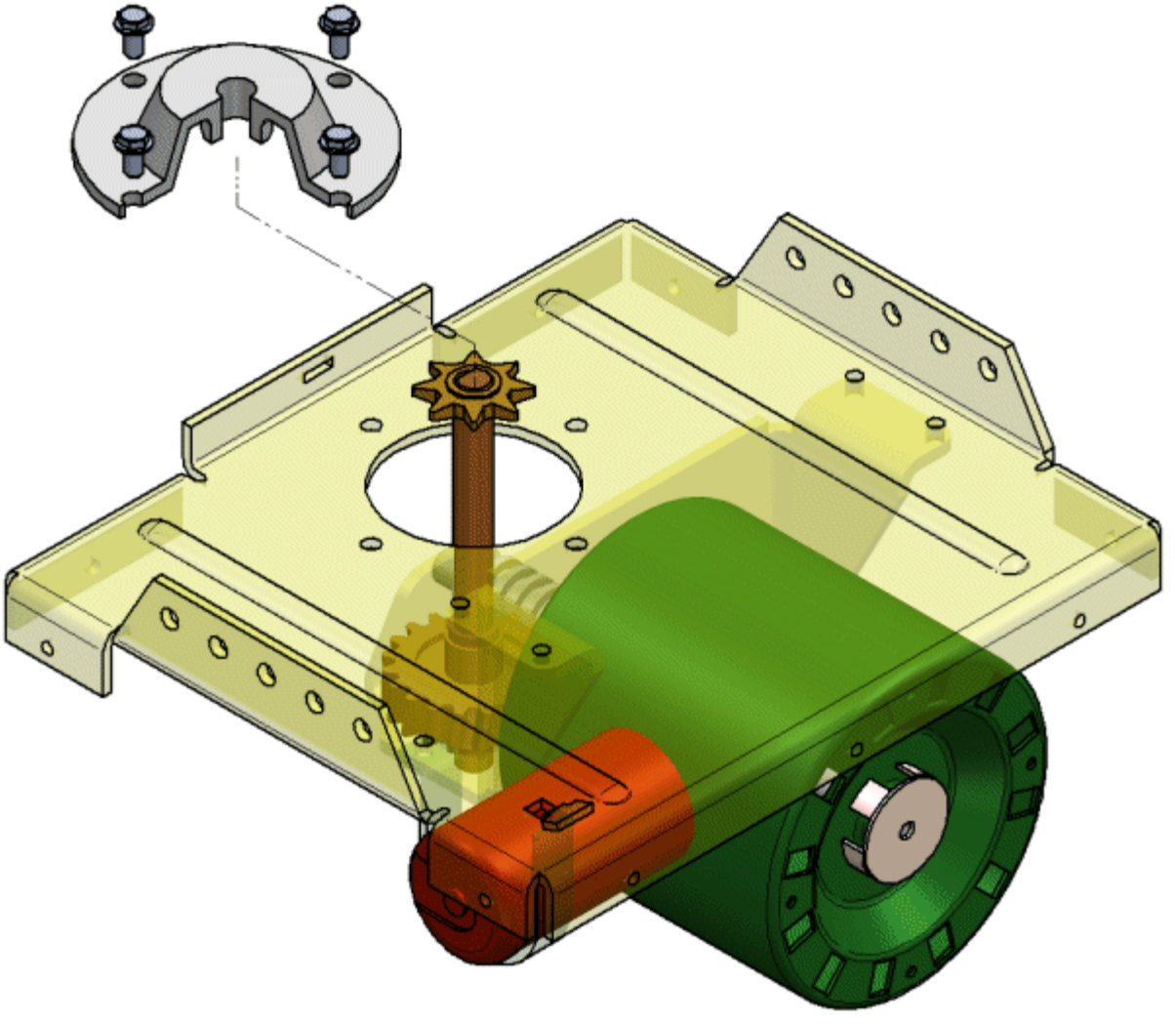
The procedures in this step-by-step guide do not require you to use Solid Edge Insight. Insight invisibly integrates design management functions within the Solid Edge user interface. At several points within the following instruct ions, we'll describe how Insight would work behind the scenes to provide product data management without disrupting the design engineer.

Chapter

2 Start Solid Edge

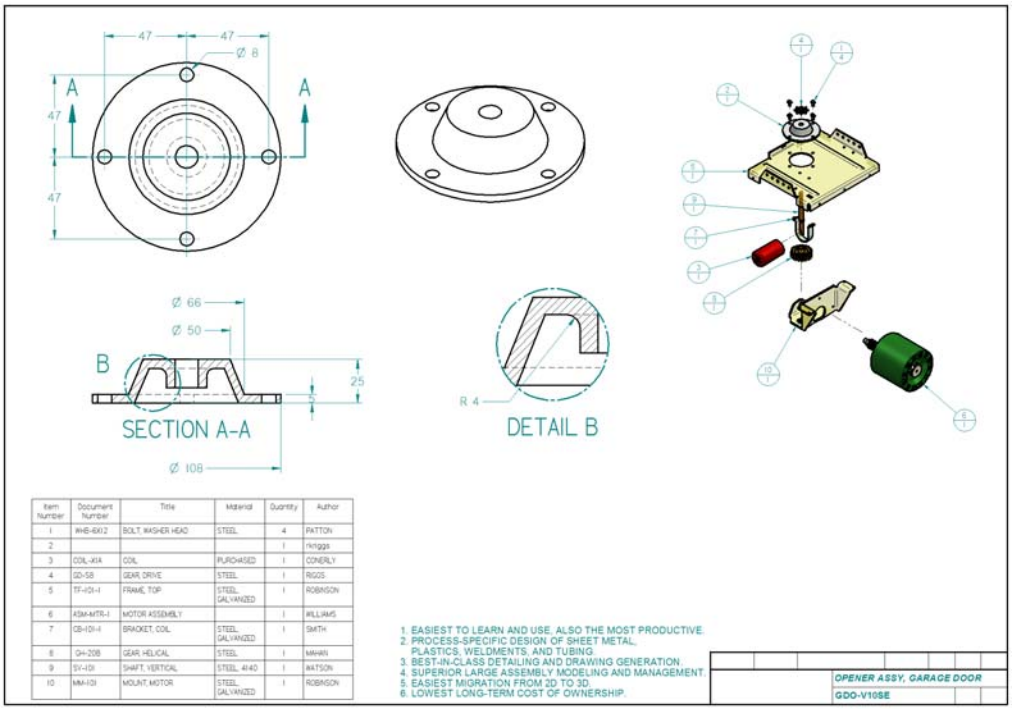
This Getting Started Guide is a work book. Expect to spend about an hour completing all the procedures. When you finish, you will have discovered for yourself why Solid Edge delivers:

- The easiest to learn and use, and the most productive mainstream CAD system available.
- Superior large assembly modeling and management.
- Best in class detailing and drawing generation.
- The easiest migration from 2D to 3D.



As you work through this booklet, you will construct the simple shaft support for the garage door opener assembly shown in the illustration. You will learn about assembly relationships as you place the part and the bolts that secure it in the assembly. Along the way, you will discover other tools and see how simple and intuitive it is to work with Solid Edge.

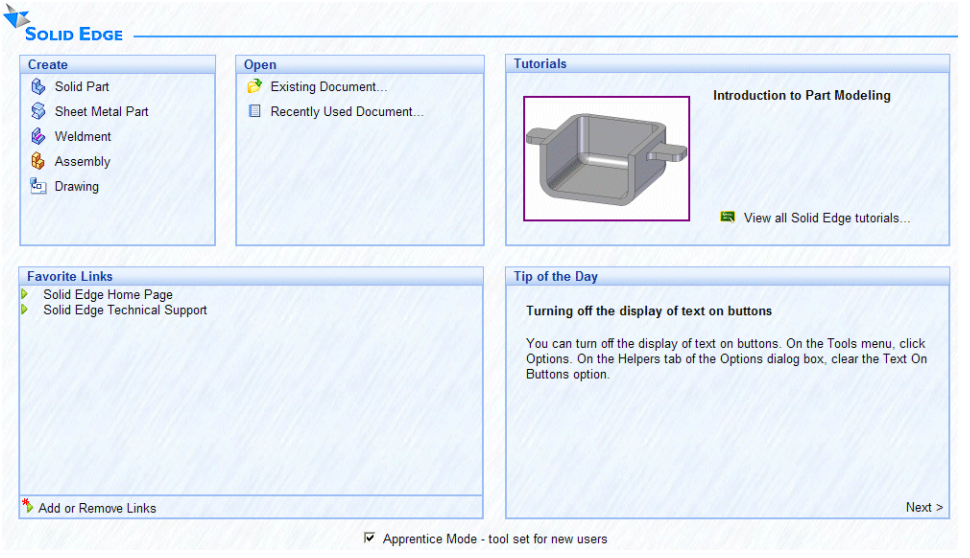
Finally, you will create a drawing of the assembly, including a complete parts list, and a detailed drawing of the part you modeled.



Step 1: Start Solid Edge

On the Start menu, click All Programs, click Solid Edge V18, and then click Solid Edge.

The startup screen is displayed.



The Startup Screen

- Solid Edge has four separate document types: Parts (.par), Sheet Metal Parts (.psm), Assemblies (.asm), and Drawings (.dft).
- Solid Edge provides many tutorials to help you learn to use the product. After working through this Getting Started Guide, you might find it helpful to continue learning more about the product by working through the tutorials.
- You can add links to favorite web sites, such as your corporate web page, sites that track inter-department status, or www.SolidEdge.com.
- Many tips have been compiled to help you discover the best ways of using the product. A new tip is displayed each time you restart Solid Edge.
- Whether you are an experienced user of other CAD systems or a novice, you might find Solid Edge easier to learn by using the Apprentice Mode for a while. With Apprentice Mode, additional text is displayed in Solid Edge to help focus your attention on the parts of the user interface that are most important to what you're doing. Included with Apprentice Mode are:

  – The startup screen you see now.
  – Command Assistant, which clarifies commands and explains the SmartStep workflow.
  – Command Finder, to help you transition from other design systems by cross-referencing Solid Edge commands with popular terminology from other systems.
  – Enhanced Command Tips to help you recognize objects and options that you give you capabilities and choices.
  – Assembly documents open with all occurrences active. For small assemblies, this makes it easier to see and work with an entire assembly at once. Advanced users who work with large assemblies will probably prefer to always open assemblies with all occurrences simplified and inactive.

  The startup screen is always available. At any time you can collapse any open Solid Edge documents to access the favorite links, commands, tutorials, tips, and and Apprentice Mode option.
  The illustrations in this book show Solid Edge with the Apprentice Mode turned off. If you leave Apprentice Mode turned on, you will see command names on the toolbar buttons, Command Assistant, and Command Tips that are not shown in the illustrations.

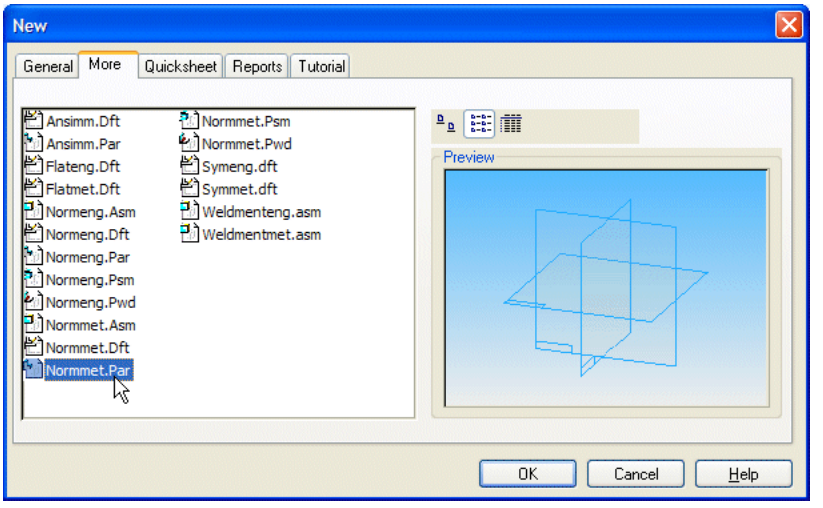
Step 2: Create a solid part

Solid Edge comes with blank documents called templates that you can customize to meet the requirements of your company. This part and the entire garage door assembly are based on metric units. If you specified English units when you loaded Solid Edge, there is no problem. Solid Edge templates make it easy to work with any units and any standard – at any time. In a global workplace, this gives your company the edge it needs to stay competitive and productive.

To ensure that you are working with a metric document, you will open a new document using the metric template.

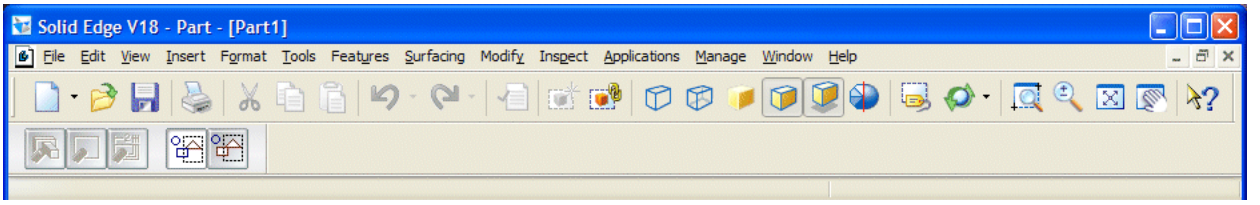


On the Main toolbar, click the New button. On the New dialog box, click the More tab, then select the Normmet.par template.



Step 3: Look around

Solid Edge is easy to use, because it was designed that way. Developed with STREAM technology, Solid Edge is designed to increase software performance with an interface that ensures maximized user productivity and return on investment. STREAM technology makes Solid Edge easy to learn, easy to use, and more productive than any other mainstream CAD system on the market. The innovative STREAM technology in tools such as the SmartStep ribbon bar; IntelliSketch, SmartSketch, EdgeBar, and more make Solid Edge the easiest-to-use MCAD package available.





At the top of the window, below the title bar, the menu bar has cascading menus for standard Windows commands such as File, Edit, View, and so forth, as well as Tool, Environment, Model, and other menus of commands that are specific to Solid Edge.

Beneath the Menu bar, The Main toolbar contains the typical Windows commands you use for opening and saving documents, cutting, copying, and pasting elements, along with other commonly-used commands for shading windows, zooming in and out, and so forth.

Beneath the Main toolbar, the ribbon bar changes as you work to provide options for the command that is currently running. Ribbon bars guide you through the steps of working through commands, let you choose options, and accept input such as names and distances.

The status bar always displays what the system expects you to do next – look here first if you are not sure what to do.

EdgeBar, which is arranged vertically along the left side of the graphic window, helps you manage and complete tasks on a design project. The Feature PathFinder tab on EdgeBar makes it easy to keep track of the features you construct when designing parts. Each feature you construct is added to EdgeBar as you work. Later, when you need to edit a feature, simply select it in EdgeBar, and the feature highlights in the graphic window. You can then quickly make the necessary changes.

When it is time to position parts in an assembly, you use the Parts Library tab on EdgeBar to select the proper parts and the Assembly PathFinder tab to track the parts already placed in the assembly. You also use EdgeBar to show and hide parts as you work to make it easy to manage large, complex assemblies. Many engineers routinely use Solid Edge to design machines containing tens- and even hundreds-of-thousands of components.

Beside EdgeBar, the vertical toolbars along the left side of the graphic window change as you work to provide commands for creating features on parts, drawing profiles, working with assemblies, generating drawings, and so forth.

Mouse Buttons

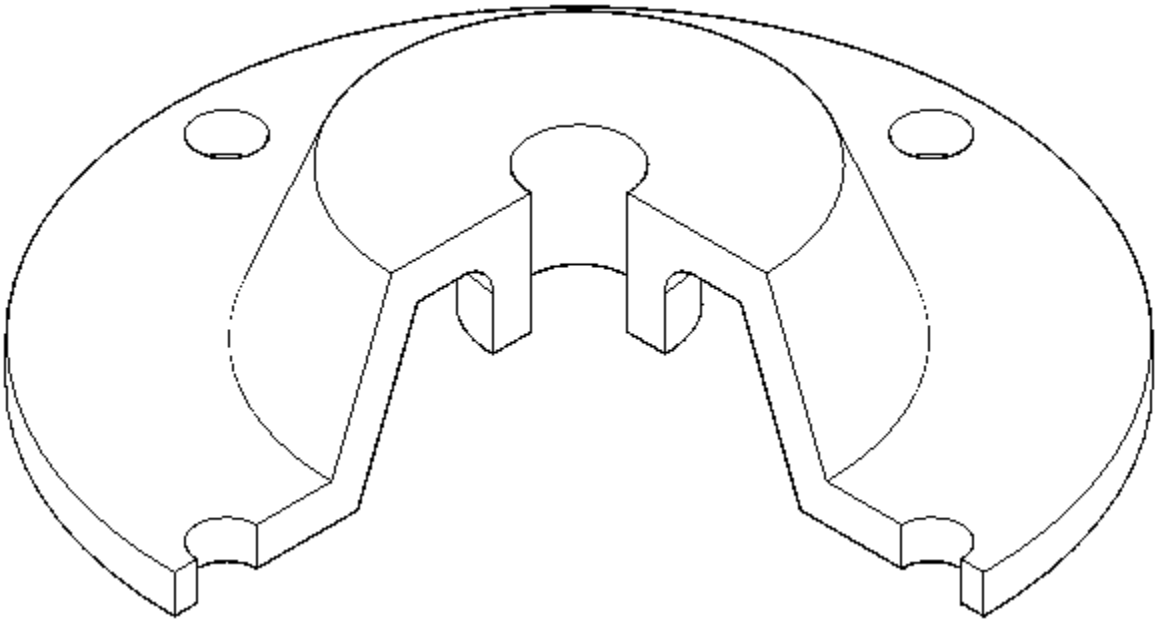
- Left button – You use this button to select geometry, command and option buttons, and entries in the EdgeBar tool. The left button is used most of the time. In fact, you can function quite well in Solid Edge using only this button.

- Right button – You use this button to activate context-sensitive shortcut menus. The contents on these menus change depending on what object the cursor is over. These shortcut menus give you quick access to frequently used commands and options and help you to work more quickly.

You can also use the right mouse button to finish commands. This is often quicker and easier than clicking the Preview and Finish buttons with the left mouse button.

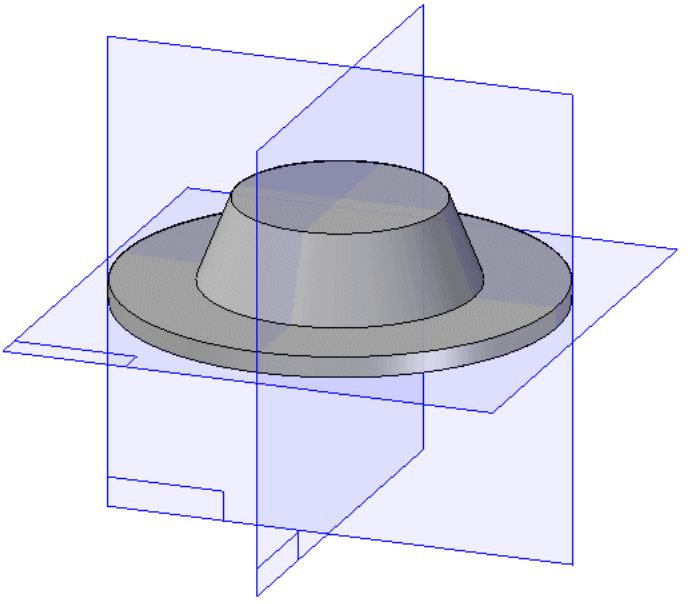
Chapter

3 Create a Part



The part you will construct is a shaft support for the garage door opener assembly.

Step 1: Construct the Base Feature

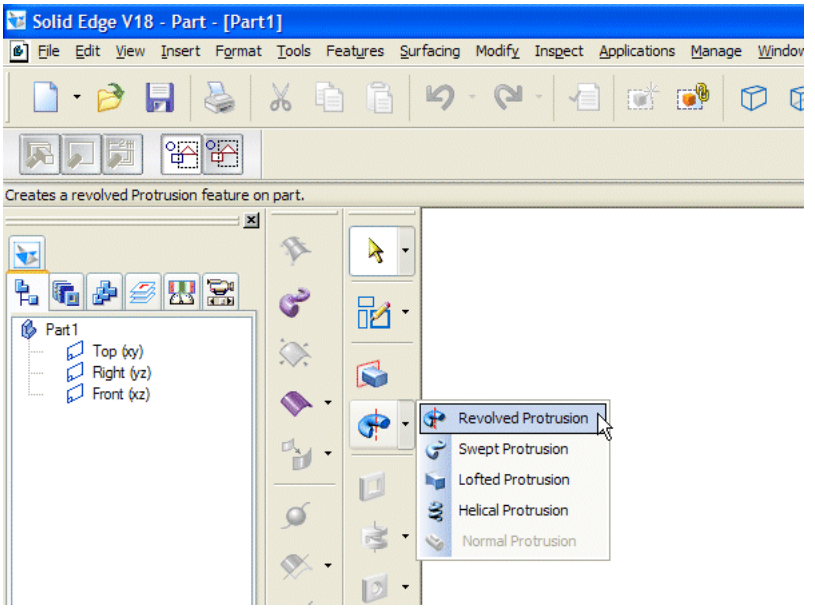


In Solid Edge, you model a part by constructing a series of features that add or remove solid material to produce the finished shape. The first feature you construct is called the base feature, and you start by deciding what type of feature you want to construct. Since this part is round and has a central axis, a revolved protrusion makes sense.

The feature construction commands are located on the Features toolbar. This is the vertical toolbar located along the left edge of the Solid Edge window.



On the Features toolbar, on the left side of the screen, you may see the Revolved Protrusion button or one of the other buttons shown in the illustration. Notice that whichever button is displayed, there is a small arrow to the right side of it. This arrow indicates that the button will display a flyout.



Position the cursor near the set of buttons shown in the illustration, and click the small arrow to invoke the flyout.

The flyout shows the related set of buttons that share the same location on the toolbar. Position the cursor over the Revolved Protrusion button, and click to select it.

The Revolved Protrusion command is now displayed on the Features toolbar, and the other buttons are hidden on the flyout.

When the command you want to select is shown, you do not need to expand the flyout in order to select it.

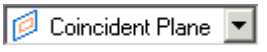
Notice that a new horizontal ribbon bar is displayed above the graphic window.



This is the Revolved Protrusion SmartStep ribbon bar. SmartStep walks you through the feature construction process step-by-step, guiding you as you proceed. SmartStep is one example of the technology in Solid Edge that makes it easier to learn and use.



The set of buttons on the left side of the ribbon bar shows that the first step of creating a revolved protrusion is drawing a profile or selecting an existing sketch of the profile.

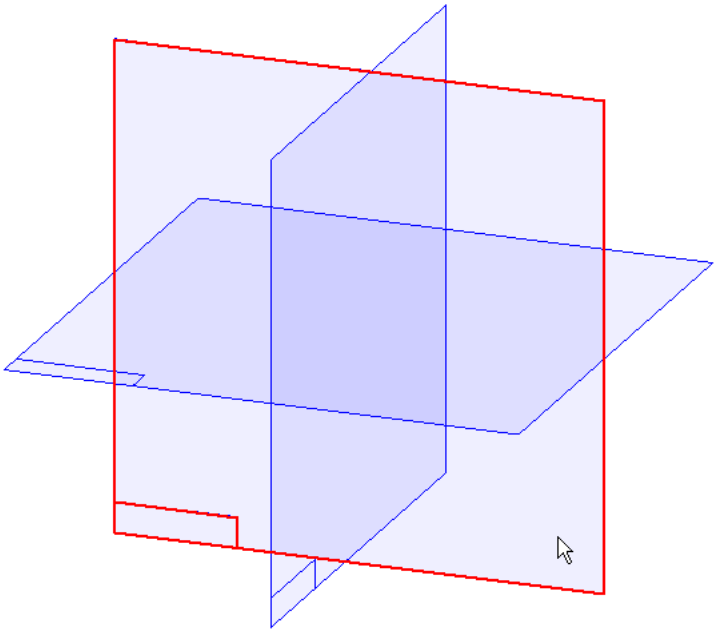


The next portion of the ribbon bar shows the options for how to define the profile plane where you will draw the shape of the revolved protrusion. You will select an existing plane, and the new profile plane will be coincident with it.

Step 2: Identify the Profile Plane

You will construct this feature by drawing a 2D sketch, called a profile, on a reference plane. A Solid Edge part document contains three default reference planes that are aligned with the top, front, and right views.

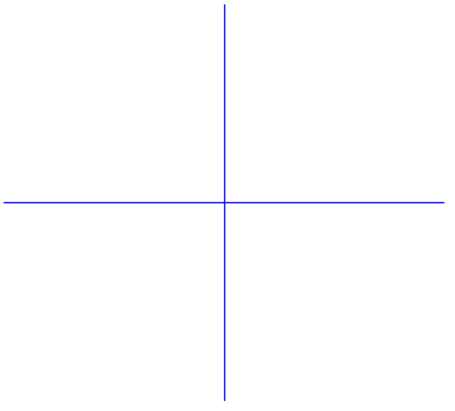
When you select a reference plane, Solid Edge displays a profile window containing a true view. A true view displays the reference plane parallel to the computer screen so you can draw the profile easily. At the same time, the toolbars and menus change. Buttons for constructing features are hidden, and buttons for drawing and dimensioning 2D shapes are displayed instead.



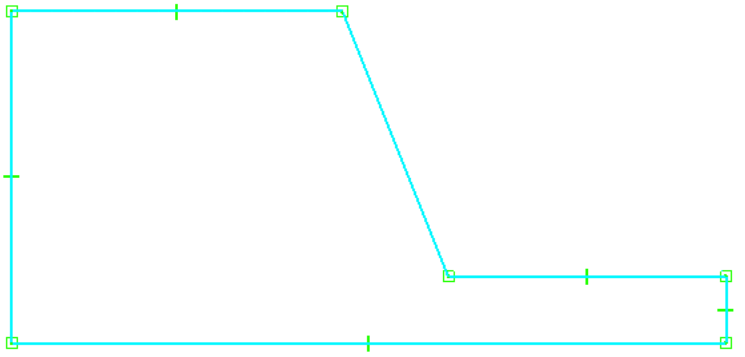
In the graphic window, move the cursor to the edge of the reference plane, as shown in the illustration. When the plane highlights, click the left mouse button.

Step 3: Draw the Profile

As the view transitions to be aligned with the profile plane, notice the horizontal and vertical lines that cross in the center of the window. These lines are actually the other two reference planes viewed on edge. When creating symmetrical parts, you can use these reference planes to help you take advantage of a part's symmetry.



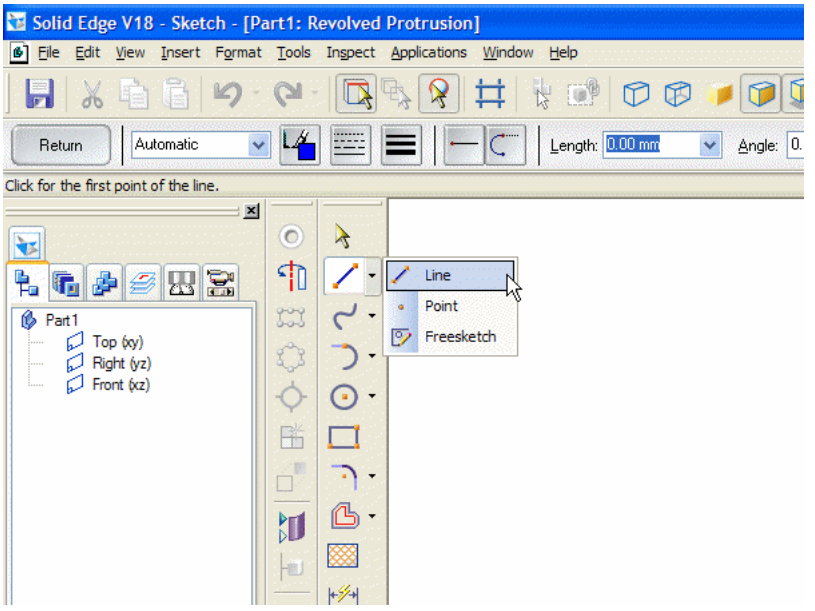
In the next few steps you will draw the profile shown in the illustration.





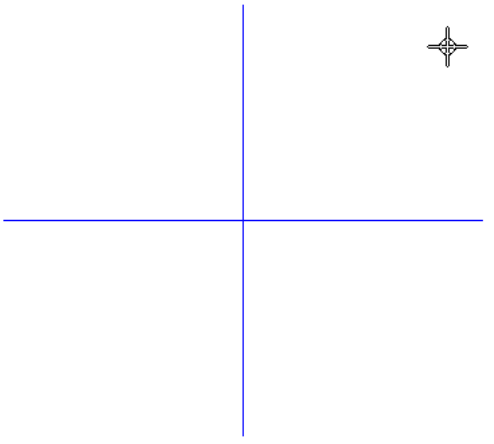
On the Draw toolbar, on the left side of the screen, click the Line button from the flyout, as shown.

When the command you want to select is shown, you do not need to expand the flyout in order to select it.



Step 4: Start the First Line

You draw lines by clicking the left mouse button once to start the line, then clicking again to complete the line. Move the cursor to the approximate location as shown in the illustration, then click to begin the line.

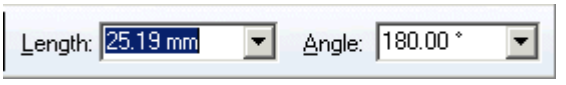


Move the cursor to the left, and notice the following:

- A line stretches to follow the cursor wherever you move it.
- When the line is vertical, a vertical relationship indicator is displayed next to the cursor.
- When the line is horizontal, a horizontal relationship indicator is displayed next to the cursor.

Step 5: Finish the First Line

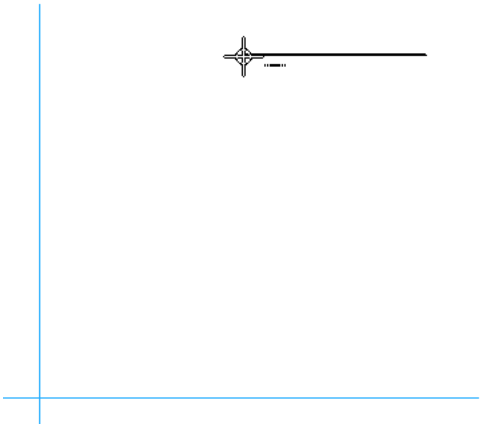
Notice the ribbon bar at the top of the window dynamically displays the current length and angle of the line. This feedback helps you quickly draw lines with precise lengths and angles.



Move the cursor to the left until:

- The horizontal relationship indicator is displayed near the cursor.
- The length displayed on the ribbon bar is approximately 25mm.
- The angle on the ribbon bar is 180 degrees.

Step 6: Click the left mouse button to finish the first line.

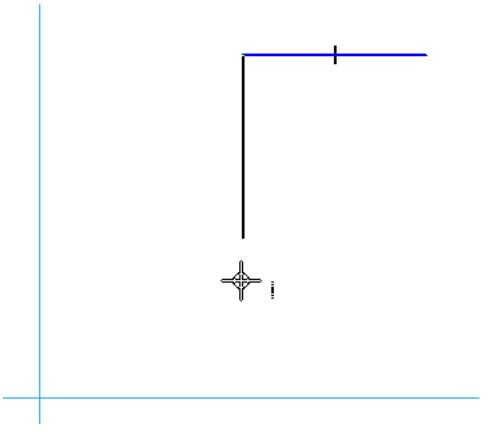


The Line command is still active, so the next line will start at the endpoint of the line you just drew.

Step 7: **Draw the Second Line**

You can also draw lines at a precise length and angle using the Line ribbon bar. If it is not already highlighted in blue, click the Length box on the ribbon bar.

- Type 25, and press the ENTER key on the keyboard. Notice that the Length box updates and the Angle box becomes active.
- Move the cursor around the screen and notice that the line length is locked at 25 mm.
- Move the cursor down until the line is nearly vertical.
- When the vertical relationship indicator is displayed , click to finish the second line.



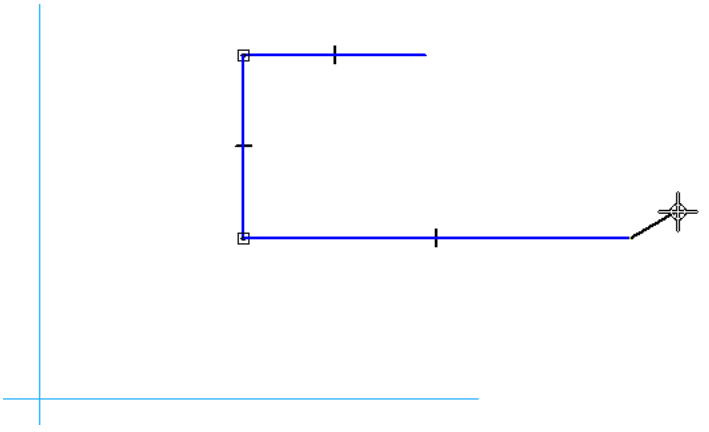
When you know the length and angle you want, this method of drawing is much faster and more accurate than using the mouse.

When typing values in a ribbon bar box, you do not have to enter the unit type, such as mm or degrees.

Step 8: Draw the Third Line

For the third line:

- Type 53 in the Length box and press ENTER.
- Type 0 in the Angle box and press ENTER.



Notice the immediate feedback showing you the type of relationship that is placed on the lines. Solid Edge displays all relationships, which allows you to easily predict the effect of design changes:

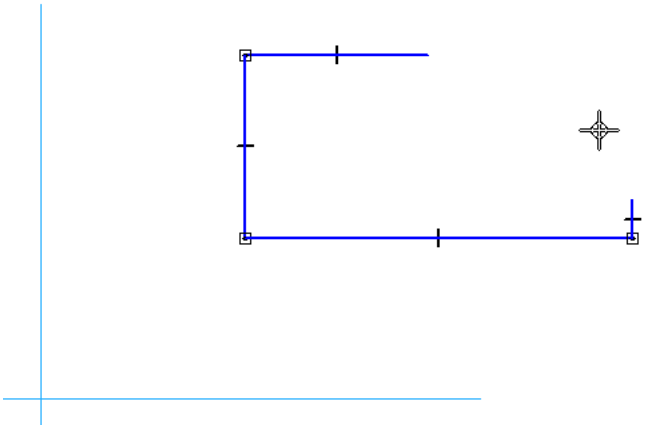
The horizontal and vertical symbols indicate that the lines will remain horizontal or vertical when the profile is edited.

The connect symbols at the endpoints indicate that the lines are endpoint connected.

Step 9: Draw the Fourth Line

For the fourth line:

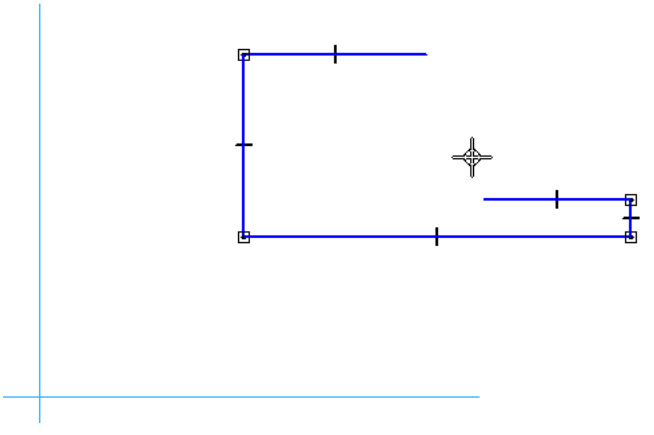
- Type 5 in the Length box and press ENTER.
- Type 90 in the Angle box and press ENTER.



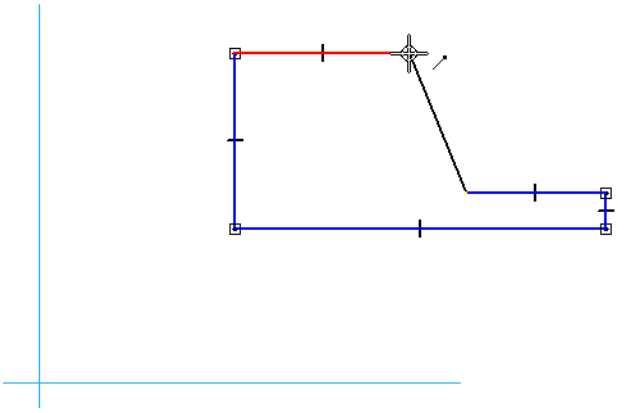
Step 10: Draw the Fifth Line

For the fifth line:

- Type 20 in the Length box and press ENTER.
- Type 180 in the Angle box and press ENTER.



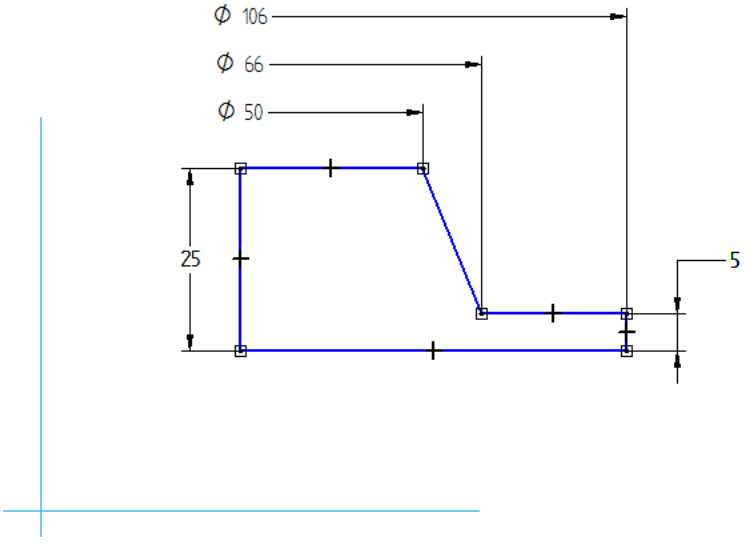
Step 11: Draw the Last Line



For the last line, move the cursor to the endpoint of the first line, as shown, and when the endpoint relationship indicator is displayed , click to place the line. The profile is now a closed shape, which is required for this feature.

Step 12: Add Dimensions to the Profile

In the next few steps, you will add dimensions to the profile.

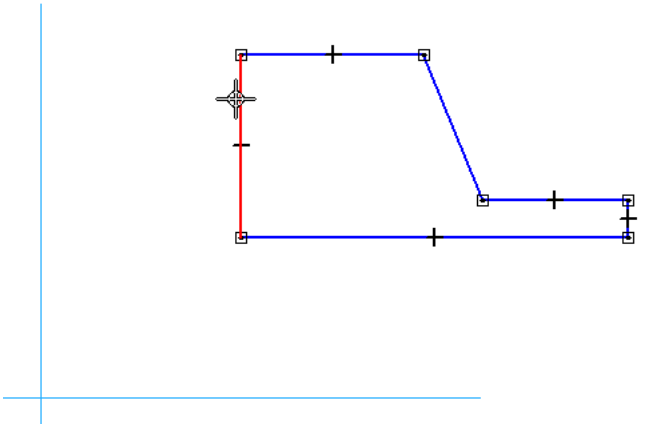




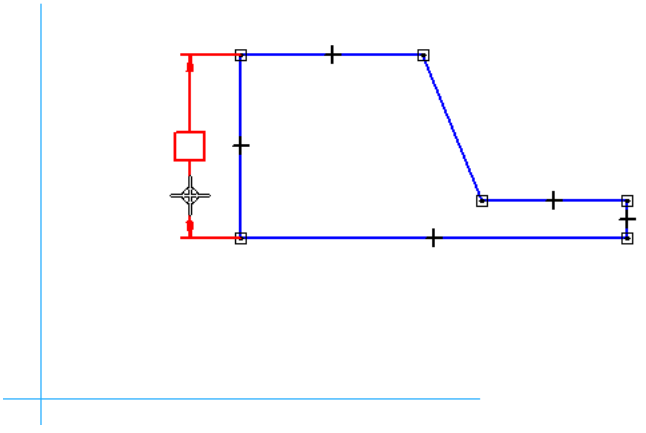
On the Draw toolbar, click the Smart Dimension button.

SmartDimension allows you to dimension lines, arcs, and circles with two clicks. The first click selects the element, and the second click places the dimension. You can also control the format of the dimension by repositioning the cursor.

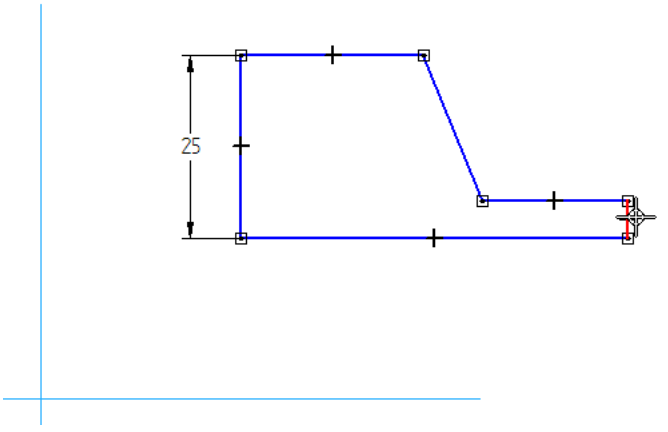
Step 13: In the profile window, position the cursor over the left, vertical line as shown, then click the left mouse button. Move the cursor to the left and notice that projection lines, dimension arrows, leader lines, and the dimension text box reposition dynamically, depending on where you move the cursor.



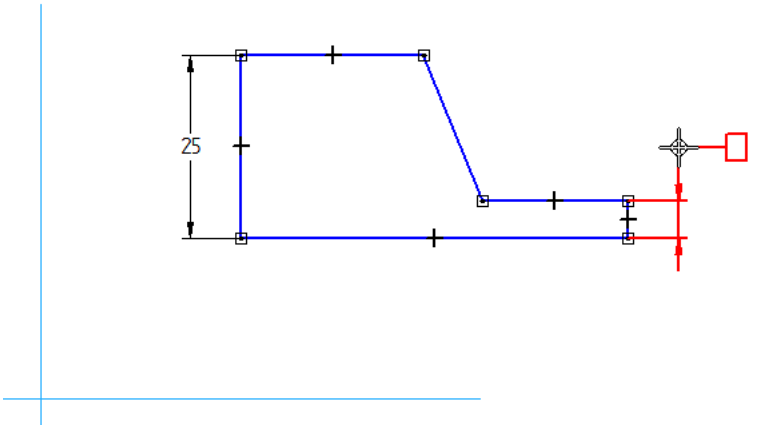
Step 14: Position the dimension as shown in the illustration, and click to place it.



Step 15: Use SmartDimension to dimension the right, vertical line. Position the cursor over the line as shown, then click to select the line.

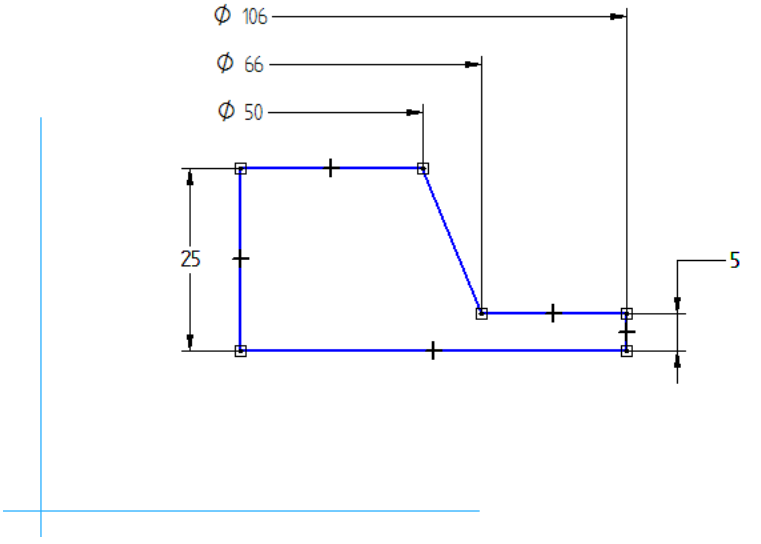


Step 16: Position the cursor above the profile and notice that the dimension format changes based on the cursor position. Click to place the dimension.



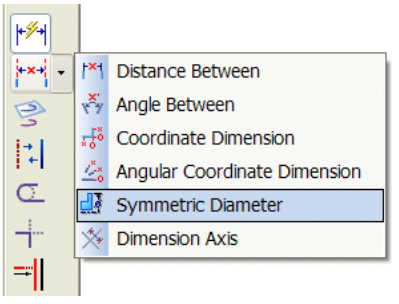
Step 17: **Add Dimensions Between Elements**

Since this feature is a revolved protrusion, you need to add dimensions that control the profile radially.



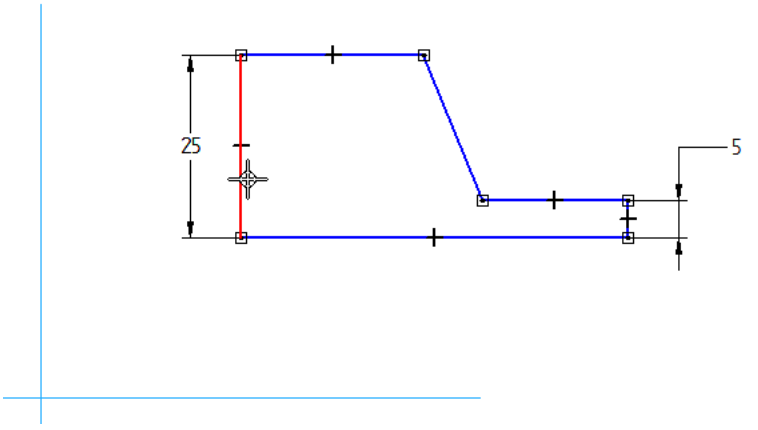


On the Draw toolbar, click the Symmetric Diameter button from the flyout. This command places radial dimensions that measure distances between elements or keypoints.

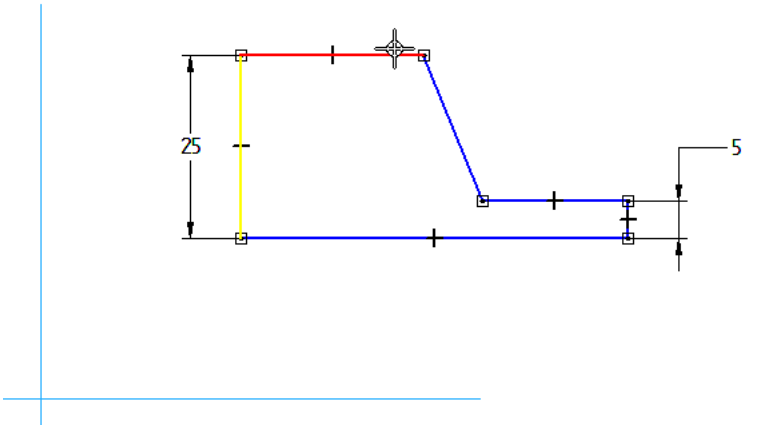


Step 18: You will use the Symmetric Diameter command to place a series of dimensions that have a common origin element.

Position the cursor over the left, vertical line as shown, since this line represents the axis of revolution. It is not necessary to select the line endpoint. When the line highlights, click the left mouse button to define the dimension origin.

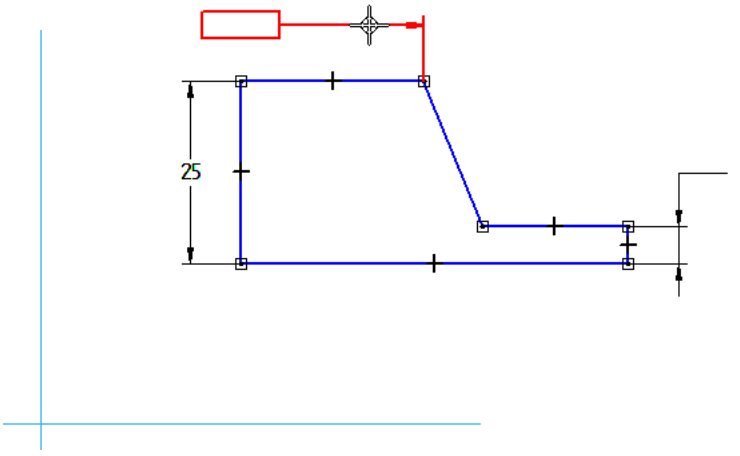


Step 19: Position the cursor near the right end of the top, horizontal line as shown, then click. Again, it is not necessary to select the end of the line. Just position the cursor near the right end of the line, and Solid Edge will know that you want to dimension the endpoint nearest to the cursor.

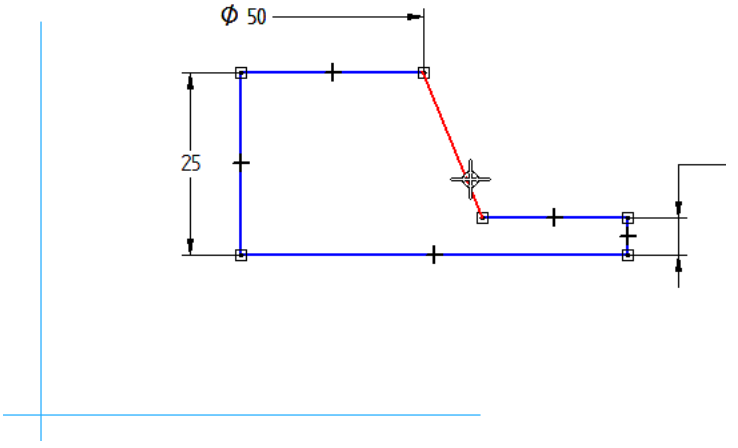


Step 20: Move the cursor above the profile, as shown, then click to place the dimension. You can edit dimensions as you place them. Notice that the Value field for the dimension you placed is active on the Dimension ribbon bar.

If the value is not exactly 50 mm, then type 50 and press the ENTER key.

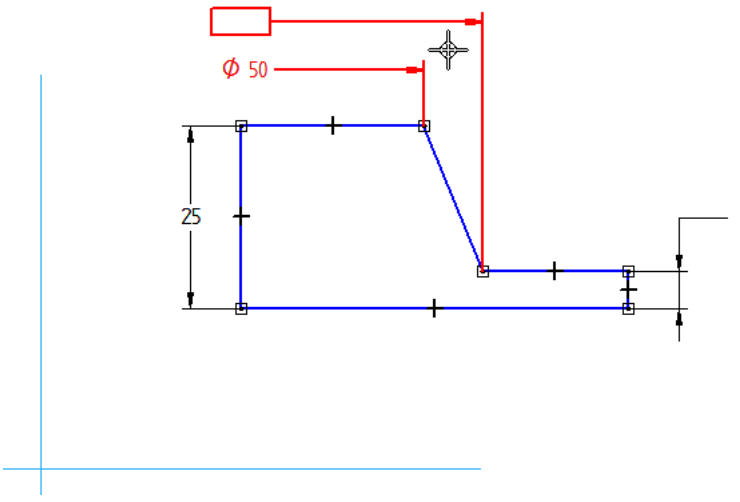


Step 21: The Symmetric Diameter command is still active. Position the cursor near the right end of the angled line as shown, then click.

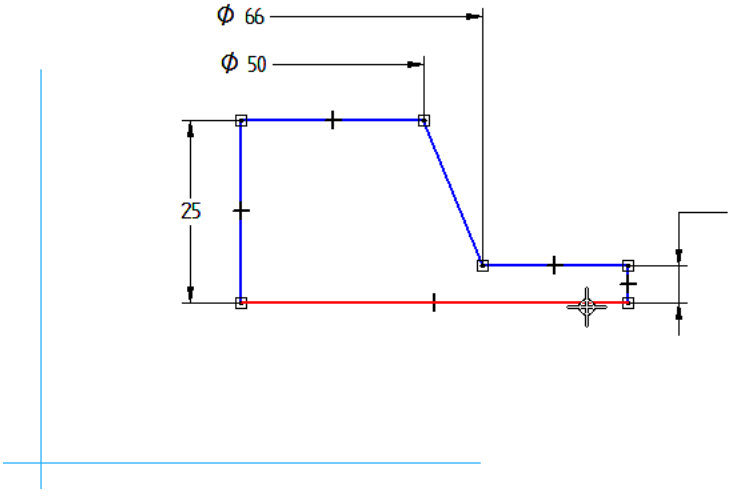


Step 22: Move the cursor above the profile, until the new dimension is above the previous dimension, then click.

If the value is not exactly 66 mm, then type 66 and press the ENTER key.

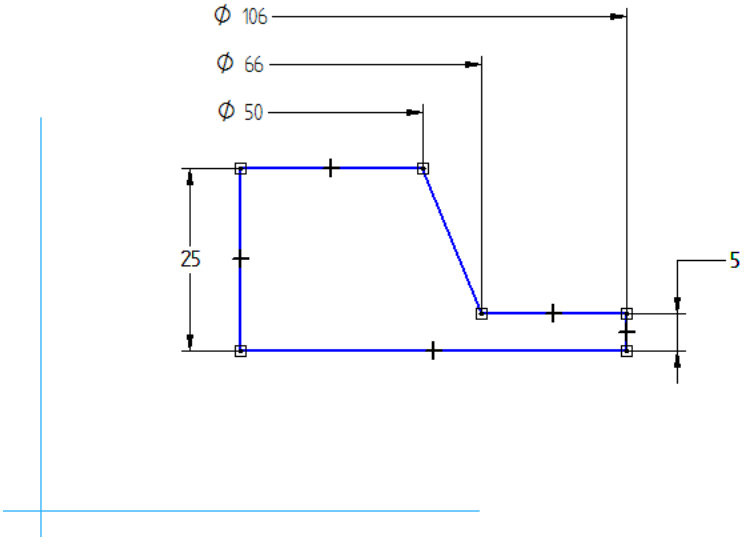


Step 23: Position the cursor near the right end of the bottom, horizontal line as shown, then click.



Step 24: Position the cursor above the 66 millimeter dimension, then click to position the last dimension.

If the value is not exactly 106 millimeters, then on the Dimension ribbon bar type 106 and press the ENTER key.



Step 25: **Assign a Variable Name to the Last Dimension**

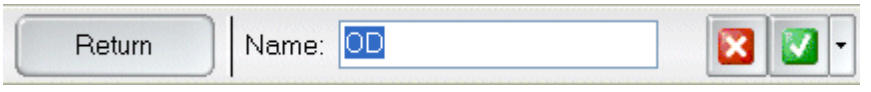
You can assign unique names to dimensions in Solid Edge. Assigning unique names makes it easier to edit your designs later. It also makes it easier to define formulas and equations to help control your designs.



On the Draw toolbar, click the Select Tool button. The Select Tool lets you edit things you have already created. You will use it to modify a dimension.

Position the cursor over the 106 millimeter dimension and click the right mouse button to display the shortcut menu. On the shortcut menu, click Edit Formula to display the Edit Formula ribbon bar.

On the Edit Formula ribbon bar, in the Name box, type OD, then press the ENTER key.



Press the ENTER key a second time, and the new dimension name is applied.

The new name will remind you later that this dimension controls the outside diameter of the part. You will edit this dimension later when you work with the rest of the assembly.

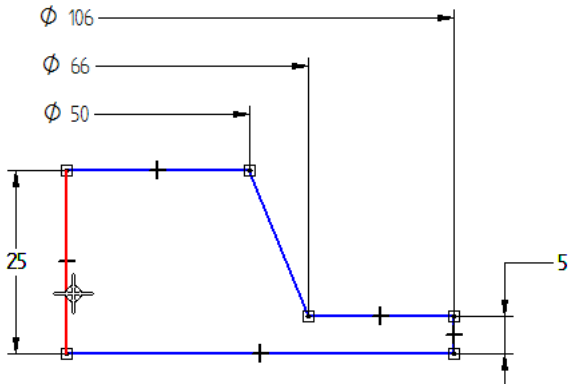
Step 26: **Define the Axis of Rotation**

The last thing you need to do is define the axis that the profile is revolved around. You can use an edge of the profile or a reference plane.



On the Features and Relationships toolbar, click the Axis of Revolution button.

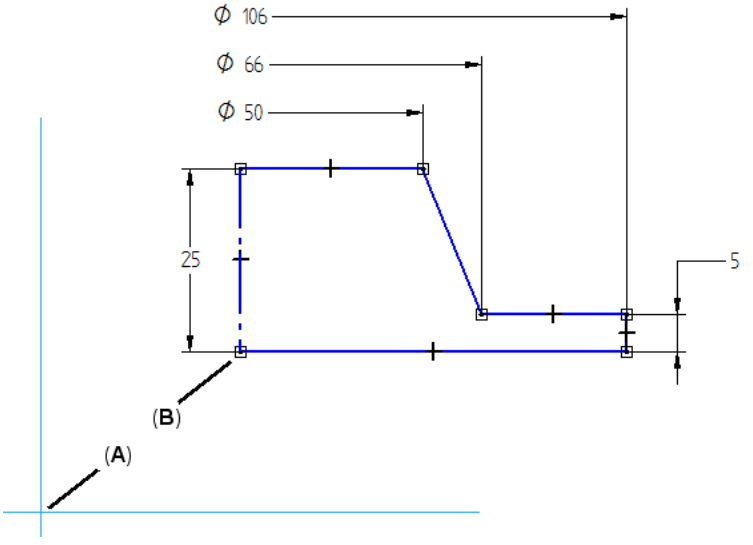
Position the cursor over the left vertical line shown highlighted in the illustration, then click to select it. Notice that the line style changes to a dashed display to indicate that this line is the axis of revolution.



Step 27: Connect the Profile to the Center of the Coordinate System

All the relationships and dimensions that define the size and shape of the profile are there, but the profile is still free to float around on the reference plane. This is not a problem, but since this part is symmetrical, it makes sense to position the profile to best take advantage of its symmetry later.

In the next few steps, you will use a geometric relationship to move the profile to the center of the reference planes. Notice the two lines that form a cross (A). As discussed earlier, these lines are actually the edges of the other two default reference planes. You will use the connect relationship to move the profile so that the bottom left corner of the profile (B) is precisely connected to the center of the reference plane origin (A).

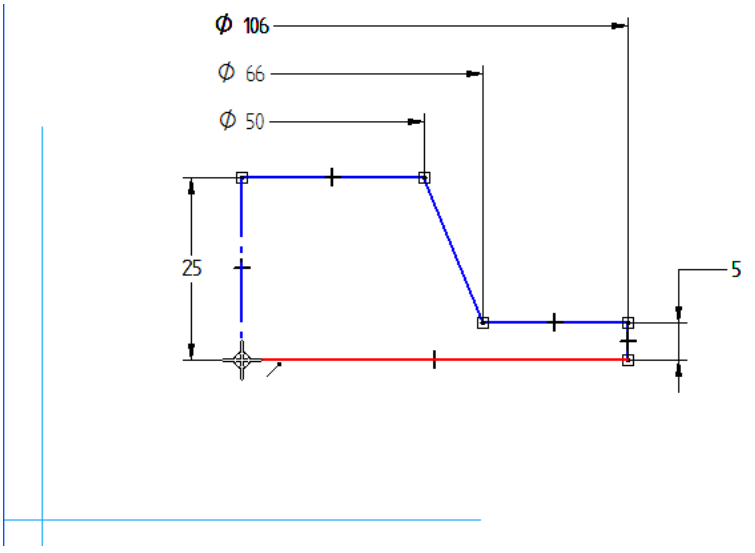




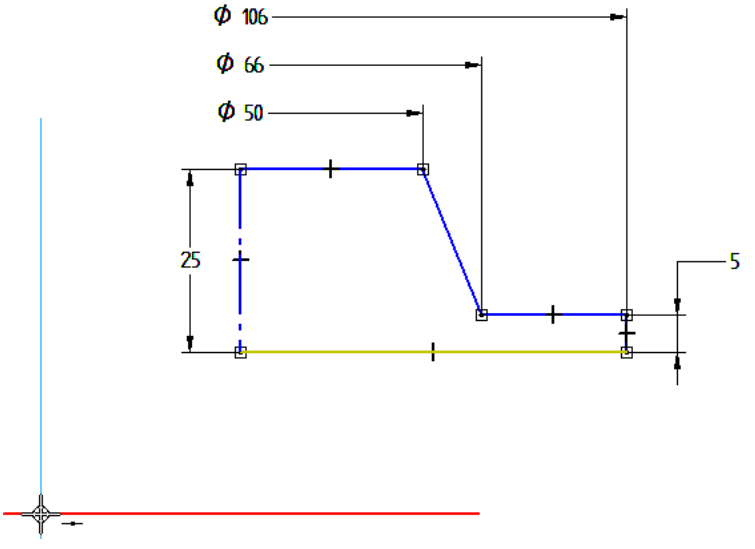
On the Features and Relationships toolbar, click the Connect button.

The Connect command places a relationship between two elements that ensures the two elements remain connected when you make other changes to the model.

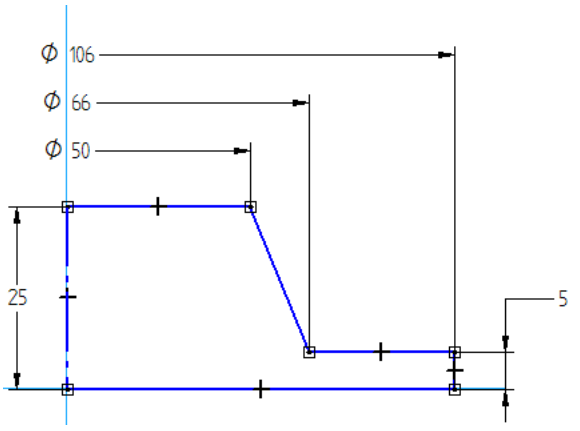
Position the cursor over the endpoint shown in the illustration, and when the endpoint relationship indicator is displayed, click.



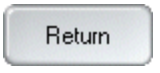
Step 28: Move the cursor to the intersection of the two reference planes. It does not matter which reference plane edge highlights, but notice the midpoint indicator symbol in the illustration. When it is displayed, click.



This moves the profile to the exact center, or origin, of the reference planes, as shown.



Step 29: Finish the Profile



The profile is now complete, so on the ribbon bar, click the Return button.

Step 30: Define the Extent for the Revolved Protrusion

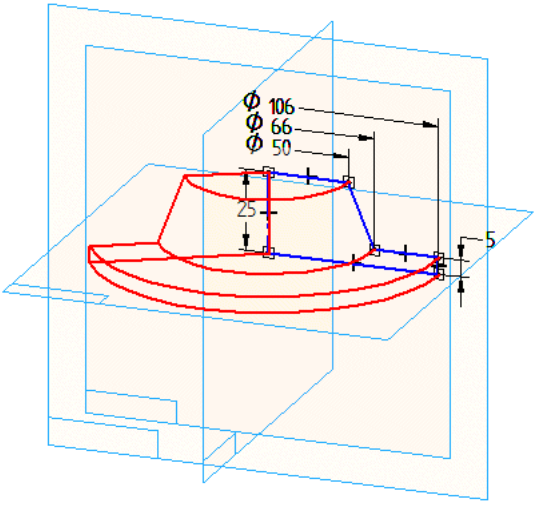
Clicking Return dismisses the profile window and returns you to the Part window.





The Revolved Protrusion SmartStep ribbon bar shows you that the next step is to define the extent of the feature.

Move the cursor around in the window and notice that you can dynamically define the extent.

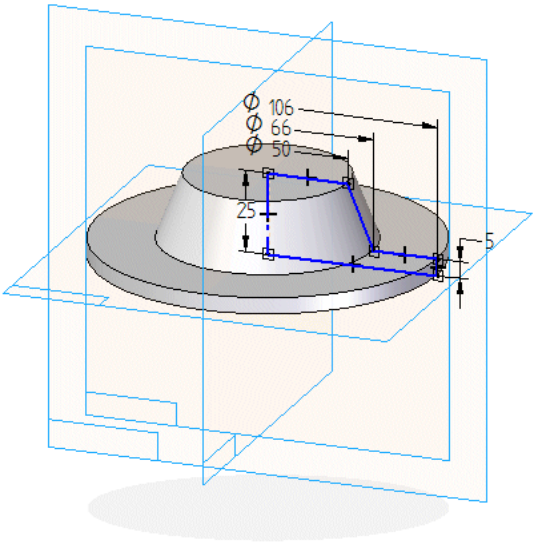




For this feature, you want a 360 degree revolution, so on the SmartStep ribbon bar, click the Revolve 360 button.

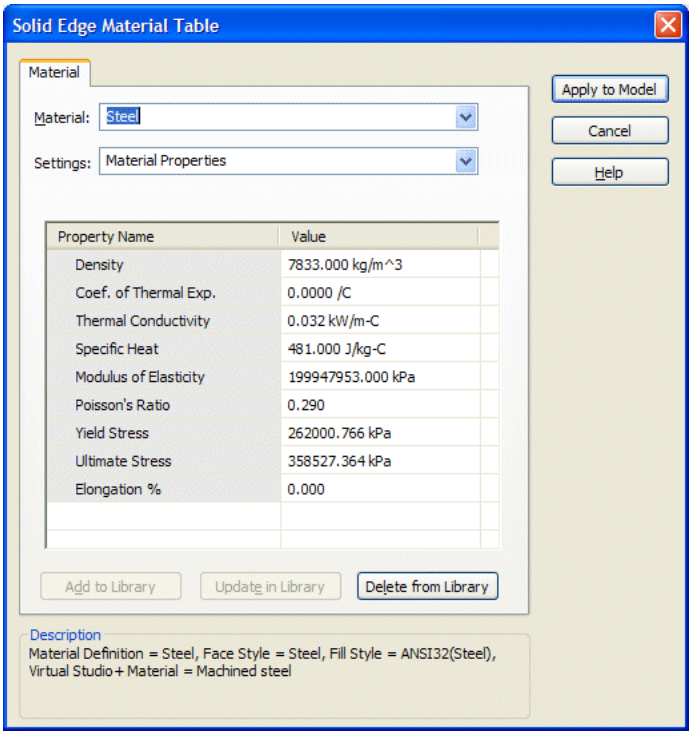


On the SmartStep ribbon bar, click the Finish button.



Step 31: Specify the Material

On the Tools menu, click Material Table.



On the Material Table dialog box, Select Steel from the Material list, and click Apply to Model.

Step 32: Save the Part



On the Main toolbar, click the Save button to save the work you have done so far.

The first time you save a new document in Solid Edge, the Properties dialog box is displayed. This allows you to add any custom properties you may want for a parts list, bill of materials, or the integrated Document Management functions in Solid Edge. Later you will create an assembly drawing with a parts list, so you will add these properties now.

- On the Summary tab, in the Title box, type: SUPPORT, SHAFT in upper case letters, then press the TAB key.
- On the Summary tab, in the Author box, type your last name in upper case letters, then press the TAB key. For example, if your last name is Smith, type: SMITH, then press the TAB key.
- On the Project tab, in the Document Number box, type SS101, then press the TAB key.
- On the Properties dialog box, click OK.



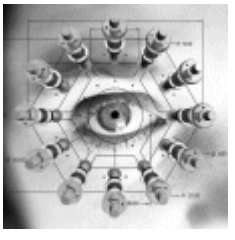
When typing values in a dialog box, you should press the TAB key. If you press the ENTER key, the dialog box is dismissed.

After you click OK, the Properties dialog box is dismissed, and the Save As dialog box is displayed. On the Save As dialog box, do the following:

- In the Save In box, set the save in location to the Solid Edge V18\Training\Evaluation folder.
- In the File Name box, type Support1.par, then click the Save button.

The default location for the evaluation folder is:

C:\PROGRAM FILES\SOLID EDGE V18\TRAINING\EVALUATION. However, your system administrator may have chosen a different location.



With Solid Edge Insight, saving new documents is the same, whether the documents are managed or not. For managed documents, you simply specify a save location that is a managed folder on your server. The document is copied to the server and marked as Checked Out and you continue working. The document properties you just defined (Title, Author, Document Number, and Material) are called metadata, which can be shared with other users of the managed workspace.

Step 33: Unshade the Part

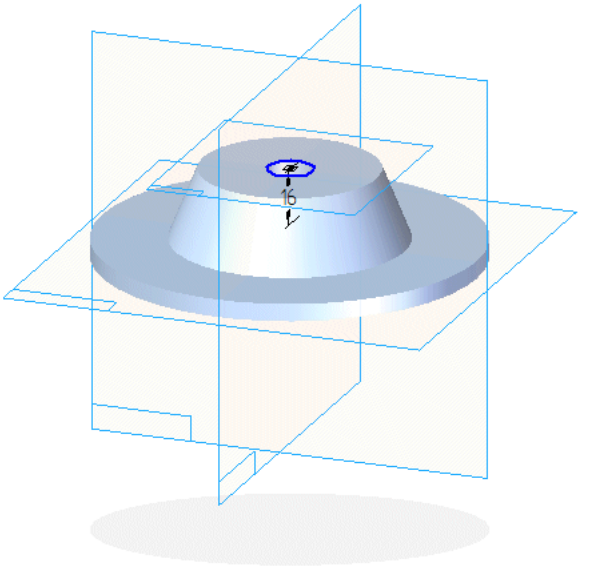
In the next few steps you will construct features in the interior of the part, and these will be easier to see with the part displayed in the Visible and Hidden Edges wireframe display.



On the Main toolbar, click the Visible and Hidden Edges button to display the part in wireframe mode.

Step 34: Construct a Hole

The next feature you will construct is a simple hole in the center of the part.





On the Features toolbar, click the Hole button from the flyout.

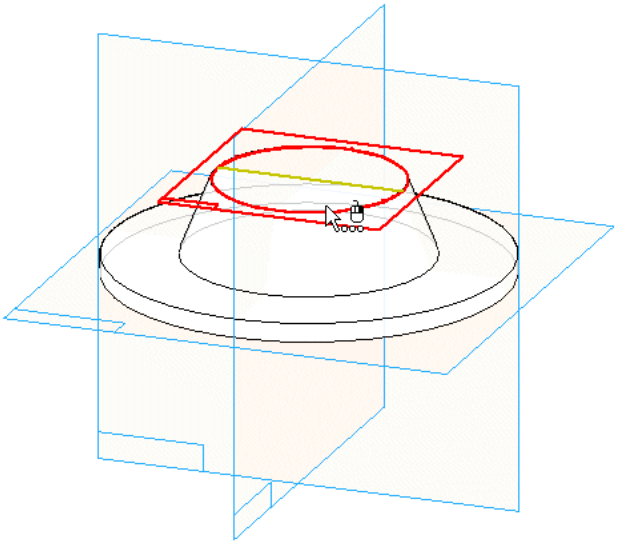


Step 35: Create a New Reference Plane



The Hole SmartStep ribbon bar shows that the Plane step is active. The first step of constructing a Hole feature is defining the plane where the hole profile will be drawn. For this feature, you will define a reference plane using the top face of the part.

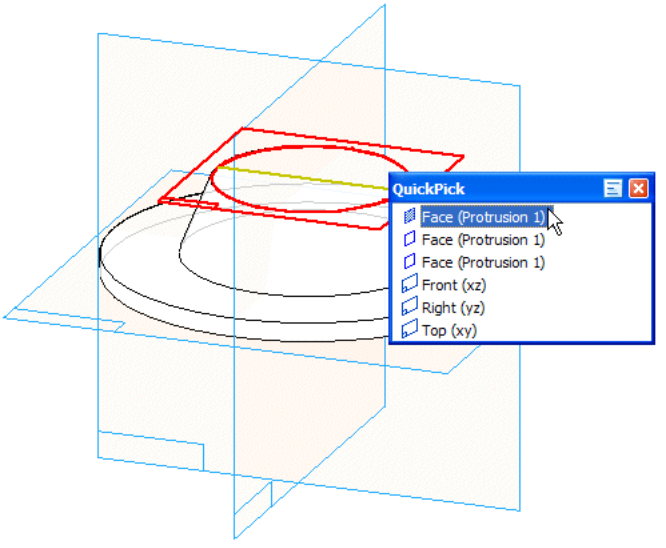
On the SmartStep ribbon bar, the Coincident Plane option is set by default. With this method of defining the profile plane, Solid Edge will create a reference plane for you, coincident with any face of the part you choose.





In the graphic window, position the cursor over the top face of the part, stop moving the cursor for a moment, and notice that the cursor changes shape.

Notice that the face highlights and a new reference plane coincident with the face is displayed. Click the right mouse button, and the QuickPick tool is displayed.



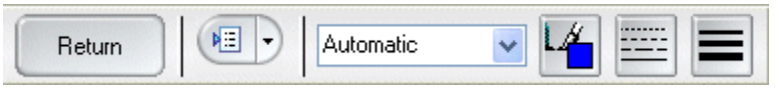
Move the cursor over the different entries in QuickPick, and notice that different faces of the model highlight. QuickPick allows you to select exactly the face you want, the first time, without having to reject unwanted faces.

Use QuickPick to highlight the face shown in the illustration. When the circular face and reference plane highlight as shown in the illustration, click the left mouse button in QuickPick to select them.

Step 36: Explore the Hole Options Dialog Box

The view is reoriented to the new profile plane, and the Draw and Feature and Relationships toolbars are displayed to provide commands appropriate for drawing the hole profile.

The Hole Circle command is started by default, and notice that a hole circle is already attached to the cursor. For the hole you are constructing, you need to specify a different sized hole.





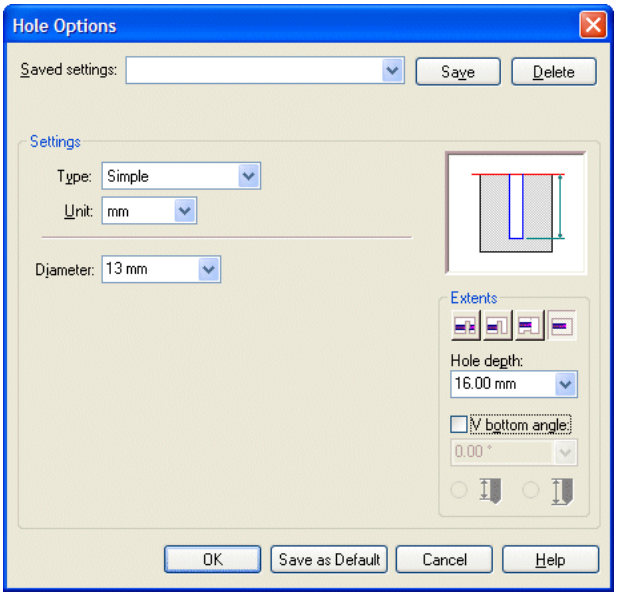
On the Hole ribbon bar, click the Hole Options button.

Take a few moments to explore the Hole Options dialog box. In the Settings frame, notice that in addition to the Simple type hole you are constructing, you can also construct more complex hole types with the Hole command.

Notice that on the right side of the dialog box, there is a Preview pane that shows the parameters of the different hole types. In the Type list, select the individual options for the various hole types listed below and notice that the Preview updates:

- Threaded
- Tapered
- Counterbore
- Countersink

Step 37: Set the Hole Type back to Simple.



As you can see, the Hole command is a very powerful tool. You can construct very complex hole geometry very easily.

On the Hole Options dialog box, set the following options:

- In the Hole Diameter box, type 13, then press the TAB key.
- Ensure that the Extents option is set to Finite Extent.

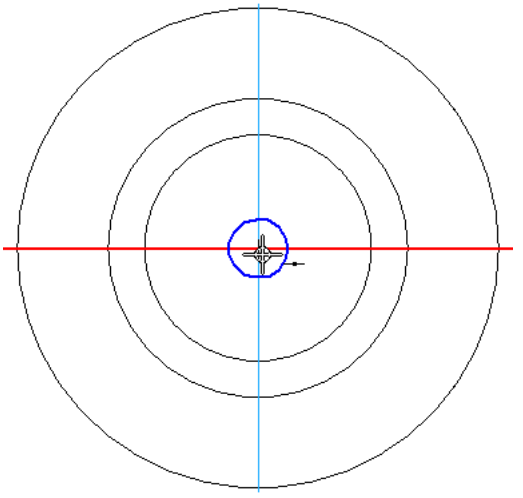


The Extents options are on the right side of the Hole Options dialog box.

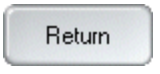
- In the Hole Depth box, type 16, then press the TAB key.

Step 38: On the Hole Options dialog box, click OK.

Position the cursor over the center of part, as shown in the illustration, but do not click. Notice that either the horizontal or vertical reference plane highlights, and that a mid-point relationship indicator is displayed. It does not matter which reference plane highlights; just click to position the circle at the center of the part.



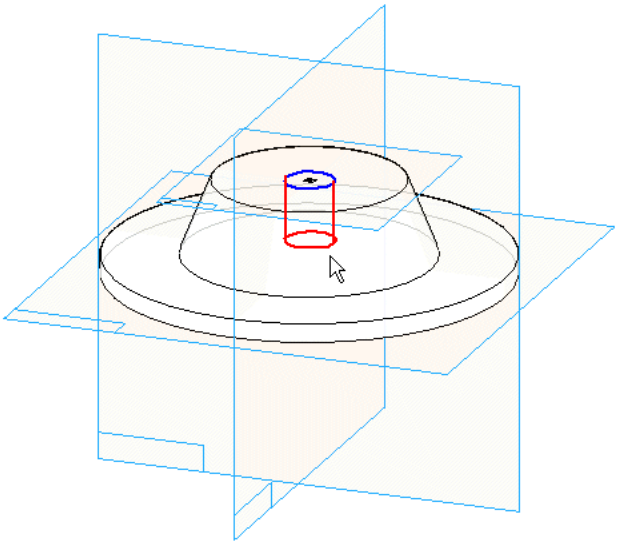
Step 39: The profile is now complete.



On the SmartStep ribbon bar, click the Return button.

The view is reoriented, and the Features toolbar is displayed. The Hole SmartStep ribbon bar shows that the next step in defining the hole is specifying the extent of the hole.

Step 40: Position the cursor so that the hole points into the part as shown, then click.



Notice that again, Solid Edge does some of the work for you. A dimension for the hole extent was added for you.

Step 41: The feature is now complete.



On the SmartStep ribbon bar, click the Finish button.

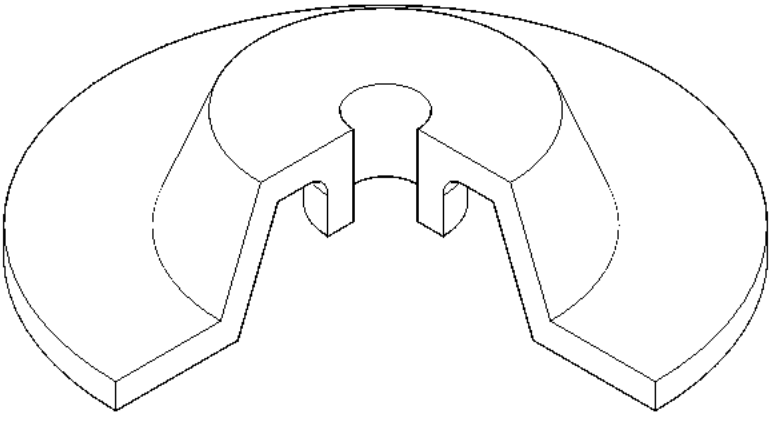
Step 42: **Save the Document**



On the Main toolbar, click the Save button to save the work you have done so far.

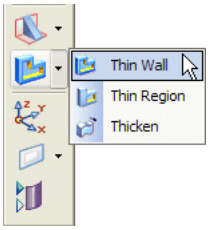
Step 43: **Thin-Wall the Part**

You will use the Thin Wall command to hollow the part out a bit, as shown in the cut-away illustration.





On the Features toolbar, click the Thin Wall button from the flyout.

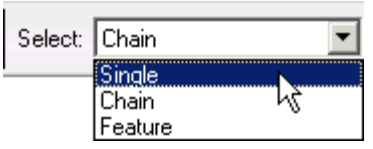


For this thin wall feature, first you will define the wall thickness, and then select the open faces.

On the Thin Wall SmartStep ribbon bar, in the Common Thickness box, type 5, then press the ENTER key.



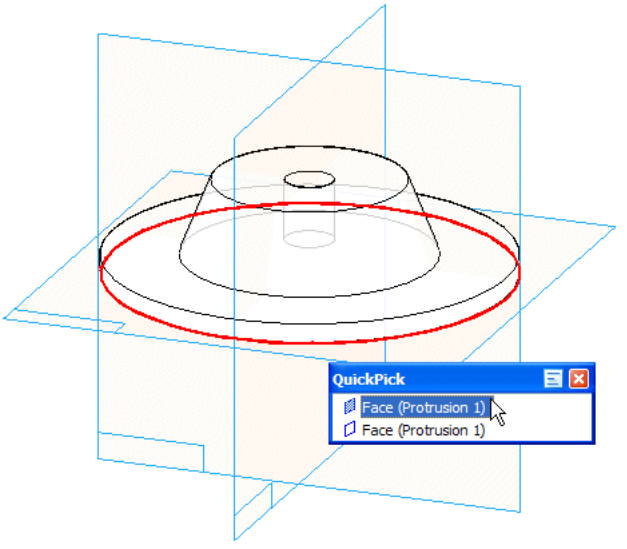
On the SmartStep ribbon bar, in the Select list, set the Single option.



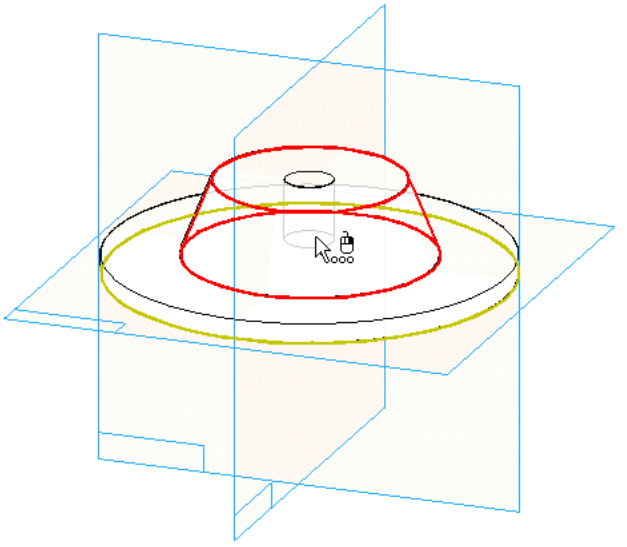
Step 44: Select Open Faces

When the Thin Wall command makes the interior of the part hollow, you have the option to remove some faces of the part, exposing the hollow area. You will make the bottom of the part hollow.

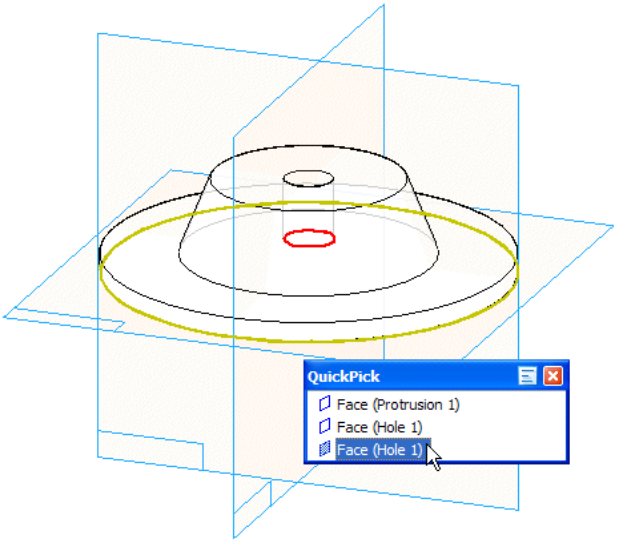
Position the cursor along the bottom edge as shown, and when the bottom face highlights, click.



Step 45: Now position the cursor over the bottom of the hole you placed earlier.



When the QuickPick prompt is displayed, click the right mouse button to display the QuickPick dialog box. Select the bottom of the hole. The command will leave this face open when the thin wall is created.



Step 46: You have provided all the information needed to complete the feature.



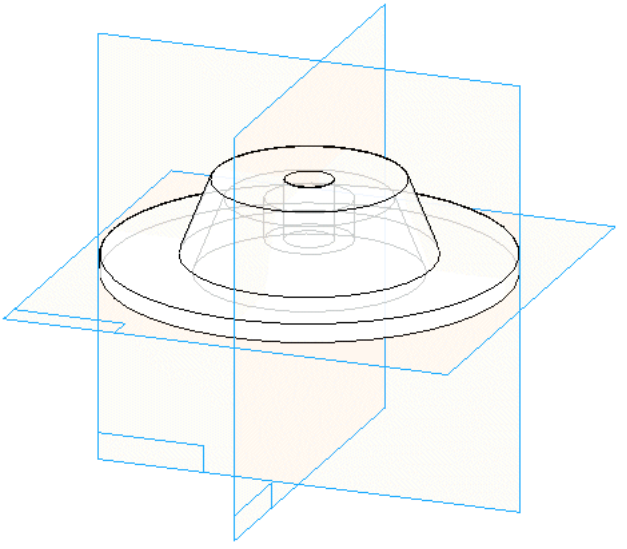
On the SmartStep ribbon bar, click the Accept button.



On the SmartStep ribbon bar, click the Preview button.



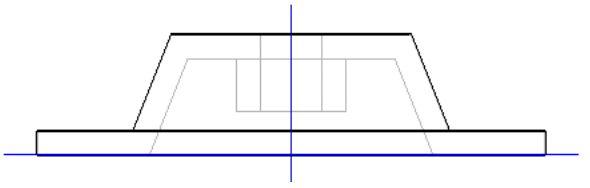
Click the Finish button to complete the feature.



Step 47: **Look More Closely at this Feature**

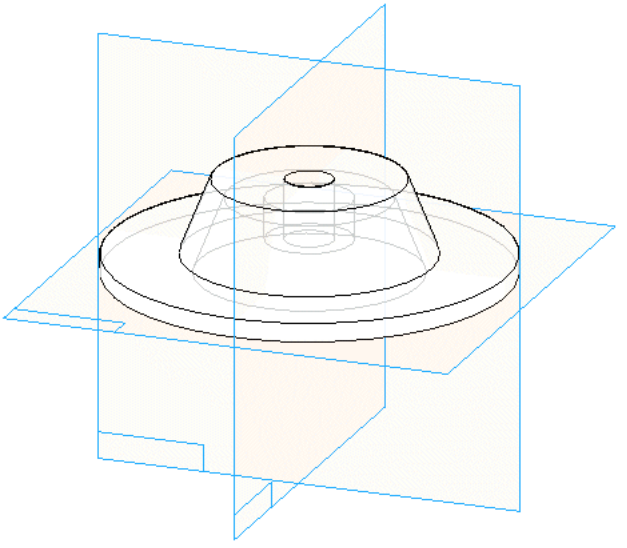
If you look closely at the thin wall feature, you may notice something. Rotating the view will make it more apparent.

On the keyboard, press and hold the CTRL key down, then press the F key. This rotates the view to the front orientation. Notice that the thin wall feature added material around the hole to form a boss, and the hole goes completely through the boss.



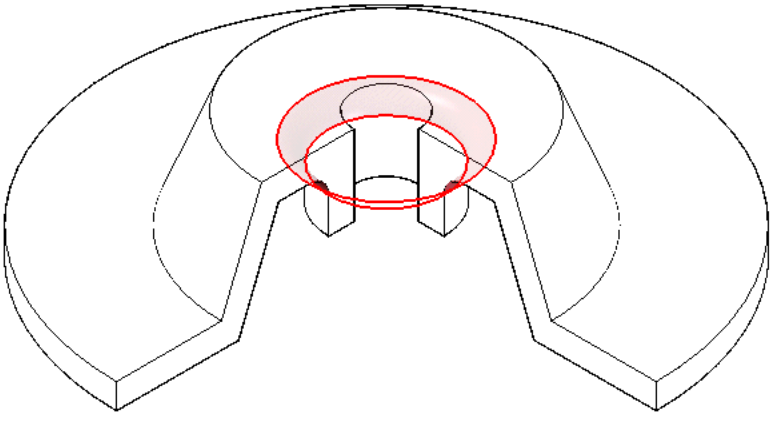
Step 48: Return to the Dimetric View

On the keyboard, press and hold the CTRL key down, then press the J key to reorient the view to dimetric perspective.



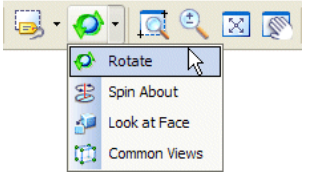
Step 49: Round an Edge

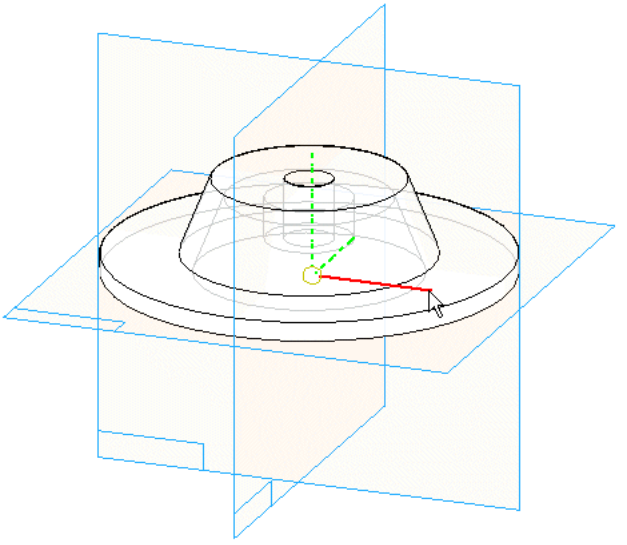
In the next few steps, you will round an edge on the part, as shown in the cut-away illustration. To make it easier to view the edge, you will rotate the view first.





On the Main toolbar, click the Rotate button from the flyout.

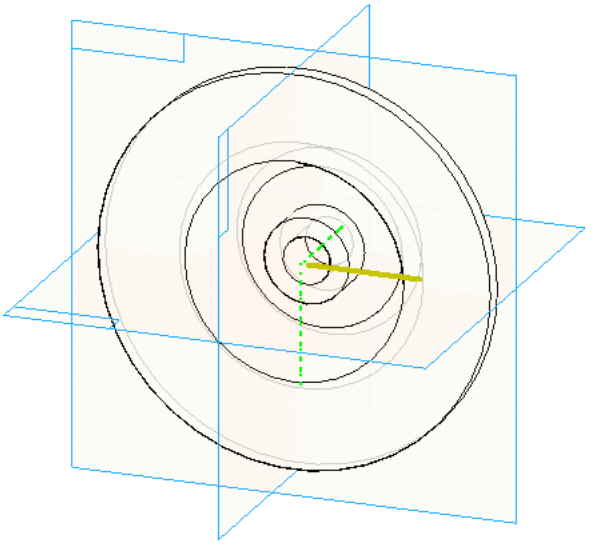




Notice that three axes display. The axes allow you to specify which axis you want to rotate the view around. You can select an axis and rotate the view dynamically or you can type a value on the Rotate ribbon bar.

Position the cursor over the axis shown, then click to select the axis.

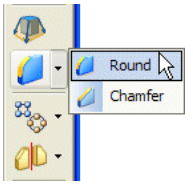
Step 50: On the Rotate ribbon bar, type —90, then press the ENTER key. The view rotates as shown in the illustration. Click the Close button on the Rotate ribbon bar to dismiss the Rotate command.



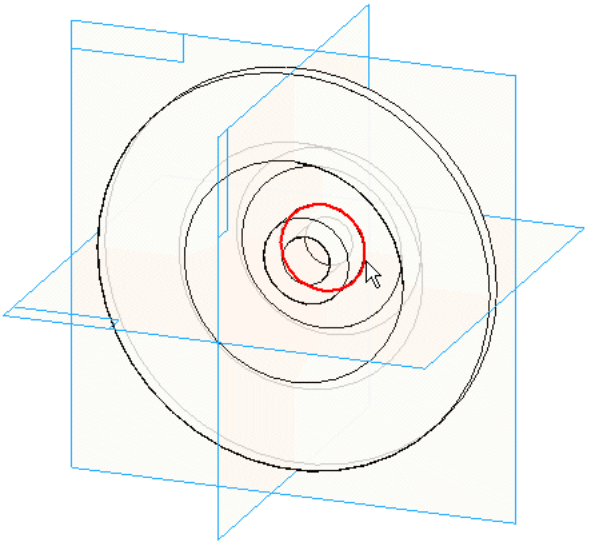
Step 51: The edge you will round is now visible.



On the Features toolbar, click the Round button from the flyout.



Use QuickPick to select the circular edge shown in the illustration.





On the Round SmartStep ribbon bar, in the Radius box, type 4, then either press the ENTER key or click the Accept button.

Step 52: You have provided all the information necessary to complete the feature.



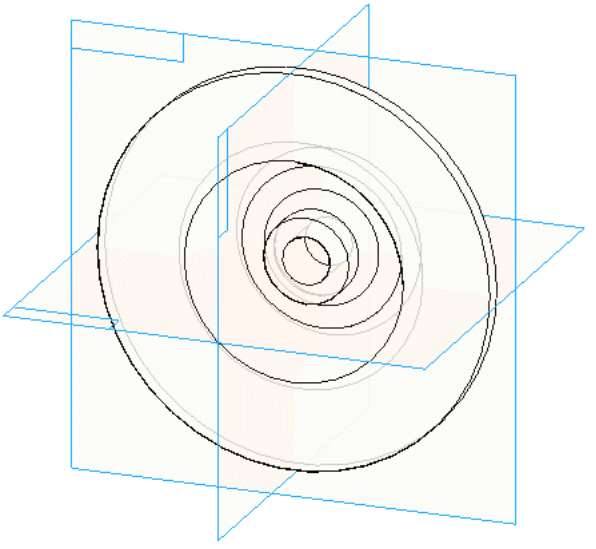
On the SmartStep ribbon bar, click the Preview button.



The feature is now complete. On the SmartStep ribbon bar, click the Finish button.

Step 53: Shade the Part

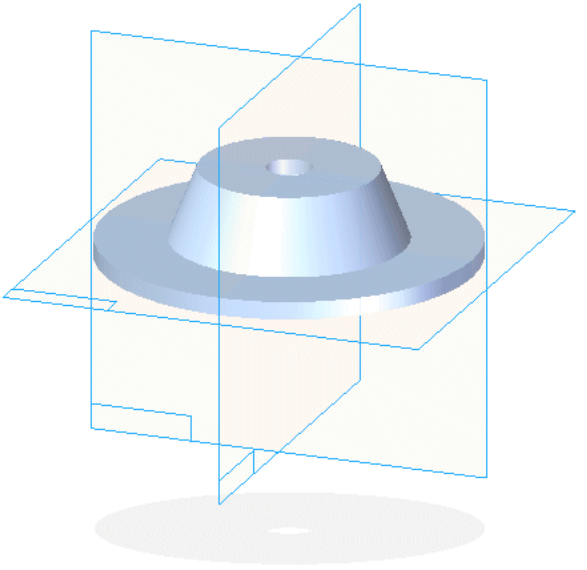
The part is now complete, but before you place the part in the assembly, you will display a shaded image of the part and return the view to the dimetric orientation.





On the Main toolbar, click the Shaded button.

Step 54: On the keyboard, press and hold the CTRL key down, then press the J key to return to the dimetric orientation.



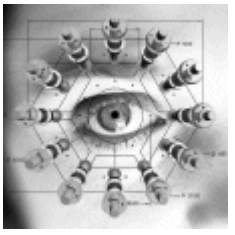
Step 55: Save and Close the Document



On the Main toolbar, click the Save button to save the work you have done so far.

On the File menu, click Close to close the Support1.par document.

Although the part still needs a set of mounting holes, you will construct those later, after you place the part into the assembly.



When you close a managed document, it can be automatically checked in to the server and published. Publishing means that the document has been made available to other users, and that its version history is maintained.

Chapter

4 Build an Assembly

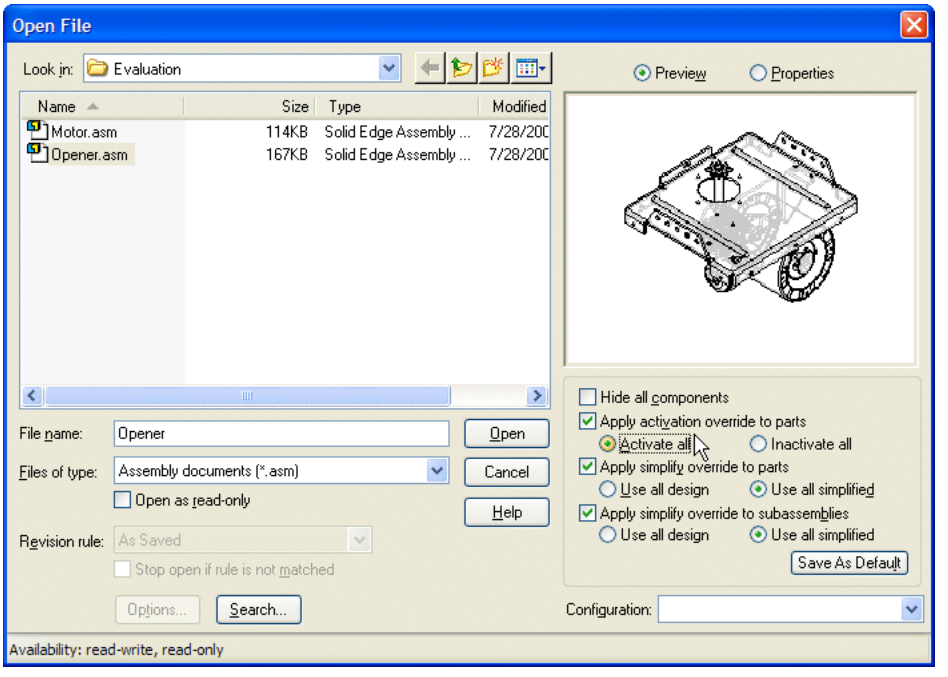
Now that you have modeled the support part, you will place it into the assembly. After you add the mounting holes using one of the other parts in the assembly, you will place a mounting bolt, and then pattern the bolt to complete the assembly. You will then create an exploded view of the assembly that you will use in the drawing.

Step 1: Open the assembly



On the Main toolbar, click the Open button.

On the Open File dialog box, do the following:



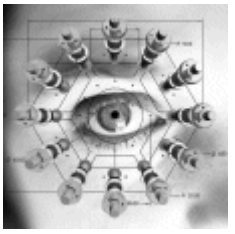
- Set the Look In location to the Solid Edge Evaluation folder. The default location of the Solid Edge Evaluation folder is:

  C:\PROGRAM FILES\SOLID EDGE V18\TRAINING\EVALUATION

- Set the Files Of Type option to Assembly Document (*.asm)

- Ensure the Apply Activation Override to Parts option is set and choose the Activate All Parts option.

- Select Opener.asm from the Solid Edge Evaluation folder, then click the Open button.



The real-world task of opening an assembly from a collaborative project involves searching for the correct and most up-to-date version of a design. Insight ensures that only the most recent versions are available. The metadata is indexed by Insight, which makes it easy to search for and quickly find the documents you need, which reduces a project's costs and time to market.

In the next few steps, you will place the SUPPORT1.PAR part into the final assembly. The support part will be used to support and align the vertical shaft that protrudes through the large hole in the sheet metal part.

Step 2: Arrange the display

In Solid Edge, you place parts into assemblies by dragging them from the Parts Library tab on EdgeBar, and drop them into the assembly. In the next few steps, you will display EdgeBar, arrange the assembly window, and then shade the window.

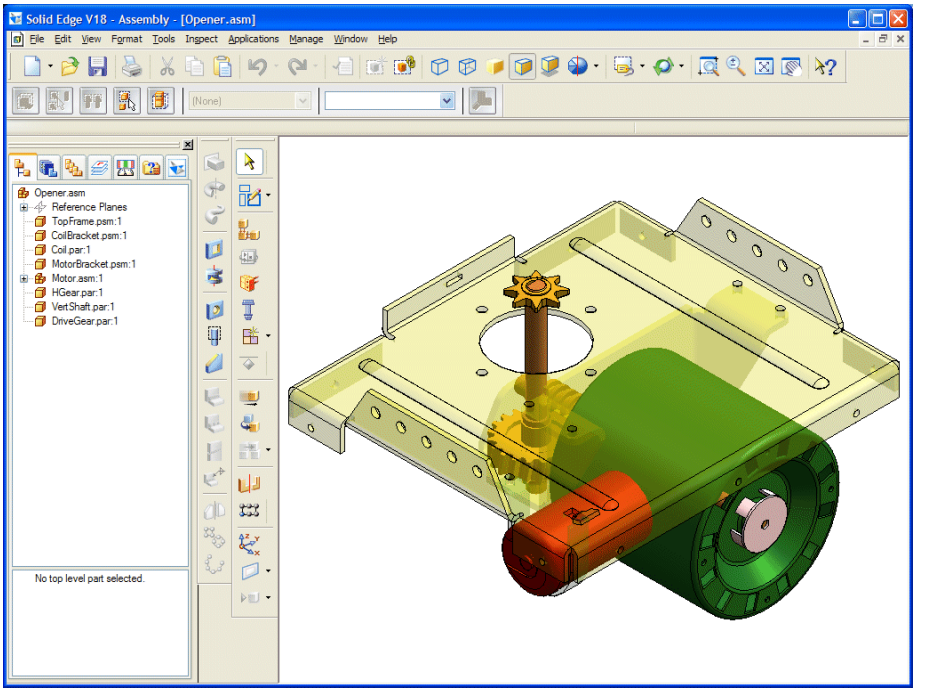
If EdgeBar is not displayed, on the Tools menu, set the EdgeBar option.

- On the Tools menu, click Color Manager.
- On the Color Manager dialog box, ensure that the Use Individual Part Styles option is set, then click OK.
- Only the edges of parts in the assembly are displayed . You can shade the faces of parts for clarity



On the Main toolbar, click the Shaded With Visible Edges button to shade the assembly window.

Your window and EdgeBar should now look similar to the illustration.



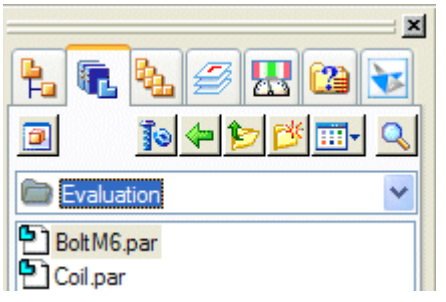
Step 3: Set the Parts Library Folder

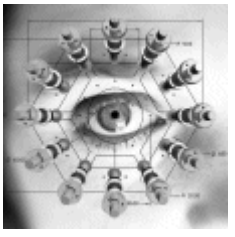


On EdgeBar (the pane at the left side of the Solid Edge window in the illustration above), click the Parts Library tab.

On the Parts Library tab, browse to the Solid Edge Evaluation folder. The default location of the Solid Edge Evaluation folder is:

C:\PROGRAM FILES\SOLID EDGE V18\TRAINING\EVALUATION

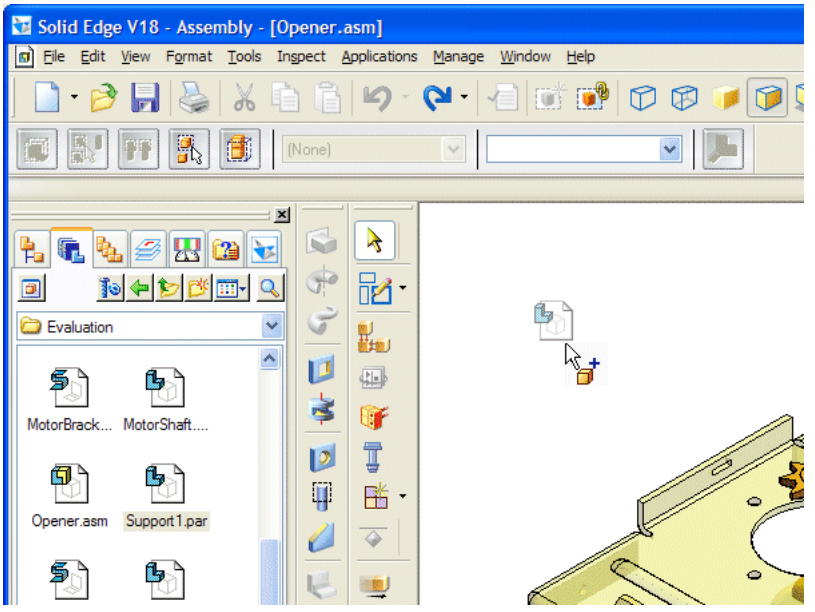




Accessing managed documents that you want to place in assemblies is easy. Since Insight is integrated within EdgeBar, you simply select a managed folder, rather than an unmanaged folder. Typical PDM systems require additional, cumbersome steps to access managed documents. With Solid Edge Insight, the workflow is identical for managed and non-managed documents.

Step 4: Place the Support Part

In the file list area on the Parts Library tab, select the SUPPORT1.PAR file, hold the left mouse button down, and drag the file into the assembly window, then release the mouse button.



Notice that the Place Part SmartStep ribbon bar is displayed above the graphic window.

Similar to constructing part features, a SmartStep ribbon bar guides you through the part placement process, and makes it easy to access and change part placement options. Solid Edge has optimized the part placement process allowing you to build your assemblies in the most efficient, understandable manner possible.

You position parts in assemblies using assembly relationships. The bottom face of your part needs to be mated the top face on the large sheet metal part, and your part needs to be axially aligned with the large cylindrical hole on the sheet metal part. You will use FlashFit to create these relationships.

Step 5: Examine the SmartStep Ribbon Bar

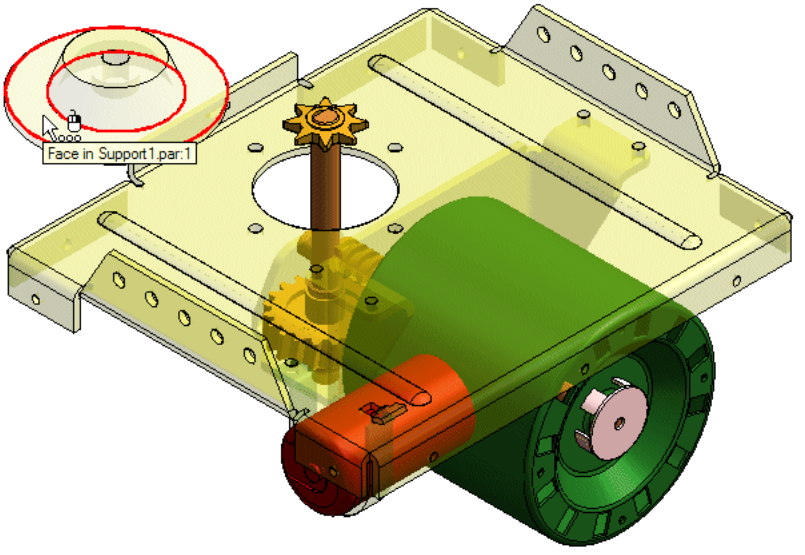


Take a few moments to examine the SmartStep ribbon bar. Working left to right, pause the cursor over the various buttons and read the Tooltips. The options you will use to apply the first relationship are the default options, so placing the first relationship will be very easy:

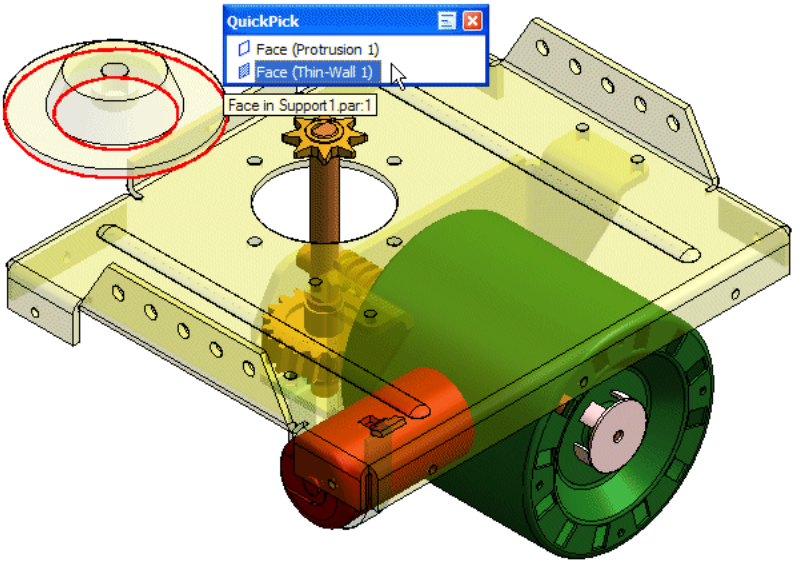


The Relationship Types option allows you to select which assembly relationship type you want to use for positioning a part. If this option is not already set, choose FlashFit from the list. The FlashFit option reduces the steps required to position parts using mate, planar align, and axial align relationships when compared to other workflows. Because many parts are positioned using these relationships, FlashFit is appropriate in most situations.

Step 6: Define the Mate Relationship

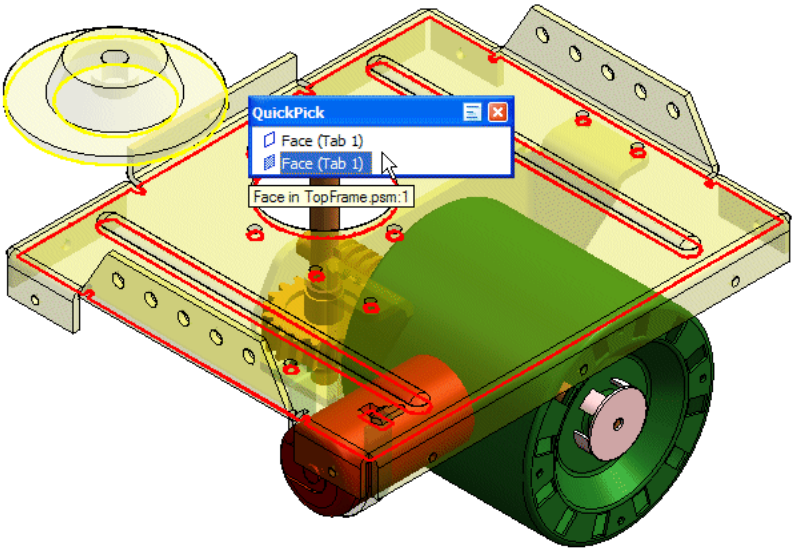


Position the cursor over the support part as shown in the illustration. Notice that the top face of the flange on the support highlights. Also notice that the part file name and the surface type is displayed to help you pick the proper face. Since you want to mate the bottom face, hold the cursor still until the QuickPick prompt is displayed, then click the right mouse button to display QuickPick.

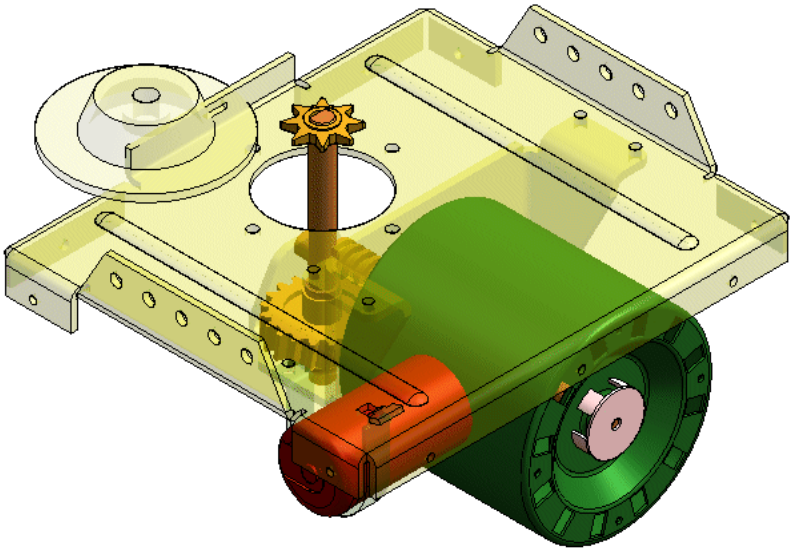


Now move the cursor over QuickPick until the bottom face shown in the illustration highlights, then click the left mouse button to select it.

Step 7: Select the Mating Face



Position the cursor and click to select the top face of the sheet metal frame, using QuickPick if necessary, as shown. The Support is repositioned so that its bottom face is coplanar with the top face of the sheet metal frame.



Depending on where the cursor was positioned when you selected the placement face in the previous step, the support part may appear in a different location than the illustration shows. This is OK, as long as the part lies on the plane of the top face of the sheet metal top frame.

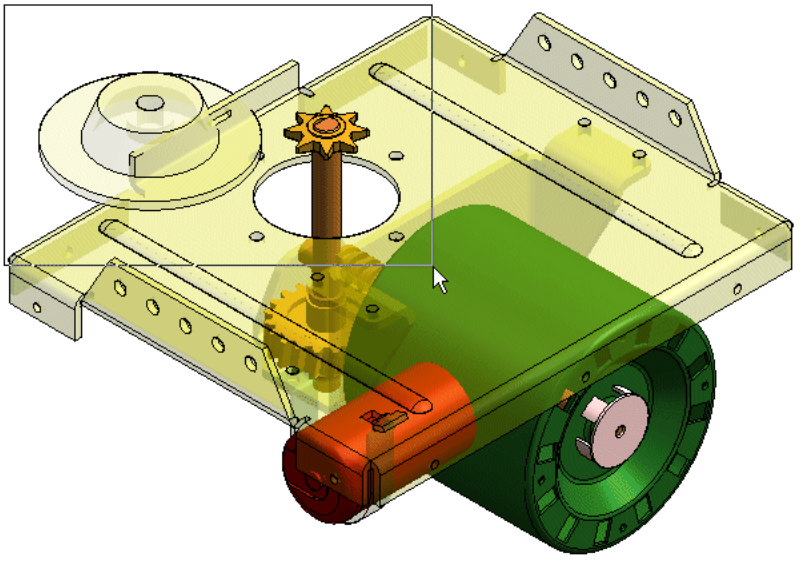
Step 8: Zoom in a Little Closer

Before you apply the next relationship, you will zoom in closer to the support part to make adding the next relationship easier.



On the Main toolbar, click the Zoom Area button.

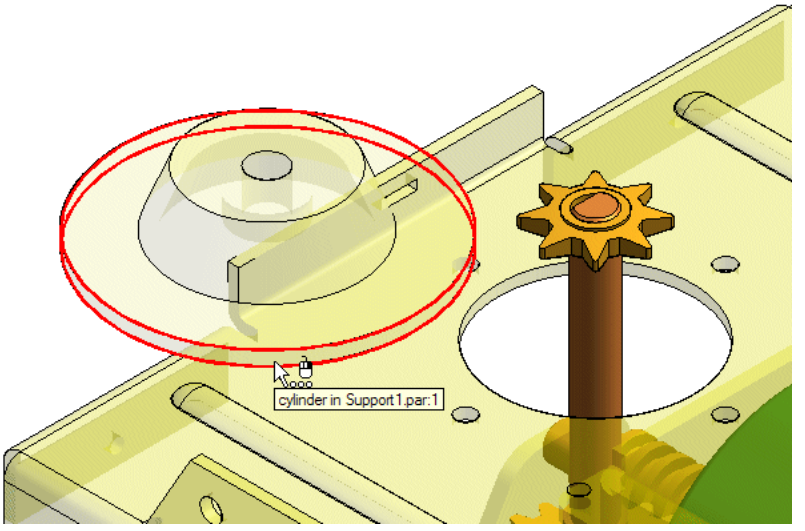
Click above and to the left of the sheet metal part, as shown, and then click again below and to the right of the support part. This defines a rectangle that becomes the view area.



Click the right mouse button to end the Zoom Area command.

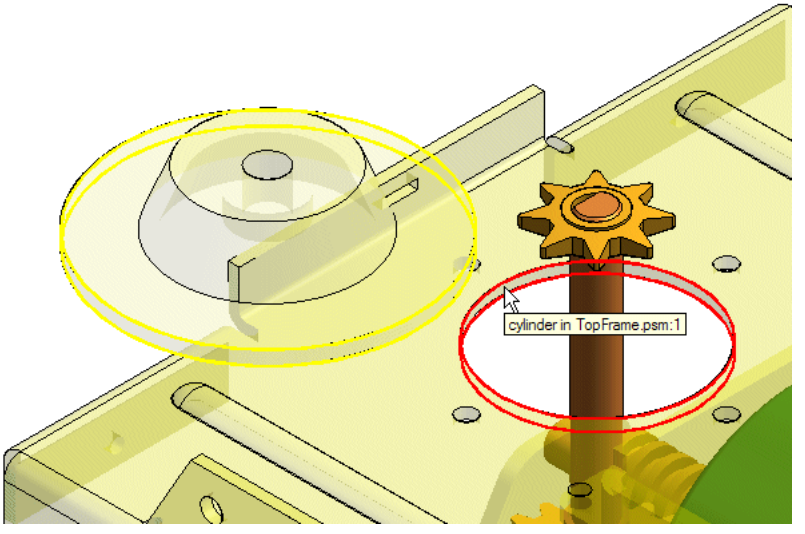
Step 9: Axially Align the Support Part

The Axial Align relationship positions a part in an assembly by aligning two cylindrical faces.



In the assembly window, position the cursor over the cylindrical face shown, then click to select it. Why this face? Any of the cylindrical faces co-axial to this face would also work. Because this face is the largest, it is also easiest to select.

Step 10: Select the Aligning Face on the TopFrame Part

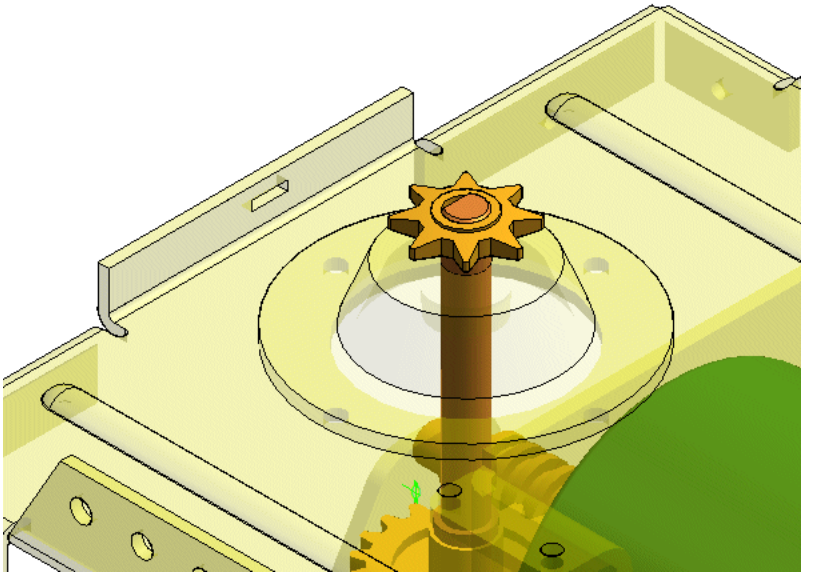


Select the cylindrical face shown.

Step 11: Examine the relationships that position the part

Notice that the support is now aligned properly in the assembly. Also notice that the Place Part SmartStep ribbon bar is still displayed and in the Relationship

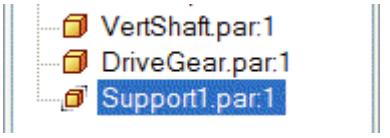
List, Relationship 3 is ready to be defined. This means the part is not yet fully positioned. In the next few steps, you will learn more about this and finish the part.





On EdgeBar, click the Assembly PathFinder tab.

On the Assembly PathFinder tab, find the Support1.par listing. Notice that its symbol is different than the other parts. It is smaller than the others, which indicates that it is not fully positioned. Also notice that other symbols are used to denote subassemblies, assembly reference planes and so forth.



It Can Rotate

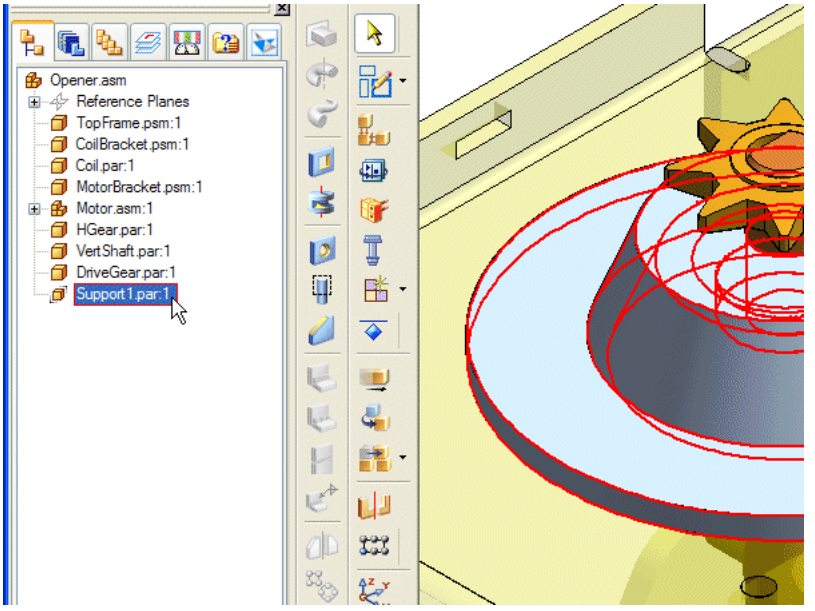
The support part is still free to rotate about its axis. Although you are not required to fully position parts, it is generally better if you do.

Step 12: Select the Support Part in Assembly PathFinder



On the Assembly Commands toolbar, click the Select Tool button.

In Assembly PathFinder, click the entry for Support1.par. Notice that when you select the part, a message is displayed on the right side of the status bar stating that Support1.par:1 is not fully positioned. Also notice that in the bottom pane of Assembly PathFinder, the assembly relationships you applied are listed, along with the assembly part that you used to position the support. All this feedback is helpful when you need to make design changes later.



Finish Modeling the Support

Until now, you have worked in Part and Assembly as separate operations. Of course, in real life, parts are generally designed within the context of the assembly. So, in the next few steps, you will finish modeling the support within the context of the assembly. You will use the mounting holes on the sheet metal part to quickly add the missing mounting holes on the support.

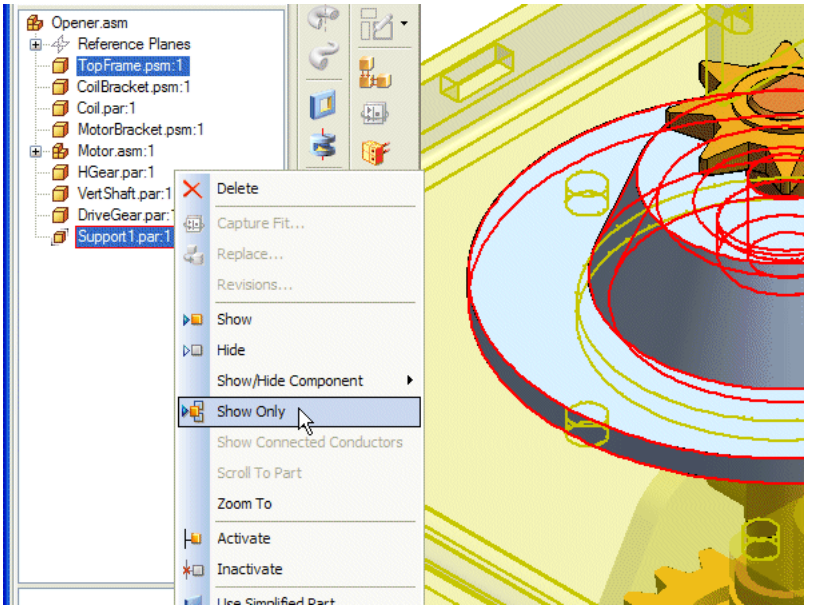
Step 13: Save the Assembly



On the Main toolbar, click the Save button to save the work you have done so far.

Step 14: Display Only the Sheet Metal Part and the Support

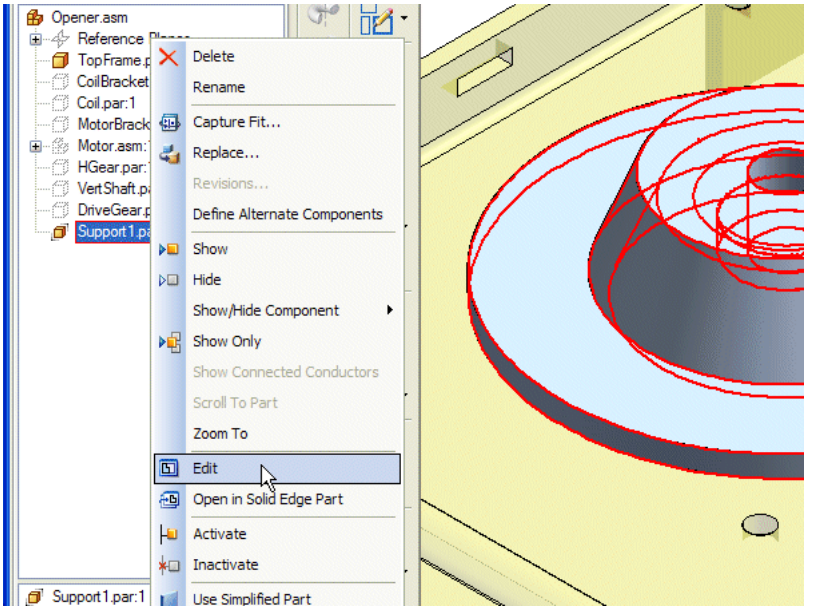
To make it easier to edit the support, you will hide everything but the sheet metal part and the support.



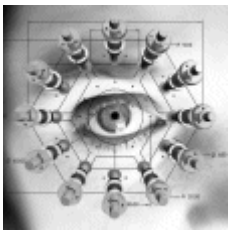
In the top pane of Assembly PathFinder, select TopFrame.psm, hold down the CTRL key on the keyboard, then select Support1.par. With the cursor still over Support1.par, click the right mouse button to display the shortcut menu. On the shortcut menu, click the left mouse button on the Show Only command.

Step 15: Edit the Support Part

Position the cursor over the Support1.par listing in the top pane of Assembly PathFinder, then click the right mouse button to display the shortcut menu. On the shortcut menu, click the left mouse button on the Edit command.



Notice that the Assembly PathFinder tab is no longer displayed. Instead the Feature PathFinder tab for the support part is displayed. Also notice that the file name bar was updated to indicate you are now working in Support1.par within the assembly.

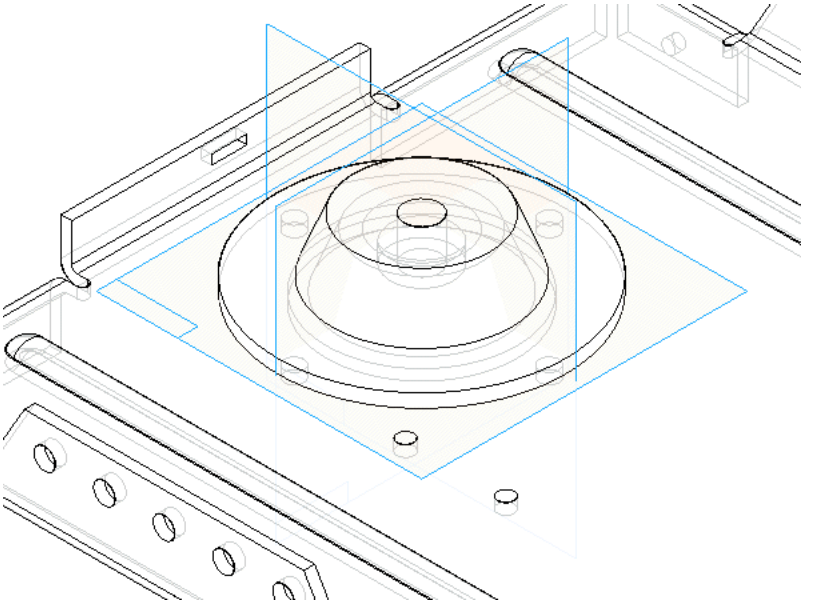


With Insight, if you were editing a managed document, then it would be checked out of the managed workspace for you.

Step 16: Unshade the Window



On the Main toolbar, click the Visible and Hidden Edges button to display the window in wireframe mode.

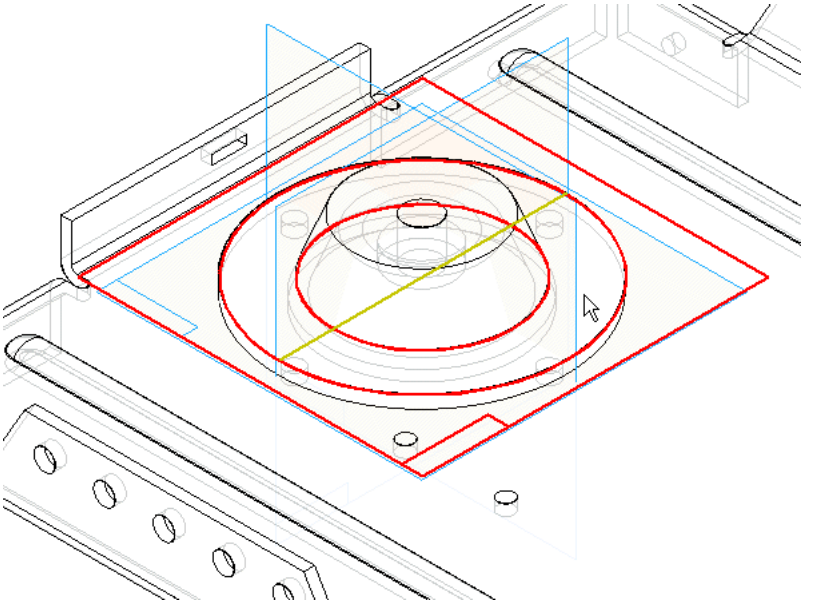


Step 17: Add the Holes

The holes in the sheet metal part were created using a single Hole feature and a separate Pattern feature. To illustrate other Solid Edge commands, you will create all four holes in one operation using the Cutout command.



On the Features toolbar, click the Cutout button.



Position the cursor over the face shown, and notice that a new reference plane coincident to the face is displayed.

When the circle and reference plane highlight, use QuickPick to select them.

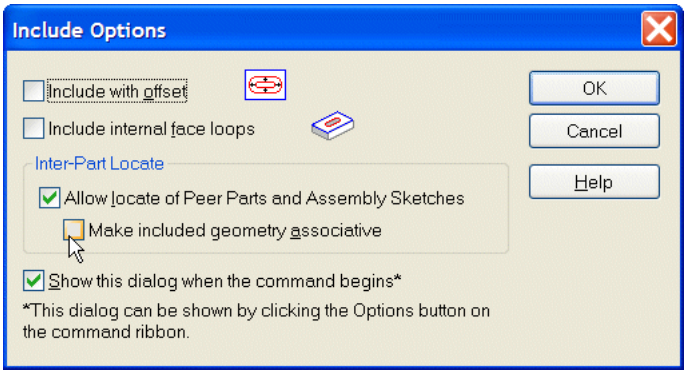
Step 18: Copy the Holes from the Sheet Metal Part

In Solid Edge, you can use the Include command to copy edges from one part to another to create profiles.



On the Draw toolbar, click the Include button.

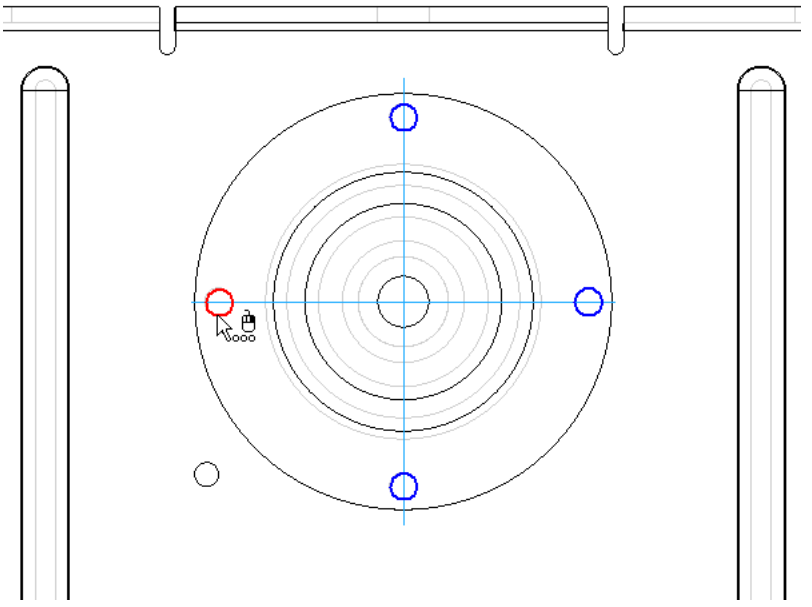
The Include Options dialog box should display. If it does not, on the ribbon bar, click the Options button to display it.



- On the Include Options dialog box, ensure that the Allow Locate of Peer Parts and Assembly Sketches option is set and that the Make Included Geometry Associative option is cleared, then click OK.

- On the Include ribbon bar, ensure that the Select option is set to Single Wireframe.

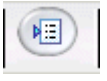
- Select the four circles shown in the illustration. You may need to use QuickPick to select them. Notice that after you select them, they change to the profile color.



Step 19: Dimension the Circles

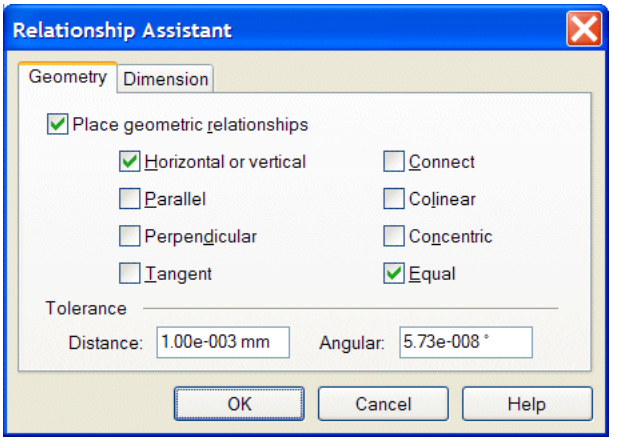
You could spend the next few minutes constraining and dimensioning the four circles, but Solid Edge has a tool that makes short work of this.

On the Tools menu, point to Dimensions, then click Relationship Assistant.

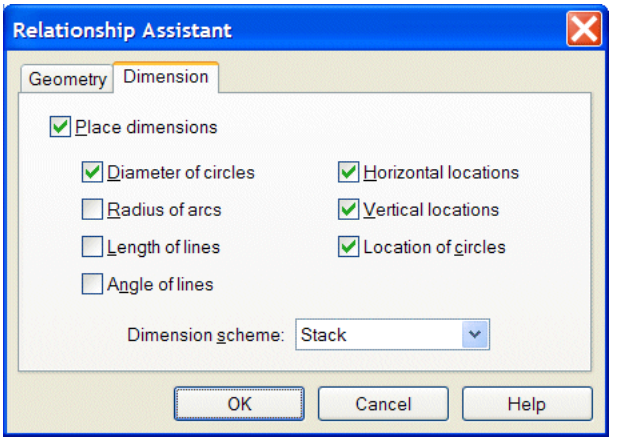


On the Relationship Assistant ribbon bar, click the Options button.

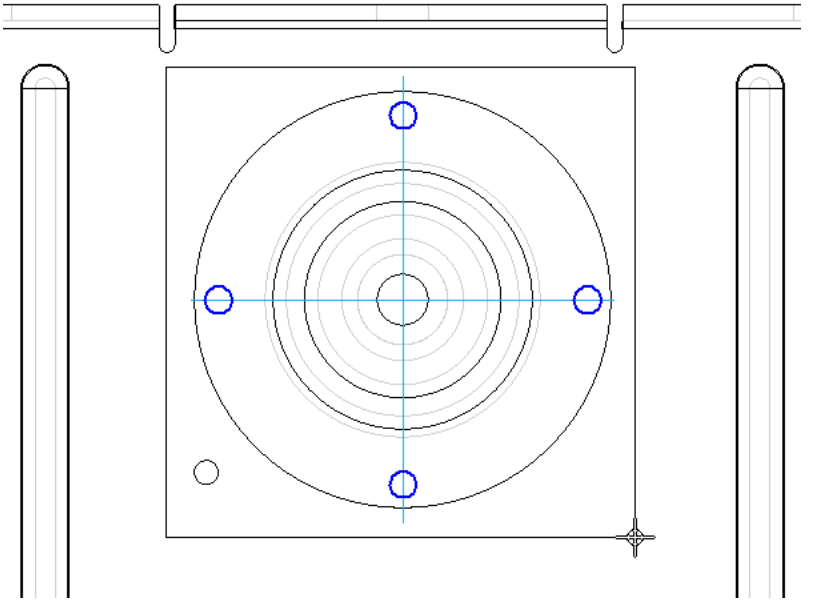
On the Relationship Assistant Options dialog box, on the Geometry tab, set the following options:



On the Dimensions tab, set the options shown in the illustration, then click OK.



Step 20: Position the cursor above and to the left of the profile circles, press and hold down the left mouse button, then drag the cursor to the approximate point shown in the illustration, then release the mouse button. The four circles should highlight.



Step 21: The circles you selected are the only elements you want to apply dimensions to.

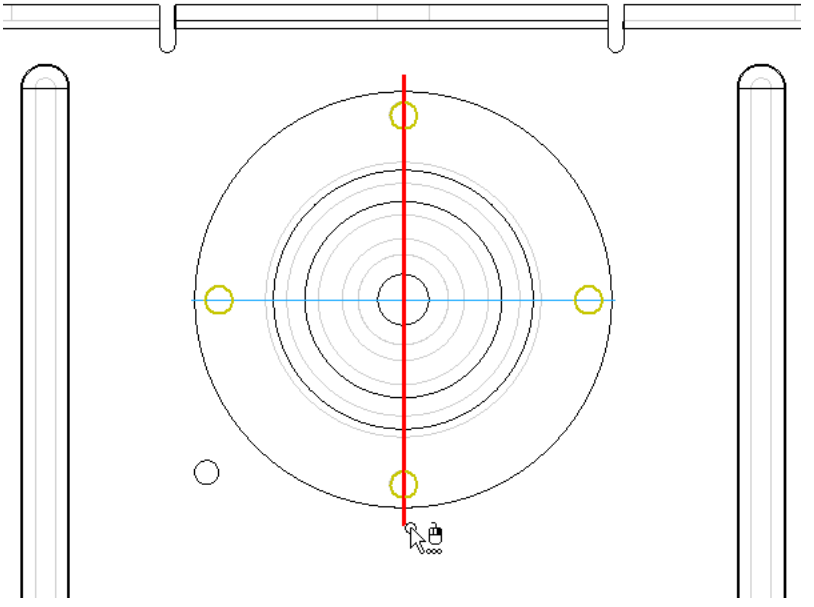


On the Relationship Assistant ribbon bar, click the Accept button.

Step 22: **Define the Horizontal Dimension Origin**

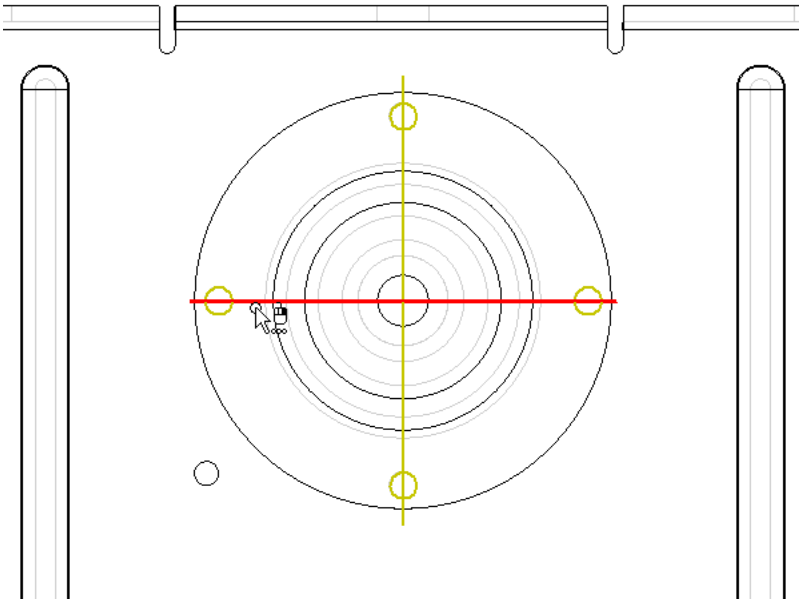
Read the message in the status bar immediately above the model work space. Relationship Assistant prompts you to define the start point for the horizontal dimensions. Since you constructed the part symmetrically about the reference planes, you can take advantage of that symmetry.

In the profile window, select the reference plane shown in the illustration. If QuickPick is displayed, either option is valid.



Step 23: Define the Vertical Dimension Origin

In the profile window, select the reference plane shown in the illustration. If QuickPick is displayed, either option is valid. The relationships and dimensions are added.



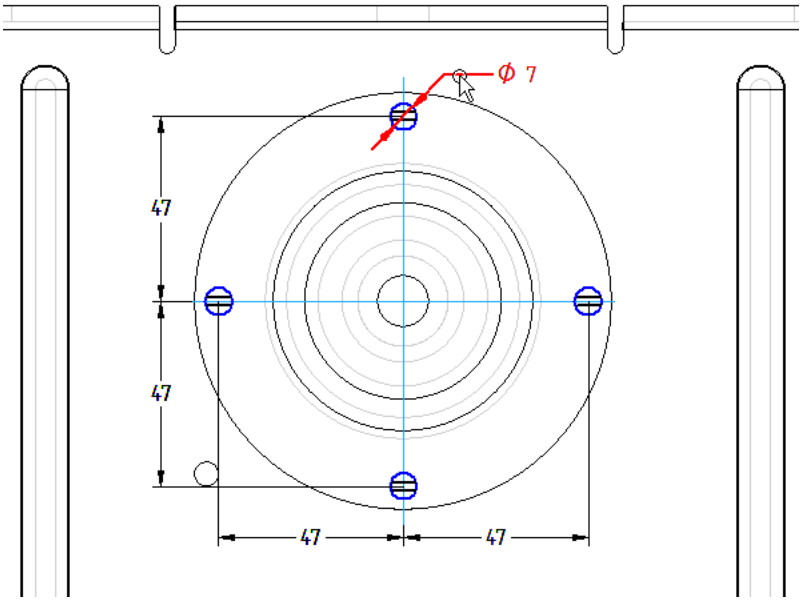
Step 24: Edit the Hole Size



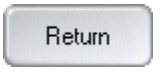
On the Draw toolbar, click the Select Tool button.

- In the profile window, select the 7 millimeter dimension shown.
- On the Dimension ribbon bar, in the Dimension Value box, type 8, then press the ENTER key.

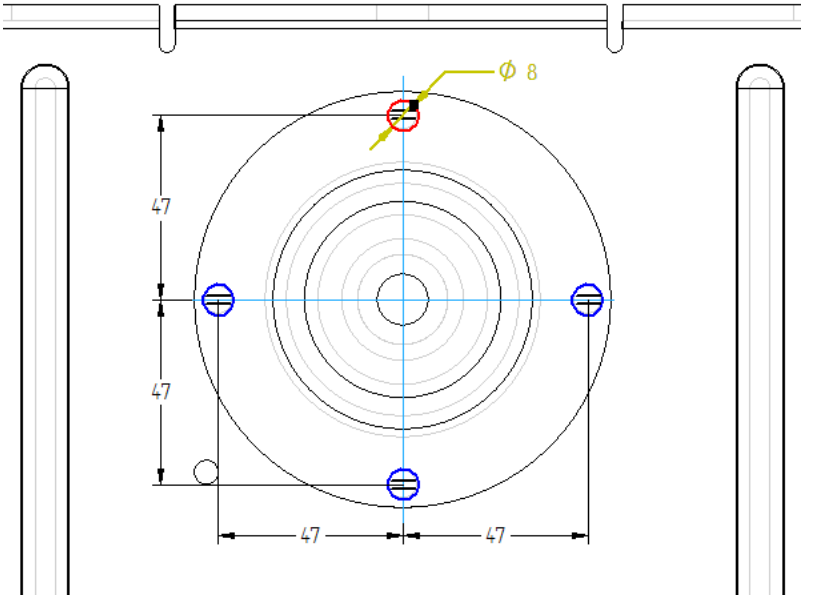
  All four holes changed size. That is because you set the Equal option in Relationship Assistant and Solid Edge knows that all four holes are the same size and applied an equal relationship between them.



Step 25: The cutout profiles are now complete.



On the ribbon bar, click the Return button to close the profile window.

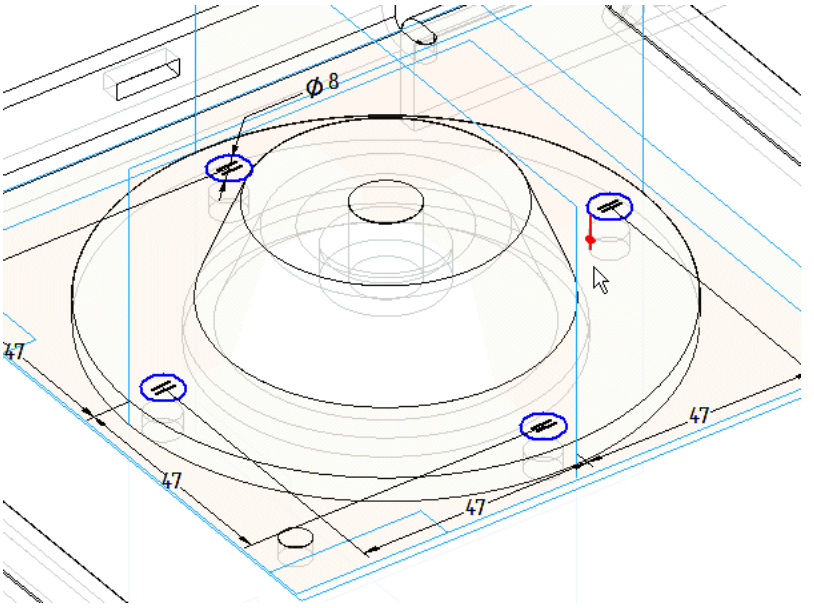


Step 26: **Define the Extent of the Cutouts**



On the Cutout SmartStep ribbon bar, click the Through Next button.

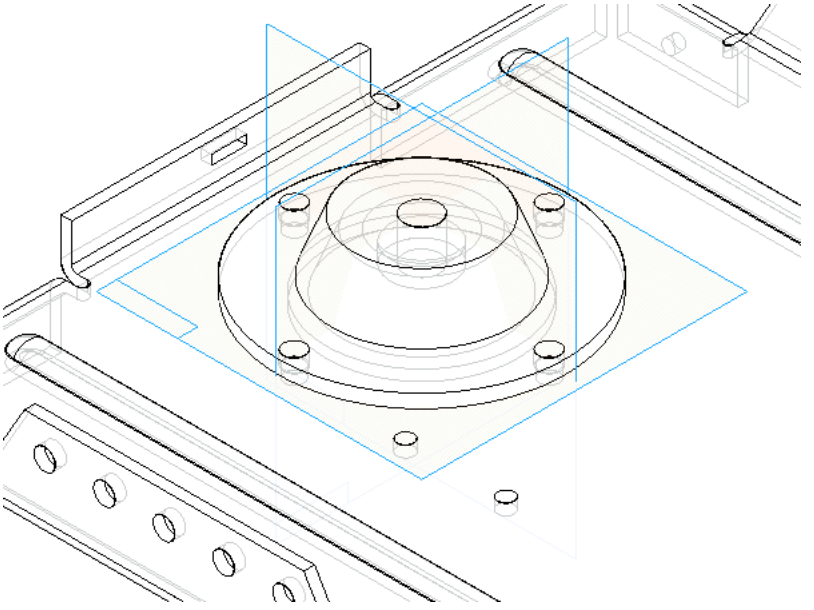
In the part window, position the cursor below the profiles, so that the red extent arrow points downward, then click. The four cutouts are constructed.



Step 27: You have provided all of the information necessary to complete the feature.



On the ribbon bar, click the Finish button to complete the feature.



Step 28: **Save the Document**



On the Main toolbar, click the Save button to save the work you have done so far.

Step 29: **Return to the Assembly**

On the File menu, click Close And Return. Notice that when you return to the assembly, Assembly PathFinder is displayed.

Step 30: Finish Positioning the Support Part

In the next few steps you will finish positioning the support part using the four cutouts you constructed.

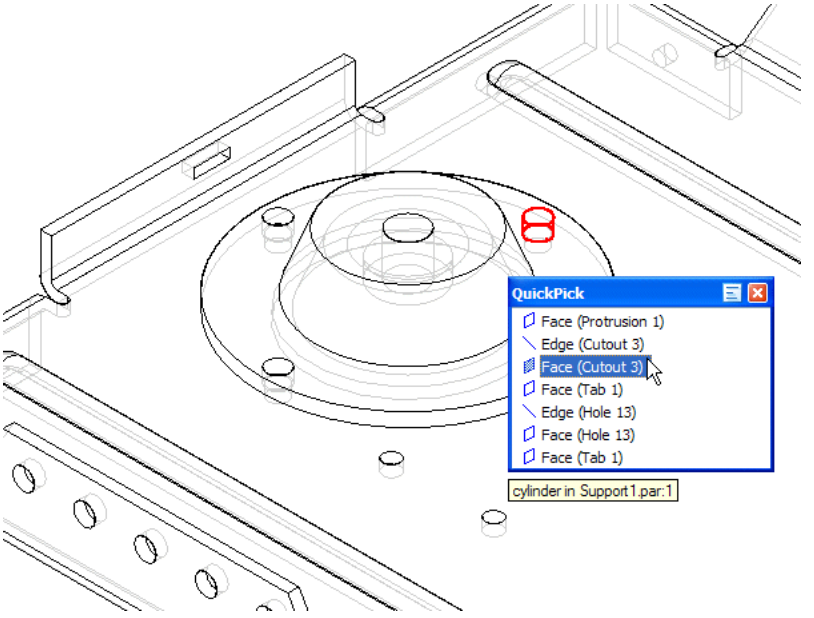


On the Assembly Commands toolbar, click the Assemble button.

When you first drag a new part into an assembly, the Place Part command runs to help you apply relationships that position the part in the assembly. At any time, you can use the Assemble command to apply relationships to parts already in the assembly.

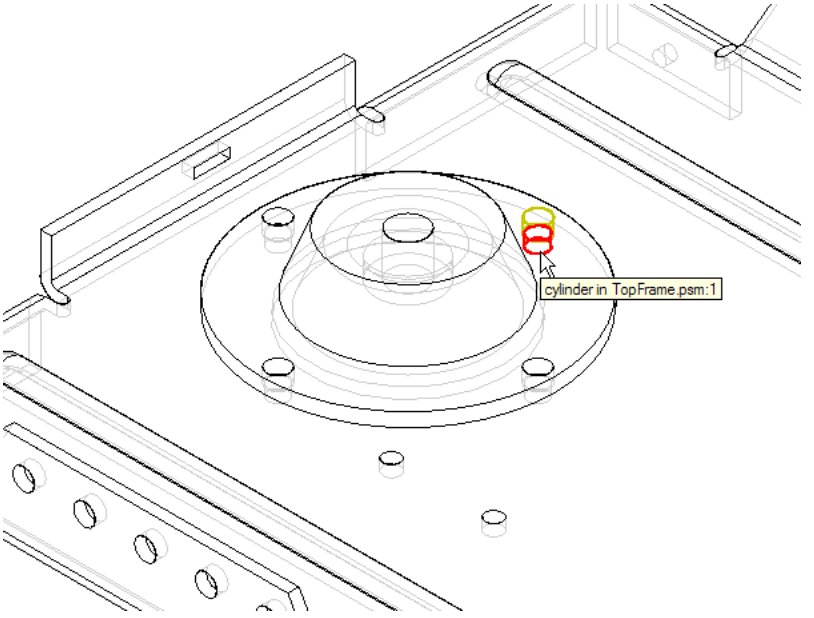
Step 31: Align the bolt holes

In the assembly window, use QuickPick to select the cylindrical face shown on the support part.



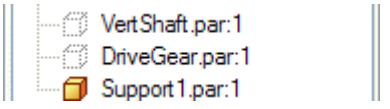
Step 32: Select the Cylindrical Face on the Sheet Metal Part

Select the cylindrical face shown.



Step 33: The Support is Fully Positioned

Recall that the display of Support1.par on the Assembly Pathfinder tab of EdgeBar earlier showed that the part was not fully positioned. Now observe that the display has changed to show that the part is fully positioned.





On the Main toolbar, click the Save button to save the work you have done so far.

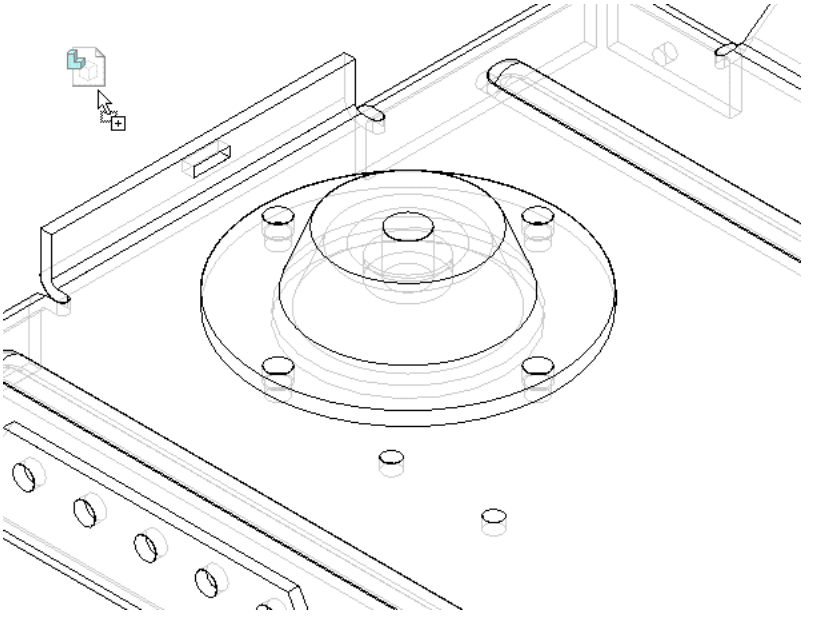
Step 34: Place Bolt M6

In the next few steps, you will place a bolt into one of the holes in the support, then pattern the bolt to fill the remaining holes.



On EdgeBar, click the Parts Library tab.

In the Parts Library, drag and drop BOLTM6.PAR into the assembly window.



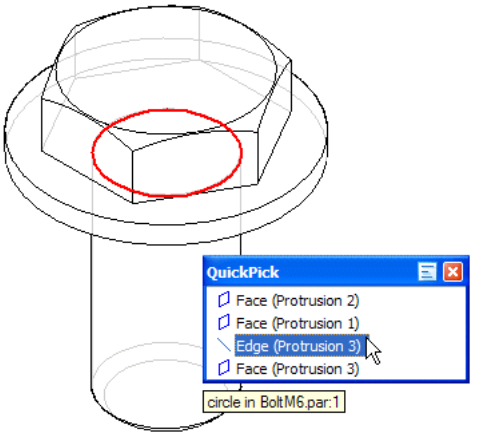


On the Place Part SmartStep ribbon bar, the Relationship Types option should be set to FlashFit.

FlashFit allows you to quickly place bolts in holes. When you select the circular edge where the shaft of a bolt meets the mating face of its head as the first input, and then select the circular edge where a hole meets a planar face as the second input, FlashFit applies a planar mate and an axial align relationship. The axial align relationship is placed using the Fixed option, which fixes the rotational orientation of the part.

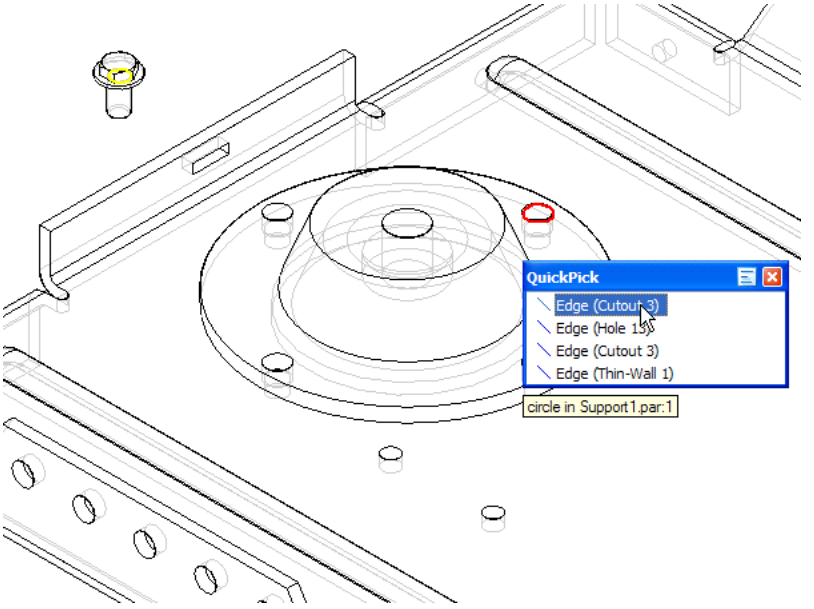
Step 35: Select a Circular Edge on the Bolt

Use QuickPick to select circular edge where the shaft of the bolt meets the mating face of its head,as shown.

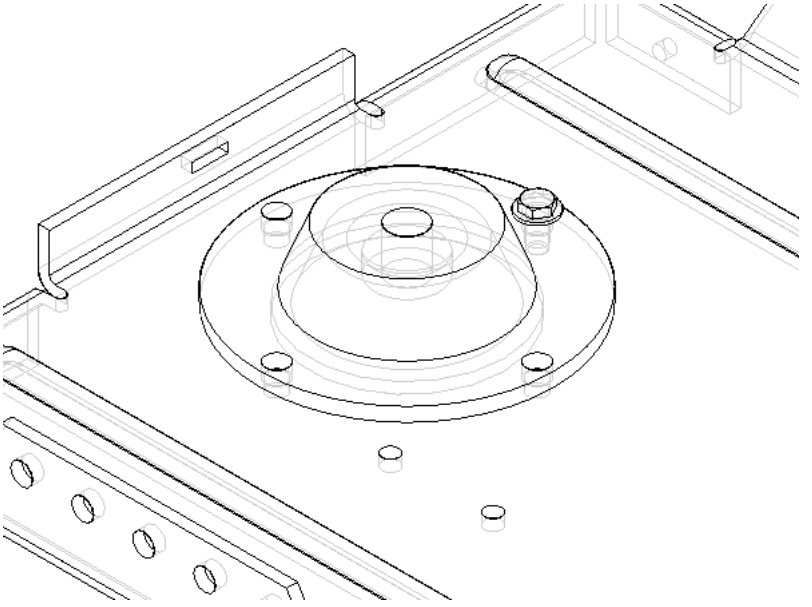


Step 36: Select a Circular Edge on the Support

Select the circular edge where the hole meets the top face, as shown.



You have provided the information necessary to assemble the bolt, and it is repositioned.



Step 37: Pattern the Bolt

In the next few steps, you will pattern the bolt.

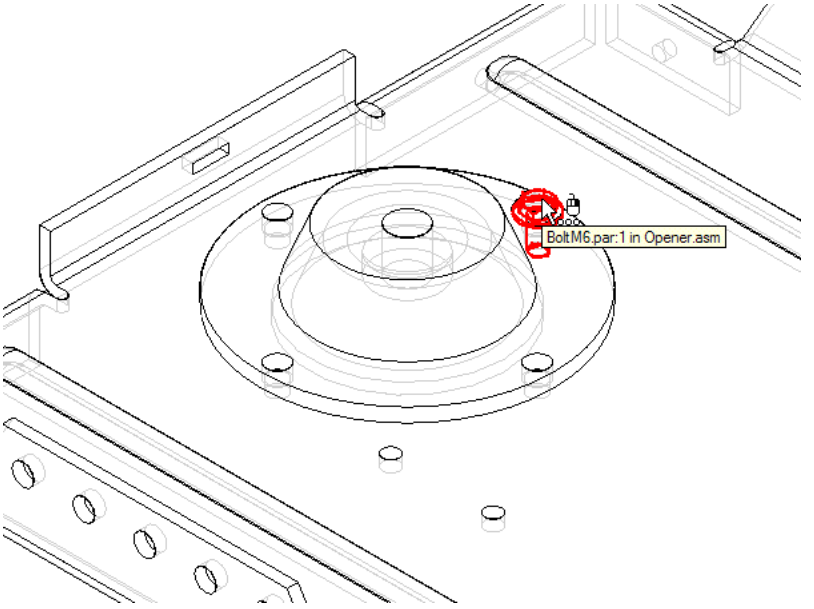


On the Assembly Commands toolbar, click the Pattern Parts button.

The Pattern Parts command allows you to create a pattern of parts in an assembly. You define the pattern by selecting a pattern feature on a part in the

assembly. The patterned parts are not positioned using assembly relationships, but are positioned using the pattern feature you select.

In the assembly window, use QuickPick to select the bolt as the part to be patterned.

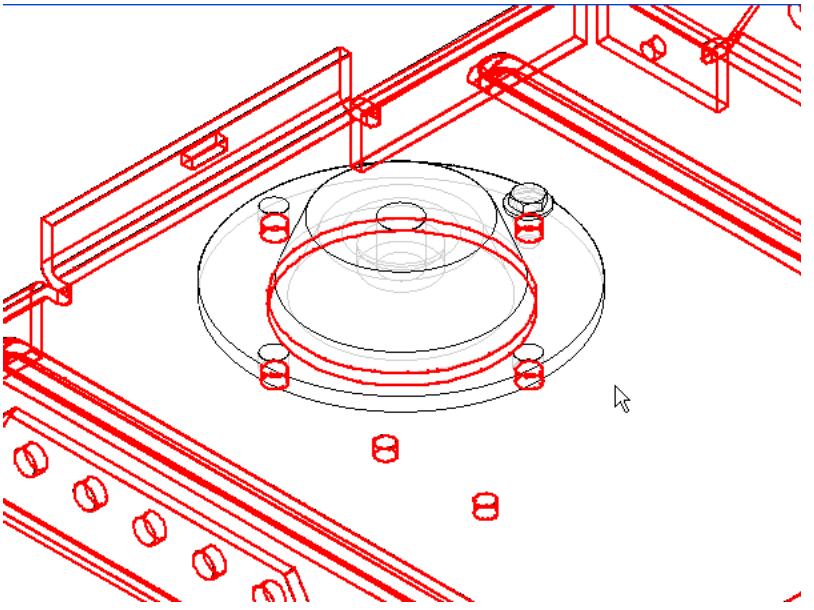




On the Pattern Parts SmartStep ribbon bar, click the Accept button.

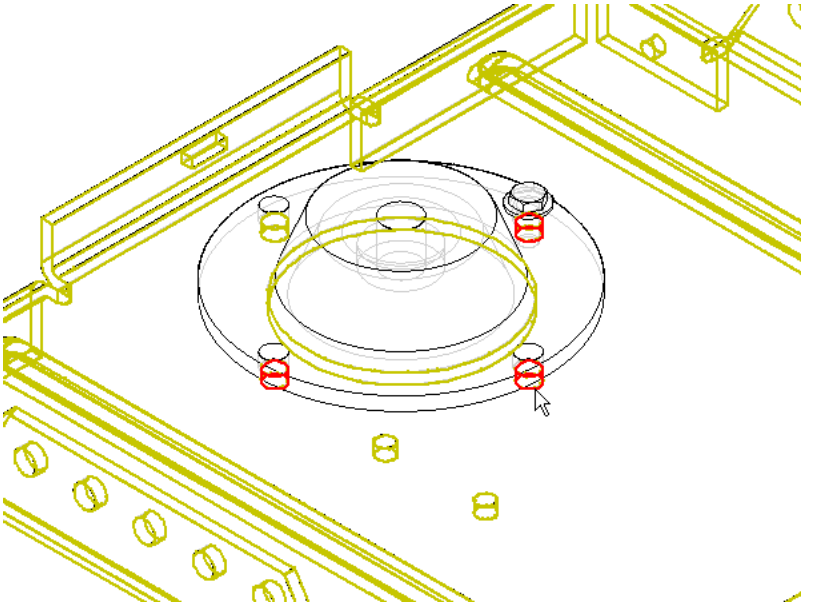
Step 38: Select the Part Containing the Pattern

In the assembly window, select the sheet metal part, as shown.



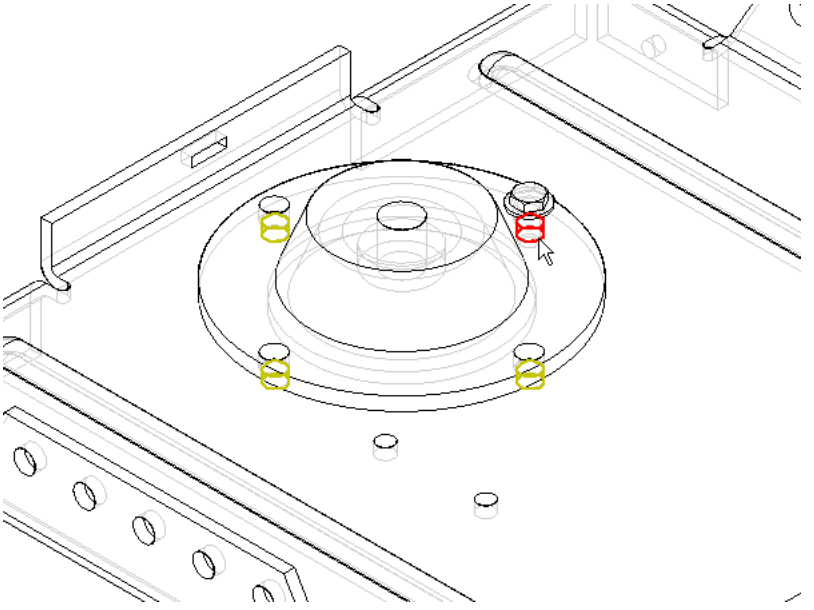
Step 39: Select the Pattern Feature

In the assembly window, select the pattern feature on the sheet metal part, as shown.



Step 40: Select a Reference Feature on the Pattern

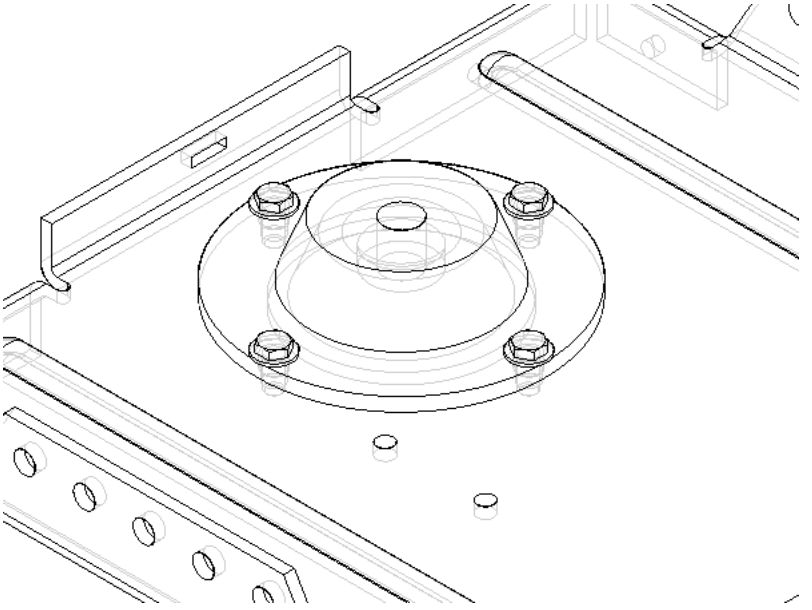
Select the feature where you placed the first bolt, as shown.



Step 41: Finish Patterning the Bolt



On the Pattern Parts SmartStep ribbon bar, click Finish.



The remaining bolts are placed into the assembly.

Step 42: Save the Document



On the Main toolbar, click the Save button to save the work you have done so far.

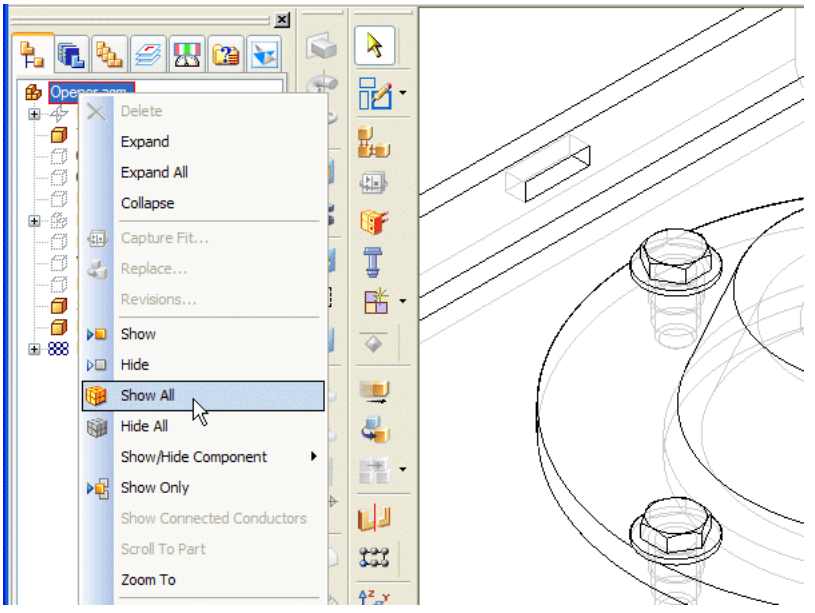
Step 43: Display all the Parts in the Assembly



On EdgeBar, click the Assembly PathFinder tab.

Step 44: Display all the Parts

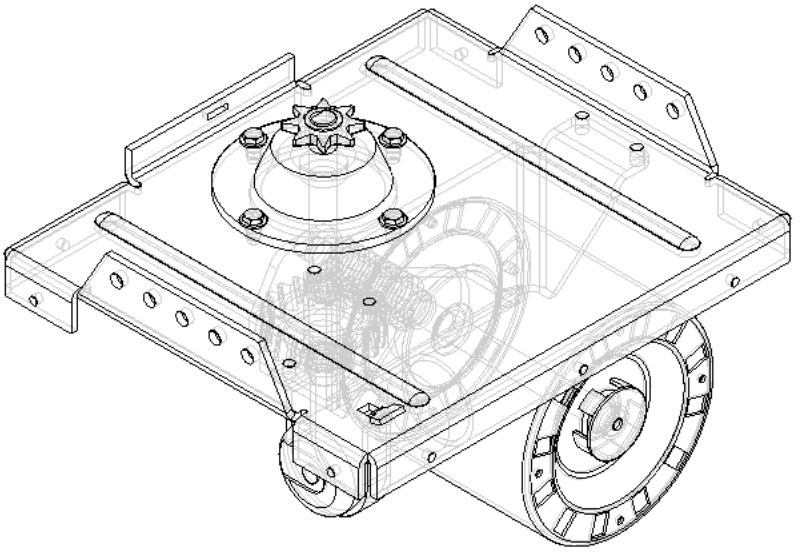
In Assembly PathFinder, position the cursor over the Opener.asm listing, then click the right mouse button to display the shortcut menu. On the shortcut menu, click the left mouse button on Show All.



Step 45: Fit the Assembly in the Window



On the Main toolbar, click the Fit button.



Step 46: Create an Exploded View

In the next few steps, you will create an exploded view of the assembly.

To make exploding an assembly easier, a separate set of commands within the Assembly document is used to explode assemblies.

On the Applications menu, click Exploded View.

The system displays the Exploded View toolbar so you can create exploded views of the assembly.

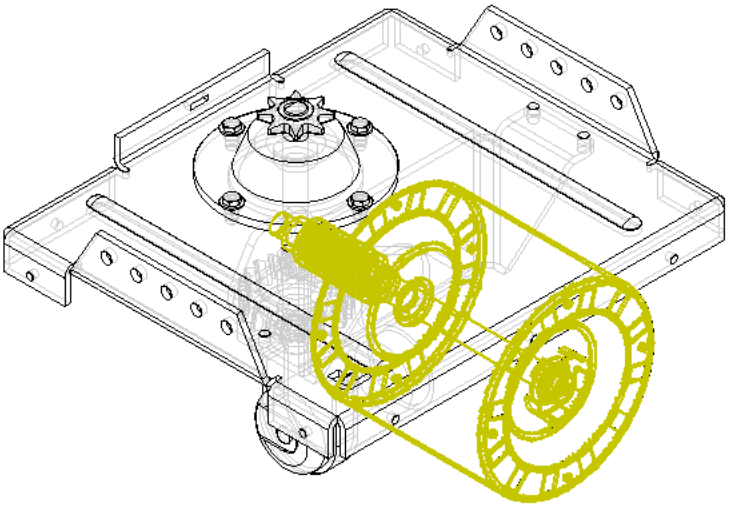
Step 47: Bind a Subassembly

For this assembly explosion, you want to maintain the Motor.asm subassembly as a single unit. You will use the Bind Subassembly command to do this.



In EdgeBar, click the Assembly PathFinder tab.

Step 48: In Assembly PathFinder, select the Motor.asm subassembly. Notice that the subassembly highlights in the assembly window.

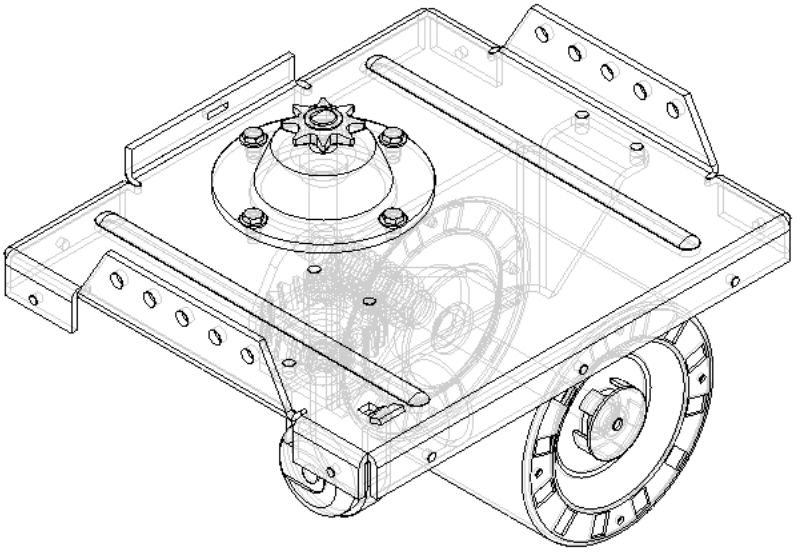




On the Exploded View toolbar, click the Bind Subassembly button.

Step 49: De-select the Subassembly

In the graphic window, position the cursor away from any graphics, then click in empty space to de-select the motor subassembly.



Step 50: Automatically Explode the Assembly

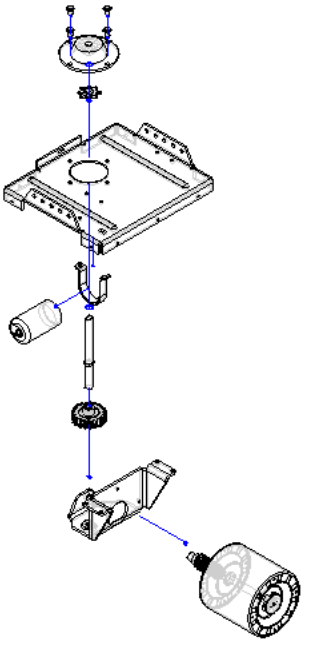


On the Exploded View toolbar, click the Automatic Explode button.



On the ribbon bar, click the Explode button.

The system processes, and then displays the exploded view.

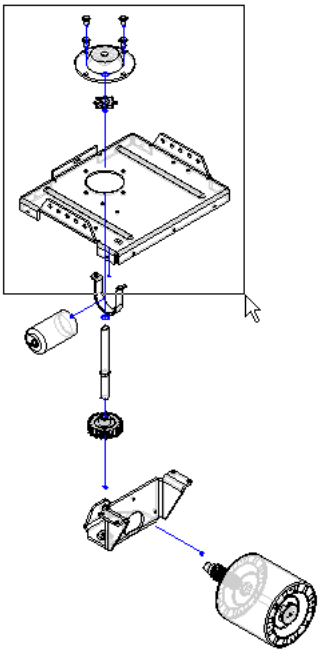


Step 51: Get a Little Closer

Although the Automatic Explode command worked well, you will use the Adjust Spread Distance command to move some of the parts a little closer together. First, you will zoom in a little.



On the Main toolbar, click the Zoom Area button.



Click above and to the left as shown, and then click again below and to the right. This defines a rectangle that becomes the view area.

Click the right mouse button to end the Zoom Area command.

Step 52: **The Parts are Too Far Apart**

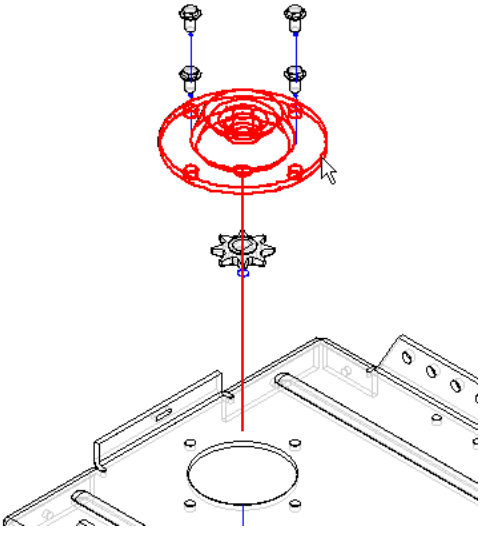


On the Exploded View toolbar, click the Adjust Spread Distance button.

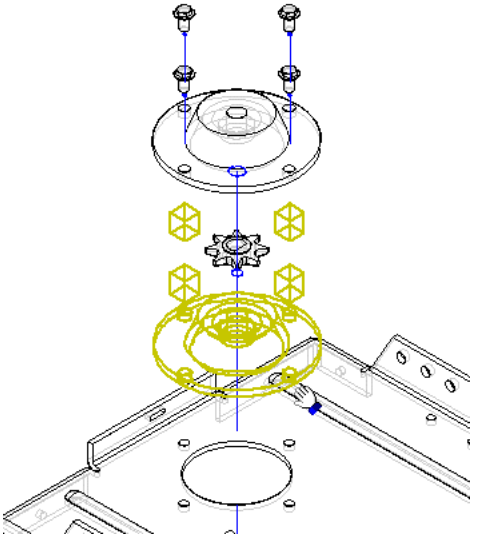


On the ribbon bar, click the Part Plus Dependents button.

The Part Plus Dependents option allows you to move several parts at one time. Position the cursor over the support part as shown, then press and hold down the left mouse button.



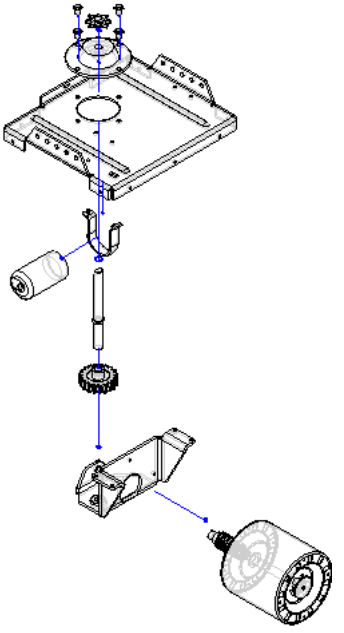
Step 53: Drag the set of parts down to the approximate location shown, then release the mouse button.



Step 54: Fit the Assembly Into the Window

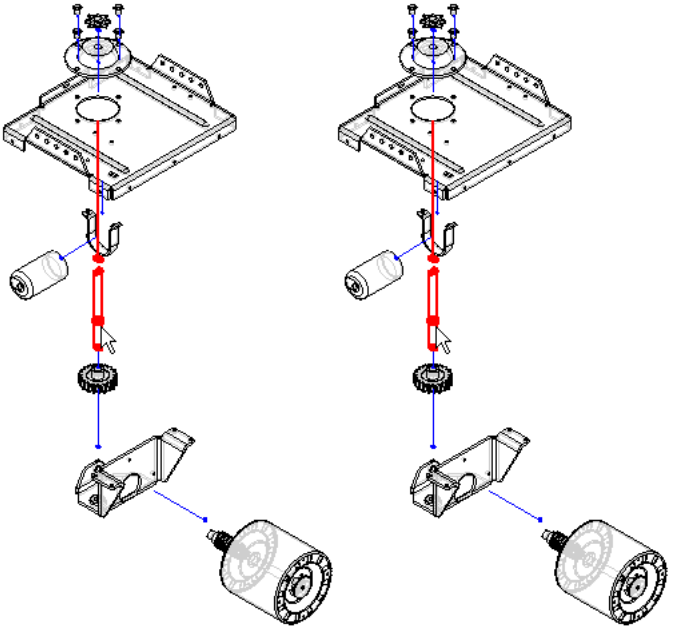


On the Main toolbar, click the Fit button.



Step 55: Move Another Set of Parts

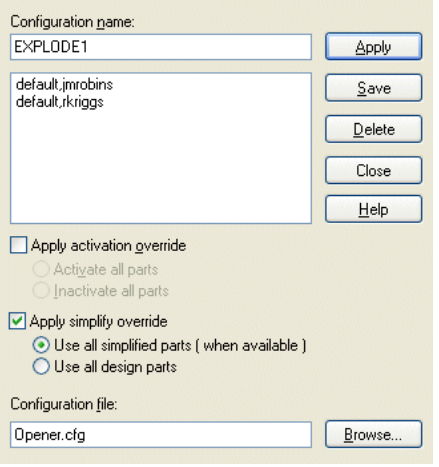
The Adjust Spread Distance command should still be active. Select the vertical shaft part as shown in the first illustration, and then drag the set of parts up to the approximate position shown in the second illustration.



Step 56: Save the Exploded Configuration

The exploded view is now complete. To take advantage of this view in a drawing you will create later, you must save the display configuration.

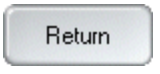
On the Tools menu, point to Configuration, and then click Display Configurations.



- In the Display Configurations dialog box, type EXPLODE1 as the name of the display configuration.
- In the Display Configurations dialog box, click the Save button

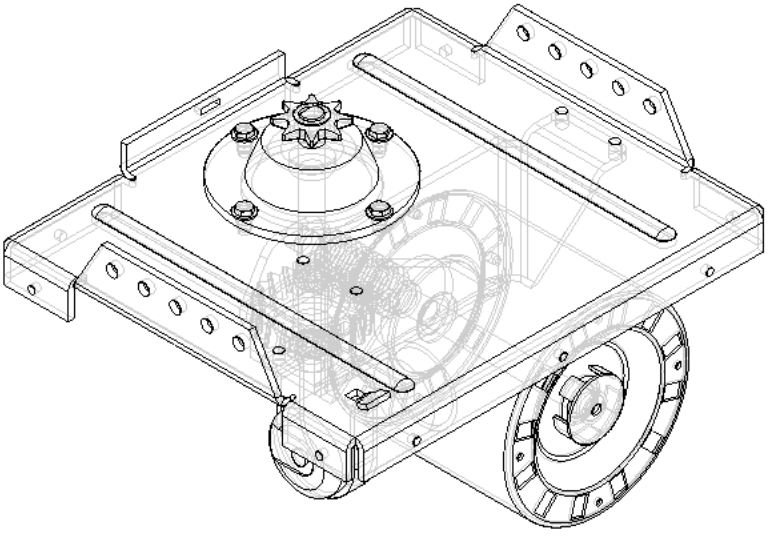
  This saves the display configuration so it can be recalled later.
- Click the Close button to dismiss the dialog box.

Step 57: Return to the Assembly



On the ribbon bar, click the Return button.

The Exploded View commands are hidden, and the commands for adding and editing assembly relationships are displayed. The assembly display returns to the unexploded condition.



Step 58: Fit the view

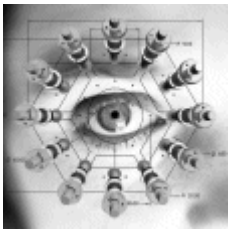


On the Main toolbar, click the Fit button.

Step 59: You Have Completed the Assembly



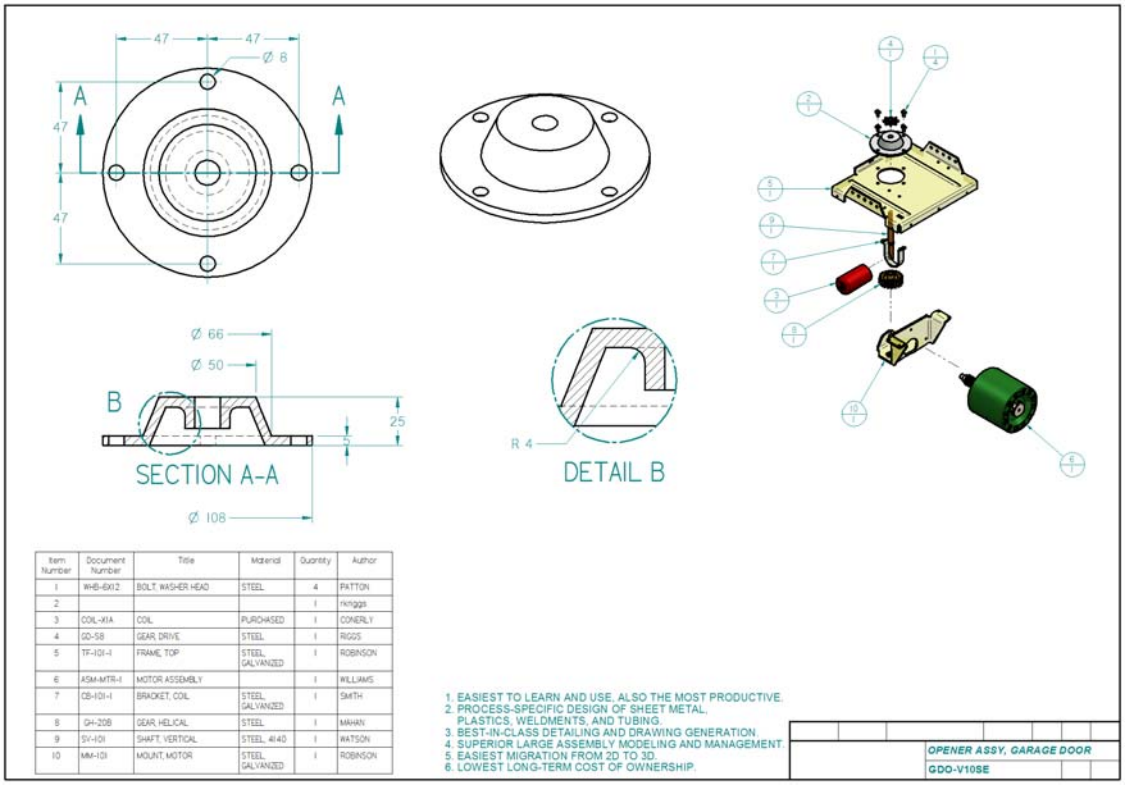
On the Main toolbar, click the Save button to save the work you have done so far. Leave the assembly file open.



When you save and close a managed assembly, the assembly and its parts are checked in for you, and the updated documents are made available to other users.

Chapter

5 *Create a Drawing*



Item Number	Document Number	Title	Material	Quantity	Author
1	WHB-6X12	BOLT, WASHER HEAD	STEEL	4	PATTON
2				1	rkriggs
3	COIL-X1A	COIL	PURCHASED	1	CONERLY
4	GD-58	GEAR, DRIVE	STEEL	1	RIGGS
5	TF-101-1	FRAME, TOP	STEEL, GALVANIZED	1	ROBINSON
6	ASM-MTR-1	MOTOR ASSEMBLY		1	WILLIAMS
7	CB-101-1	BRACKET, COIL	STEEL, GALVANIZED	1	SMITH
8	GH-20B	GEAR, HELICAL	STEEL	1	MAHAN
9	SV-101	SHAFT, VERTICAL	STEEL, 4140	1	WATSON
10	MM-101	MOUNT, MOTOR	STEEL, GALVANIZED	1	ROBINSON

Drawings in Solid Edge are fully associative to the parts and assemblies they reference. When you make changes to your designs, the drawing tools in Solid Edge allow you to assess the impact to the drawing and update the views and dimensions.

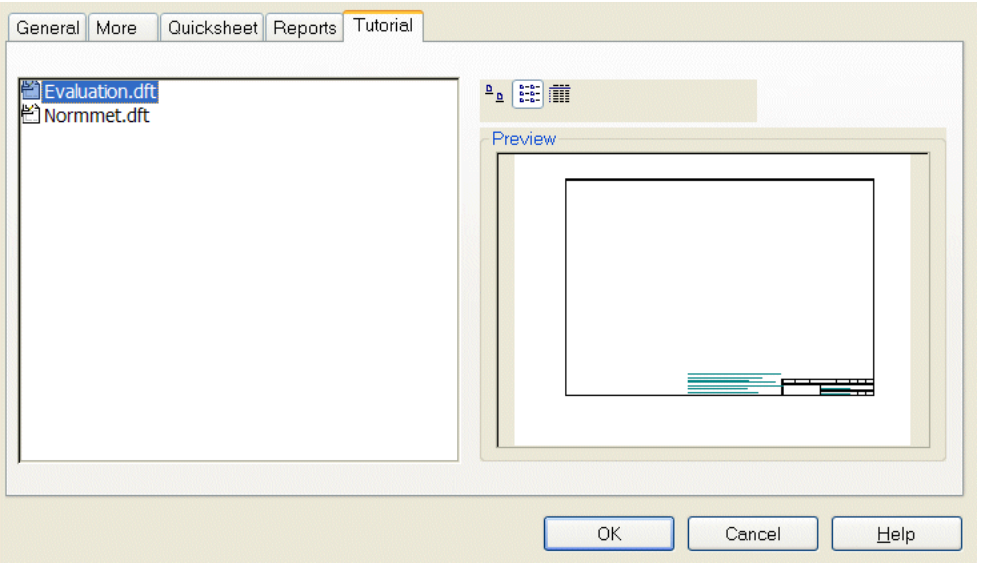
The Drawing View Wizard makes creating drawing views easy. Simply select the part or assembly you want, and the Wizard walks you through the process of defining the drawing views, their scale and display characteristics.

You can customize the drawing templates delivered with Solid Edge to establish and enforce your company's drawing standards. To illustrate this, you will be using a custom template to create the drawing.

Step 1: With the Opener.asm assembly still open, on the File menu, click Create Drawing.

Step 2: **Choose a template**

On the Create Drawing dialog box, click the Browse button to look for an appropriate template to base the new drawing upon.

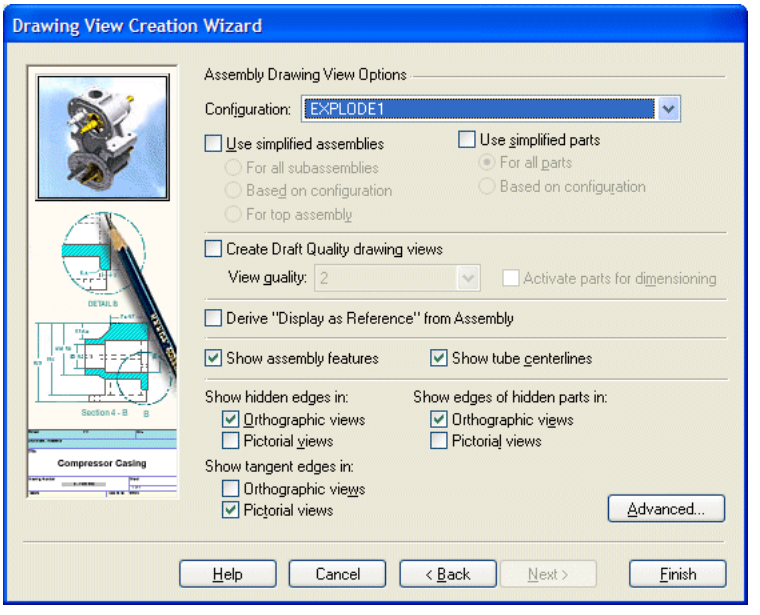


In the New dialog box, click the Tutorial tab, then select the Evaluation.dft template.

On the New dialog box, click OK, then click OK again on the Create Drawing dialog box to create a new Draft document.

Step 3: Place an Exploded View of the Opener Assembly

As the new Draft document opens, the Drawing View Wizard runs. In the next few steps, you will place an exploded drawing view of the Opener assembly, using the Drawing View Wizard.



On the Drawing View Creation Wizard dialog box, choose EXPLODE1 from the list of configurations available for Opener.asm. Recall that this is the configuration you created to save the exploded view of the assembly.

Leave the rest of the options set as they are shown in the illustration.

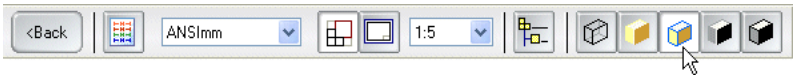
Step 4: Finish Placing the View

The wizard is ready to place exploded view for you.

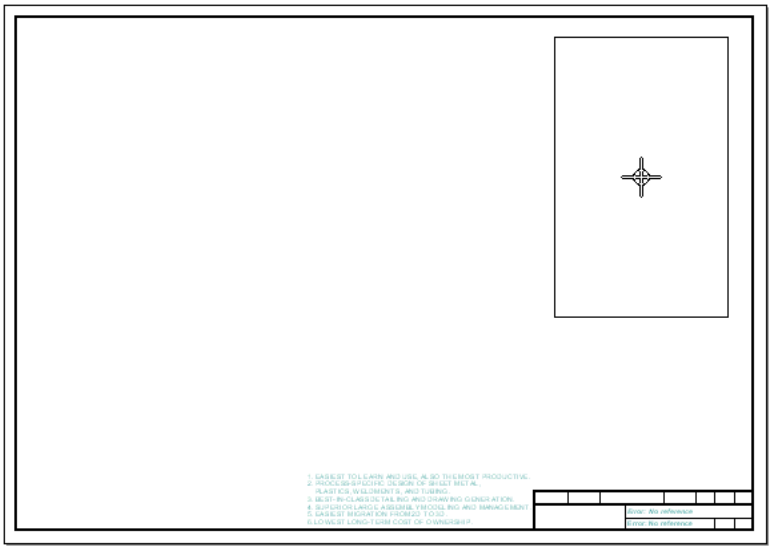


On the Wizard, click the Finish button.

On the ribbon bar, set the drawing view scale to 1:5, and set the Shaded With Visible Edges option.



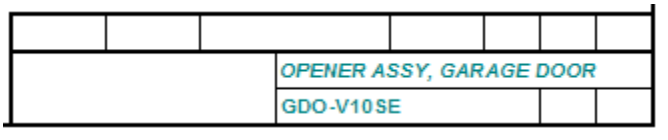
Notice that there is a rectangle attached to the cursor. This rectangle represents the approximate size of the drawing view. Position the rectangle in the top, right corner of the drawing sheet, approximately as shown in the illustration, then click to place the drawing view.



The drawing view of the assembly is placed.

Step 5: Examine the Draft Document

Take a few moments to examine the new Draft document. Notice that a set of standard notes has been placed and that there is text in the title block area. The notes and text were added to the template to illustrate how you can customize the templates delivered with Solid Edge. The text in the title block is called property text, and when you place the drawing view of the assembly, the title and drawing number update based on properties that were defined in the assembly.

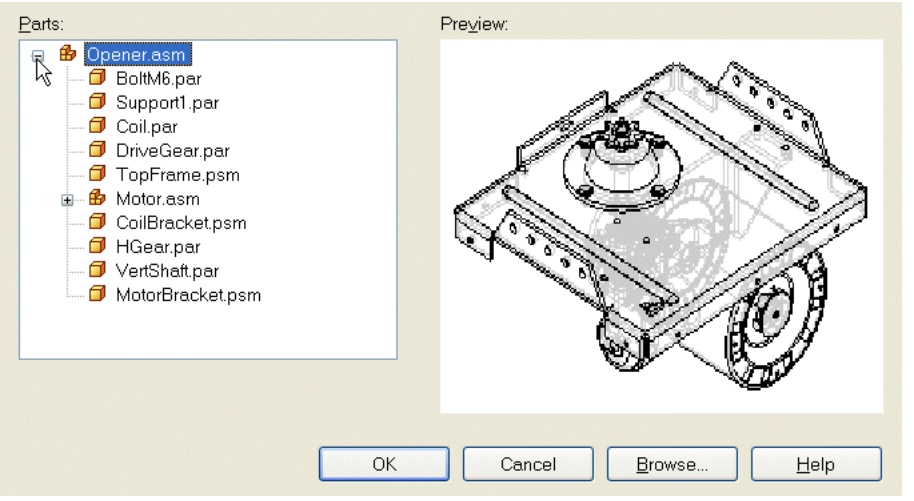


Step 6: Place Drawing Views of the Support

In the next few steps you will use the Drawing View Wizard to place several views of the support part.



On the Drawing Views toolbar, click the Drawing View Wizard button.

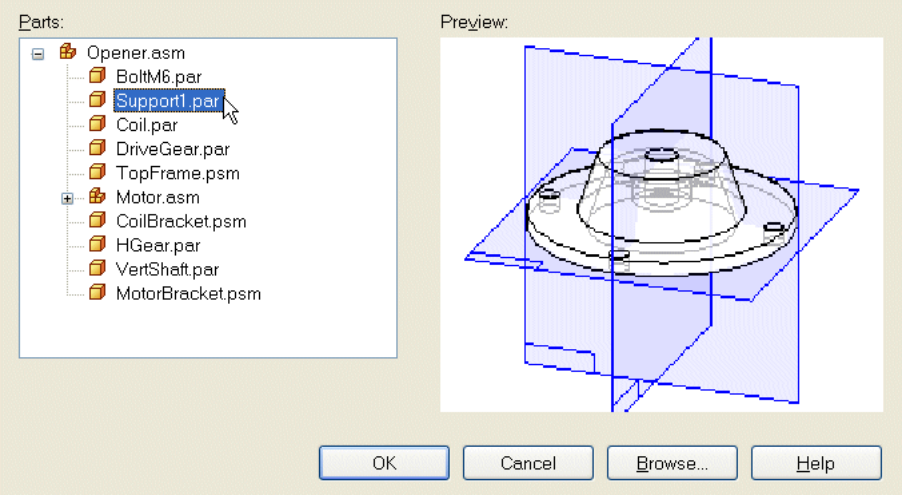


Since you have placed a drawing view of the assembly, the Select Attachment dialog box is displayed, so you can select a part in the assembly.

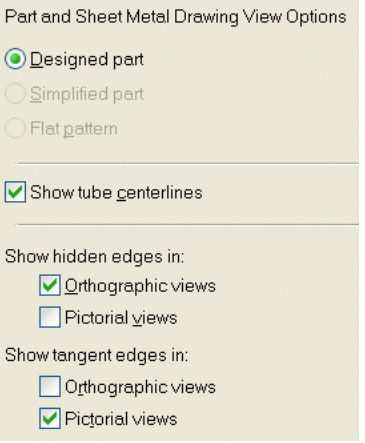
If required, on the Select Attachment dialog box, click the Plus (+) button adjacent to the Opener.asm listing to display the list of parts.

Step 7:

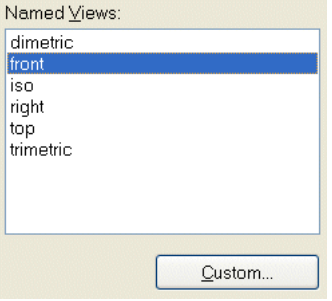
Select the Support1.par entry, then click OK.



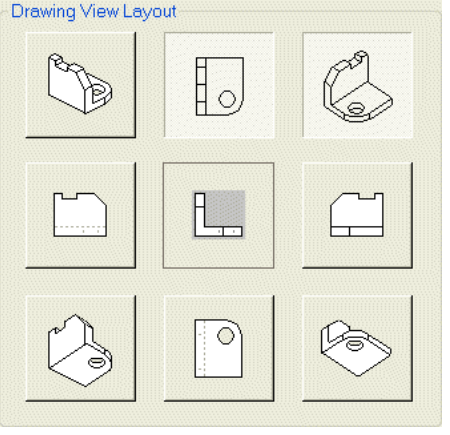
Step 8: The Drawing View Creation Wizard is displayed, so you can set the display options. For this part, the default options are fine, so click the Next button.



Step 9: The Drawing View Orientation page allows you to select the view orientation for the primary view. Select the FRONT option, then click Next again.



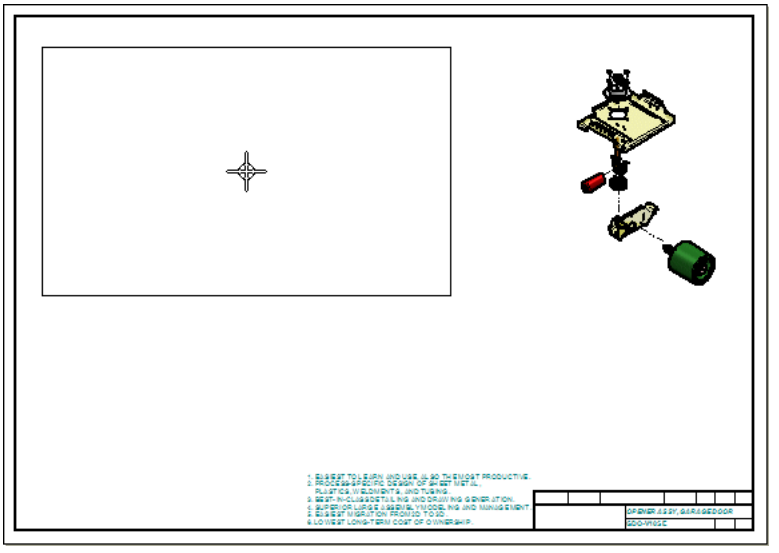
The Drawing View Layout page allows you to specify the additional views you want. The view you specified in the previous step is in the center. Click the two buttons shown to specify a top and an isometric view, then click the Finish button.



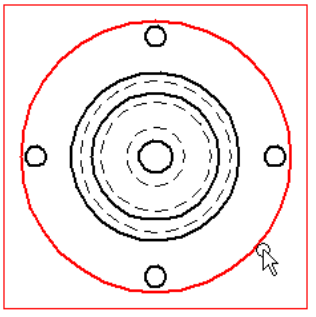
A rectangle is attached to the cursor, making it easy to position the views on the sheet. Also, notice that on the ribbon bar, a default scale is selected. The scale is chosen based on the available room on the sheet. You can select a different scale, but this scale will work just fine. Solid Edge eliminates much of the guesswork required with other CAD systems.

Step 10: Position the Views

Position the views approximately as shown, then click.



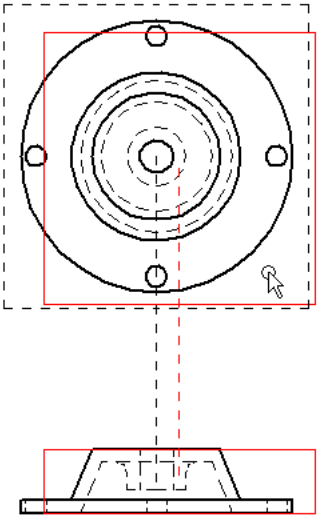
Step 11: Select the Top View



Position the cursor over the large circle as shown, but do not click yet. Notice that a red box is displayed, which represents the extent of the view.

Step 12: Reposition the Views

Press and hold down the left mouse button, then drag the cursor to the right and notice that a dashed line connects the top and front views, and that both views move in unison. The views are orthographically aligned and stay aligned when you move them. Release the mouse button to reposition the views.



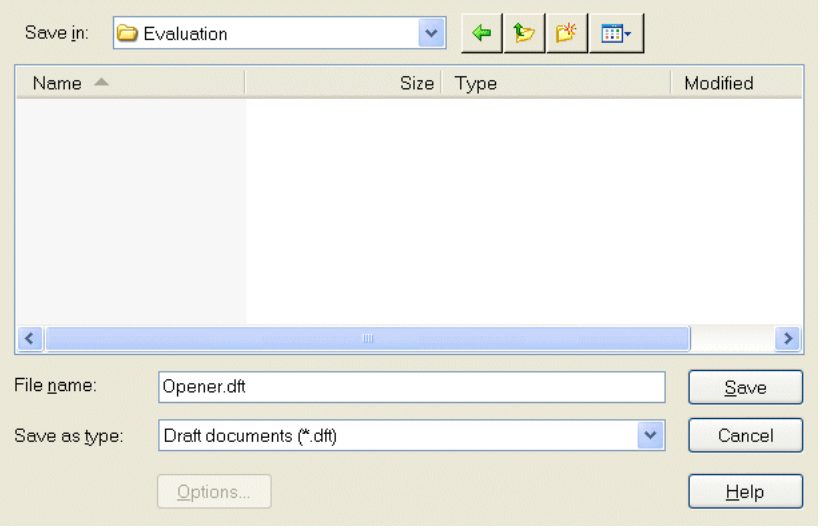
Step 13: Save the file



On the Main toolbar, click the Save button.

On the Save As dialog box, save the document to the default name and location as shown below:

- Set the Save In location to the Solid Edge V18\Training\Evaluation folder.
- Set the Filename to Opener.dft.
- On the Save As dialog box, click Save.



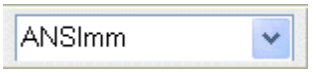
Step 14: Create a Section View of the Support

It appears that a section view would be more appropriate than the front view shown, so in the next few steps you will create a section view. In Solid Edge, creating section views is a simple, two-step procedure. First, you create a cutting plane line, then you fold off the section view using the cutting plane line.

Step 15: Create the Cutting Plane Line

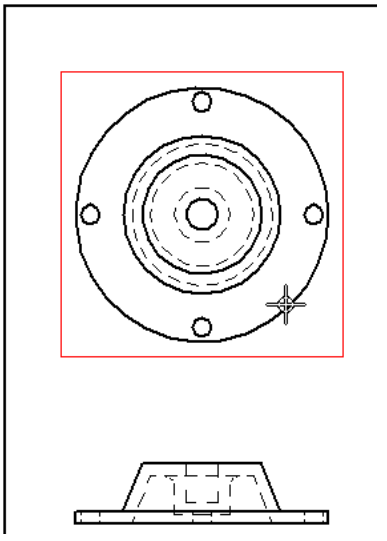


On the Drawing Views toolbar, click the Cutting Plane button.



On the Cutting Plane ribbon bar, in the Dimension Style list, notice that the ANSImm option is set. This setting controls the display properties of the cutting plane line.

Position the cursor over the top view, as shown, then click to select the view. After you select the view, 2D drawing commands are displayed so you can draw the cutting plane line.

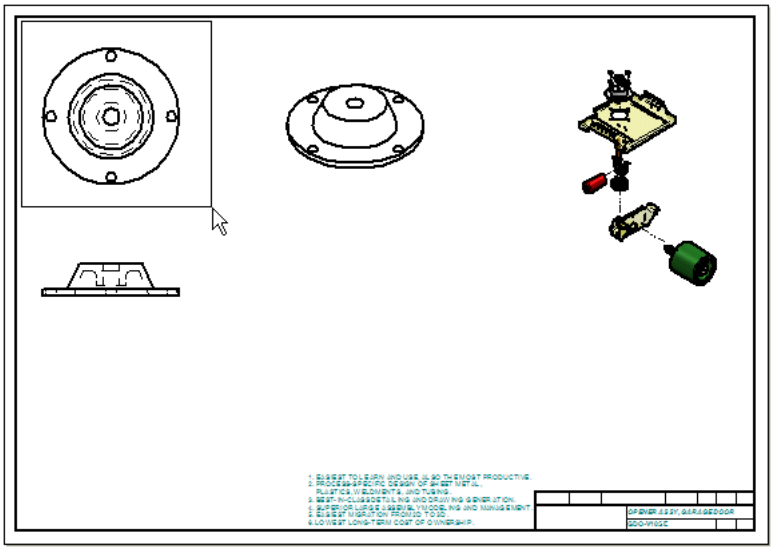


Step 16: Zoom in to the Top View



On the Main toolbar, click the Zoom Area button.

- Click once above and to the left of the top view, and then click again below and to the right of the view. The view zooms into the area enclosed by this rectangle.
- Click the right mouse button to exit the Zoom Area command.

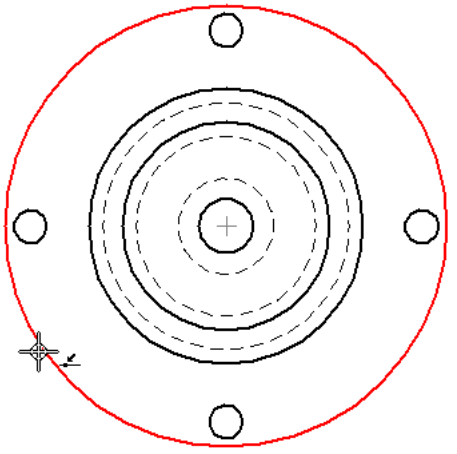


Step 17: Draw the Cutting Plane Line



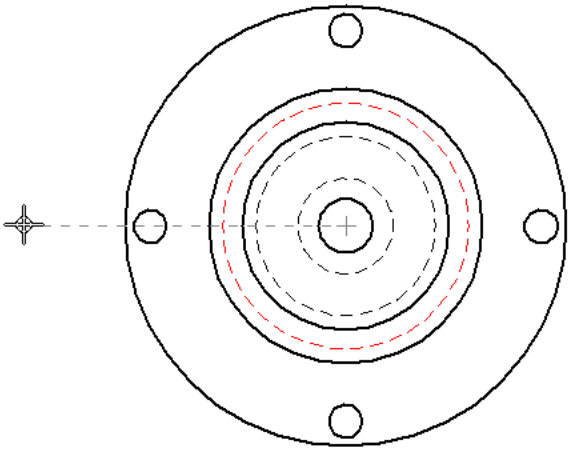
On the Draw toolbar, click the Line button.

Positioning the cutting plane line exactly at the center of the part is easy. First, touch the large circle with the cursor, but do not click.



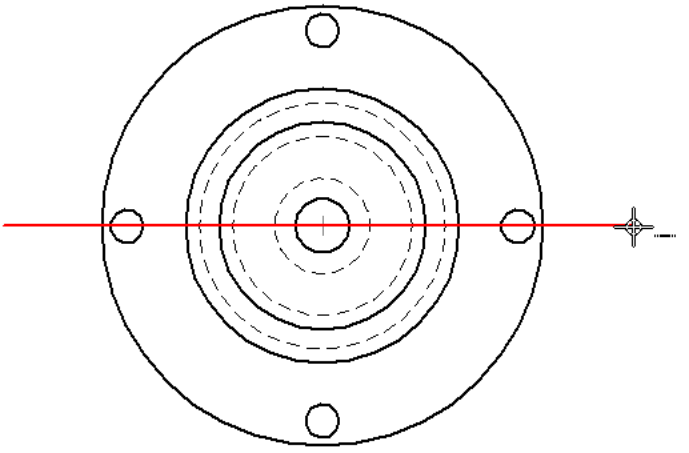
Notice that the large circle highlights and that a center-mark appears at the center of the circle.

Step 18: Now move the cursor to the position shown, and notice the dashed line extending from the center of the circle to the cursor. Click the left mouse button to place the first endpoint of the cutting line.



Step 19: Finish the Cutting Plane Line

Move the cursor to the right as shown, and when the horizontal indicator is displayed adjacent to the cursor, click the left mouse button to place the second endpoint of the line. Now click the right mouse button to restart the Line command.



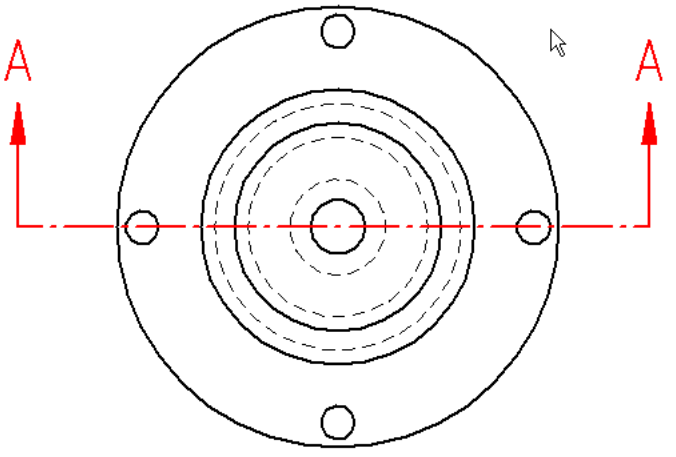


On the Cutting Plane ribbon bar, click the Finish button.

Step 20: Define the Section View Direction

When you click Finish, the Cutting Plane creation commands are hidden and the Drawing View commands are displayed again.

In the top view of the drawing sheet, move the mouse cursor above and below the cutting plane line, and notice that the section view direction arrows flip as the cursor crosses the line. Position the section view direction arrows as shown in the illustration, and click to place them. Notice that the proper line style and arrow heads are created for you. These settings are controlled by the ANSImm Dimension Style mentioned earlier.



Step 21: Fit the Drawing Sheet

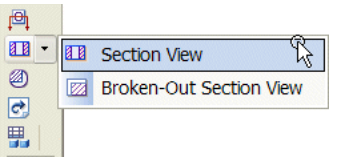


On the Main toolbar, click the Fit button.

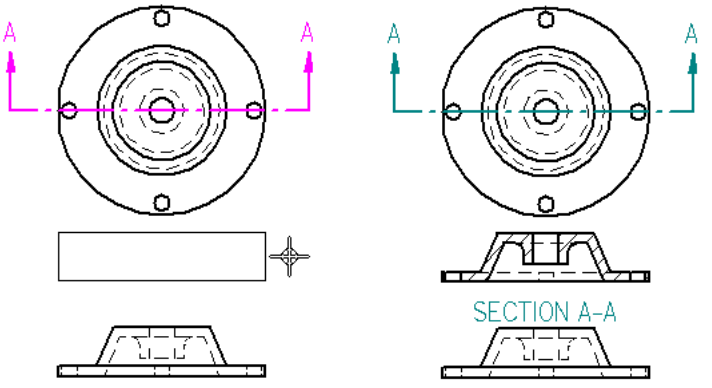
Step 22: Create the Section View



On the Drawing Views toolbar, click the Section View button from the flyout.



Click the cutting plane line you drew in the previous steps, and then click below the top view, as shown in the illustration on the left, to place the section view. The section view is orthographically aligned to the top view, as shown in the illustration on the right.

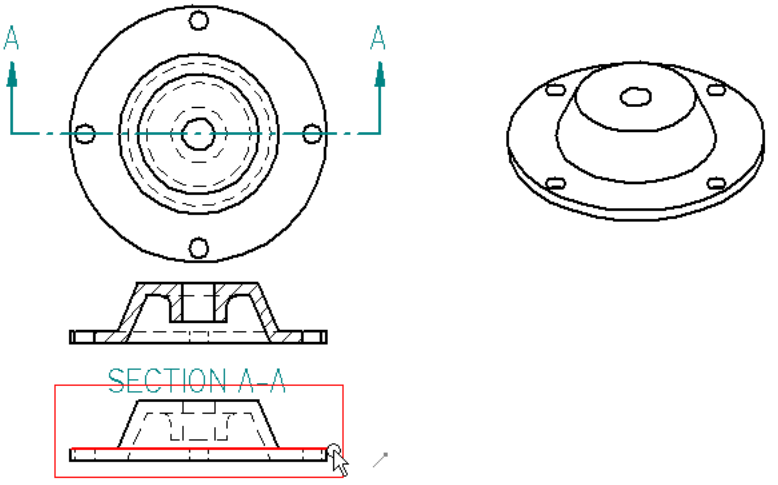


Step 23: Delete the Front View

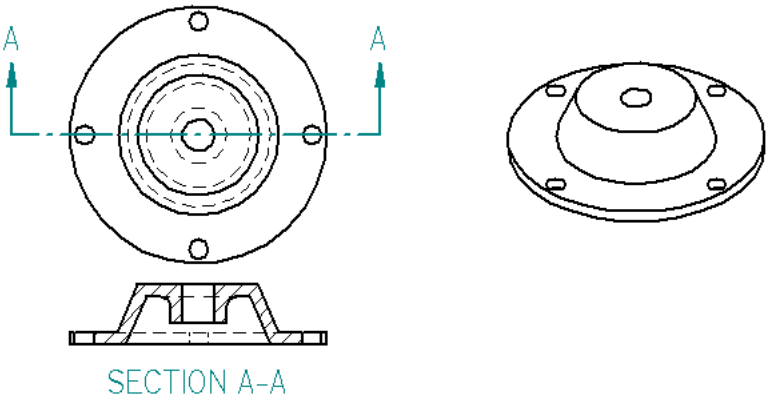


On the Drawing Views toolbar, click the Select Tool button.

Select the front view.

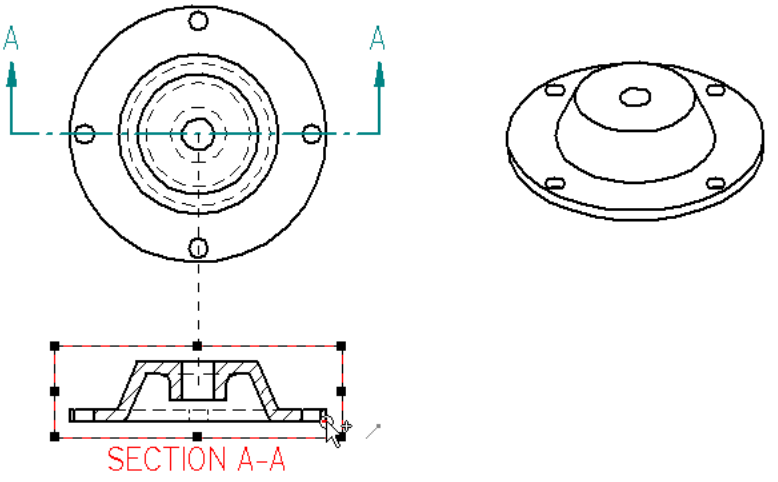


On the keyboard, press the DELETE key. A dialog box is displayed, asking if you want to delete the view. Click the Yes button to delete the view.



Step 24: Move the Section View

To make room for dimensions, select the section view and drag it lower as shown in the illustration.

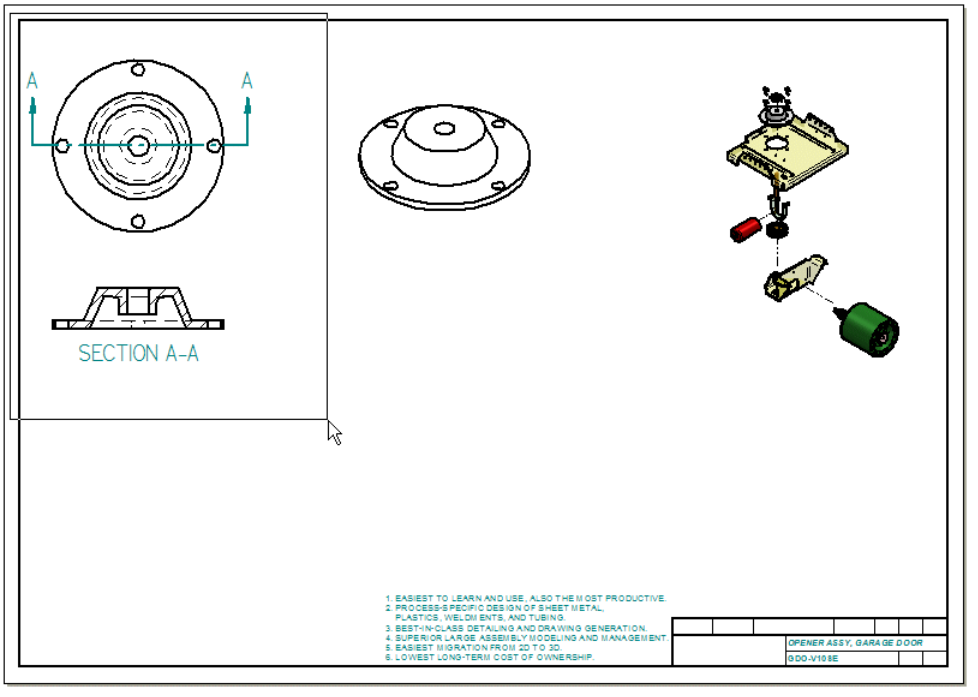


Step 25: Zoom in to the Section and Top Views



On the Main toolbar, click the Zoom Area button.

Click once above and to the left of the top view, and then click again below and to the right of the section view. Click the right mouse button to exit the Zoom Area command.



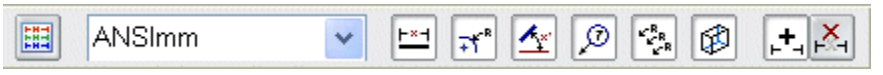
Step 26: Retrieve Dimensions from the Model

When creating drawings, you can place dimensions from scratch or retrieve dimensions from the model. For most designs, using both approaches saves time and effort.

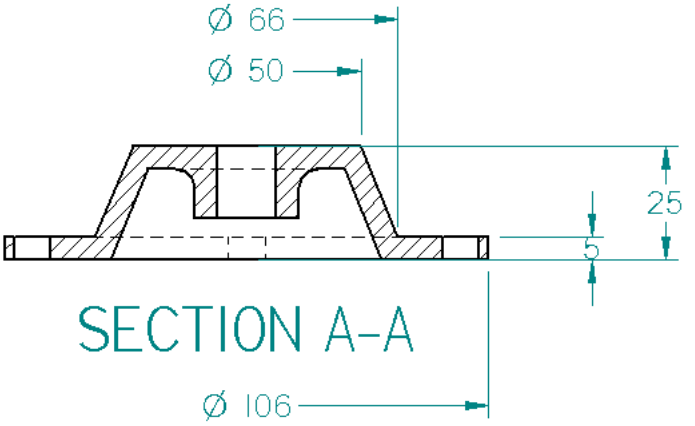


On the Drawing Views toolbar, click the Retrieve Dimensions button.

Take a moment to examine the options on the Retrieve Dimensions ribbon bar. These options give you flexibility in which dimensions are retrieved and their placement options. For the section view, you will use the default options.

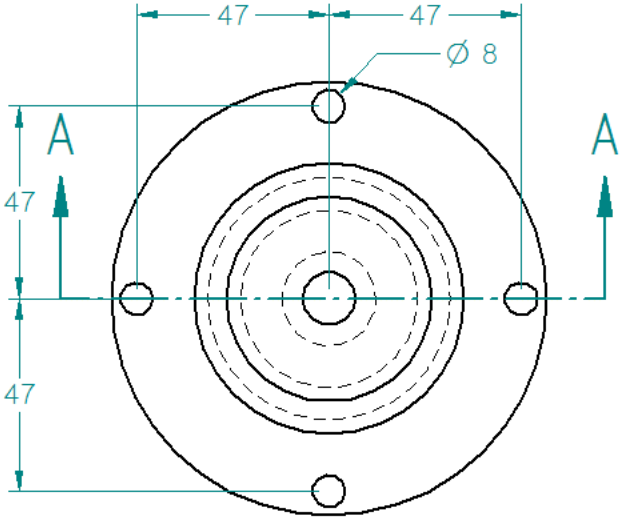


Click the section view, and the appropriate dimensions are retrieved from the model and placed on the section view. The display of your dimensions may be slightly different than the illustration.



Step 27: Retrieve Dimensions for the Top View

Click the top view to retrieve its dimensions. Again, your display may be slightly different, and you may choose to use the Select Tool to reposition the retrieved dimensions for clarity.

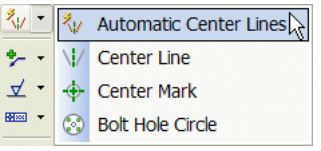


Step 28: Place Center Lines

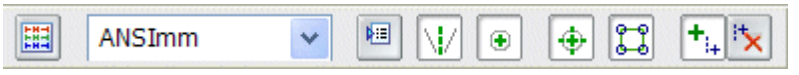
The top view should have center marks drawn for all of the holes. The Automatic Center Lines command can quickly draw them for you.



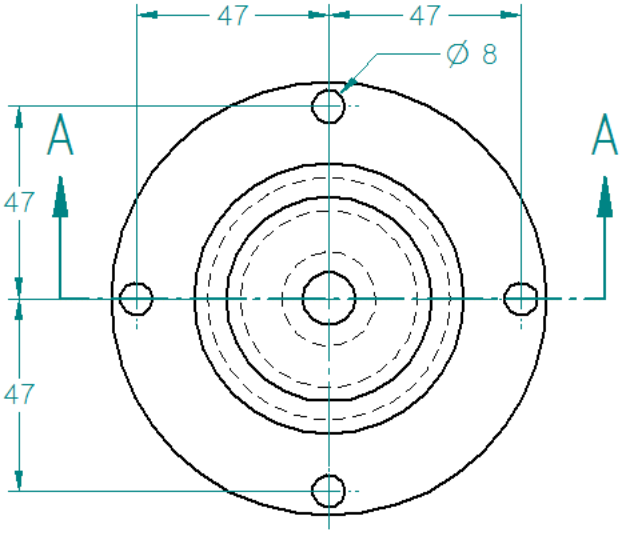
On the Drawing Views toolbar, click the Automatic Center Lines button from the flyout.



Ensure the options on the Automatic Center Lines ribbon bar are set as shown in the illustration, then click the top view to draw the center lines.



The center marks are drawn.



Step 29: Save the File



On the Main toolbar, click the Save button to save the work you have done so far.

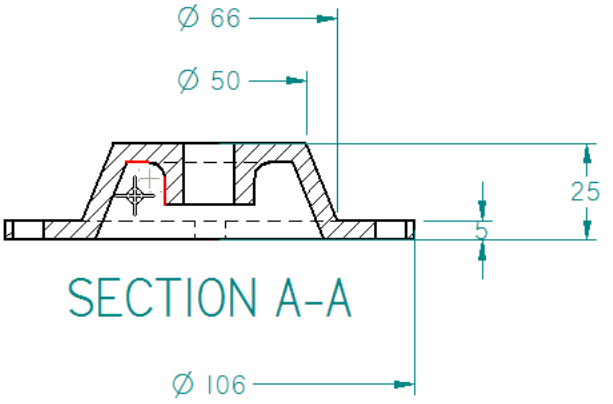
Step 30: Place a Detail View

You can quickly create a detail view from an existing view in Solid Edge.

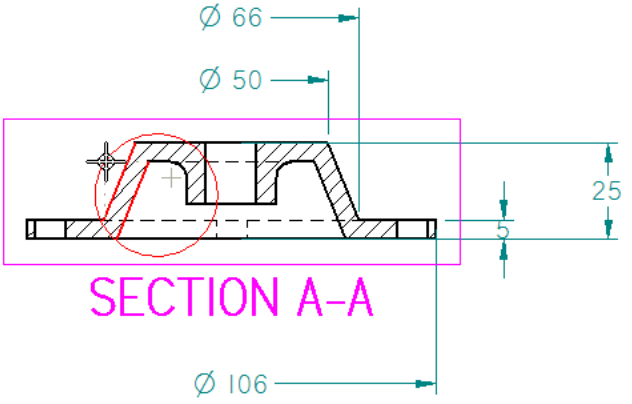


On the Drawing Views toolbar, click the Detail View button.

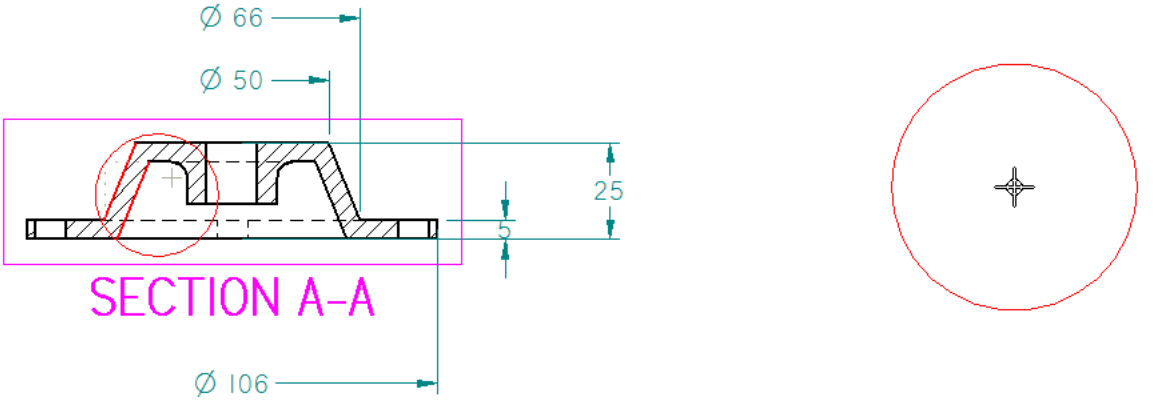
Position the cursor approximately as shown, then click to define the center of the detail envelope.



Step 31: Move the cursor and click again to define the size of the detail envelope.



Step 32: The cursor now has the detail envelope circle attached, so you can position the detail view. Move the cursor to the right and click to place the detail view.

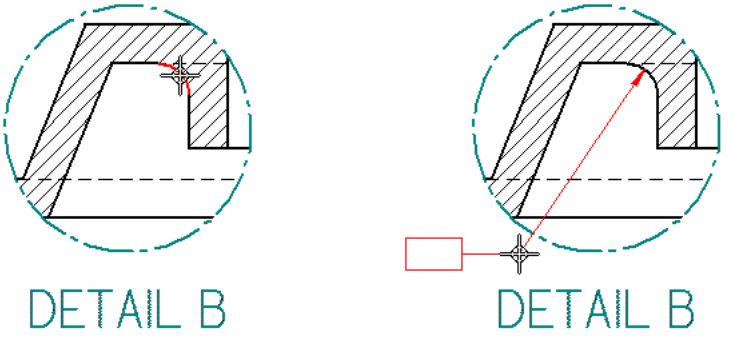


Step 33: **Dimension the Detail View**



On the Drawing Views toolbar, click the SmartDimension button.

In the detail view, click the arc shown in the first illustration, then move the cursor and click to position the dimension.



Step 34: Save the File



On the Main toolbar, click the Save button to save the work you have done so far.

Step 35: Fit the Drawing Sheet

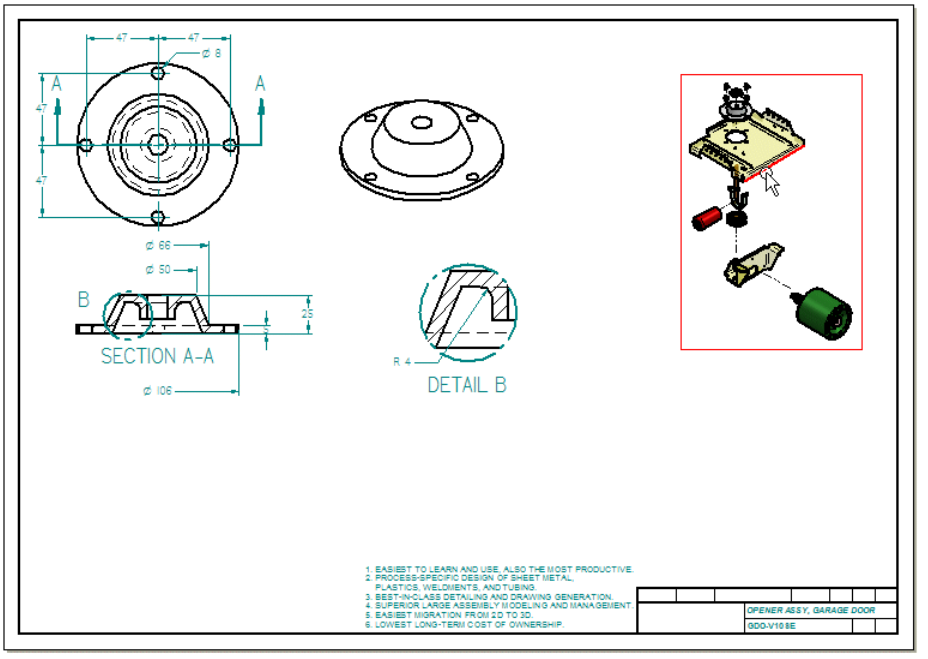


On the Main toolbar, click the Fit button.

Step 36: Modify the Support Part from the Assembly

In Solid Edge, all design changes are done in the part or assembly documents that the drawing references. In the next few steps, you will open the assembly from the draft document, then modify the support part. You will then return to the draft document to update the drawing views and the associated dimensions.

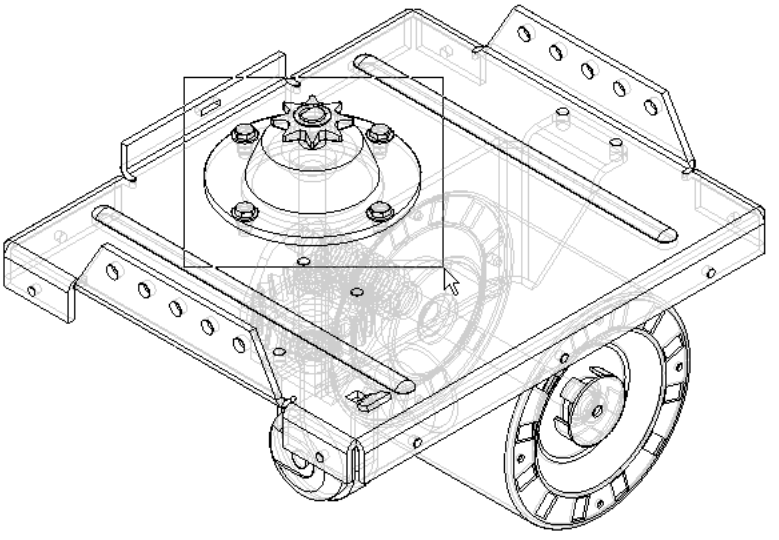
Position the cursor over any element in the exploded view of the assembly, then double-click the left mouse button to open the assembly document.



Step 37: Zoom in to the Support Part



On the Main toolbar, click the Zoom Area button.

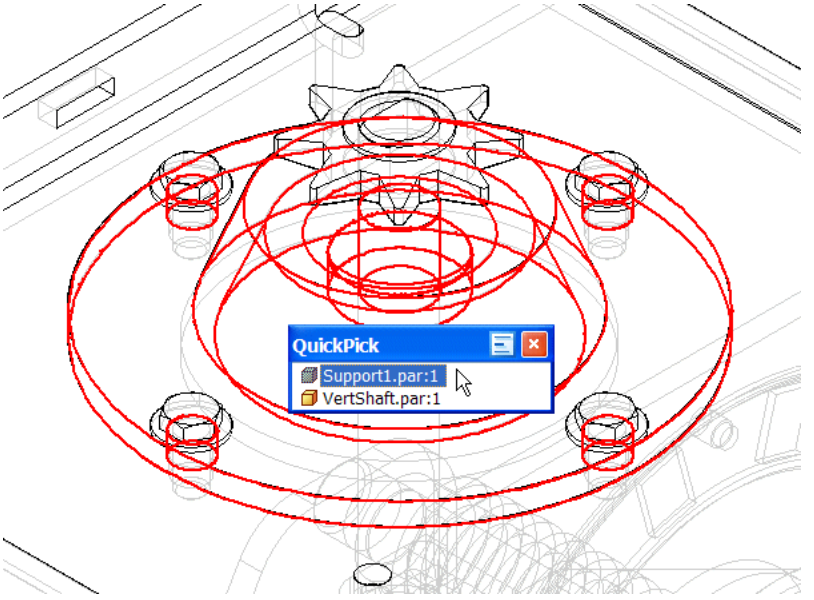


Click once above and to the left of the support part, and then click again below and to the right of the part. Click the right mouse button to exit the Zoom Area command.

Step 38: **Modify the Support Part**

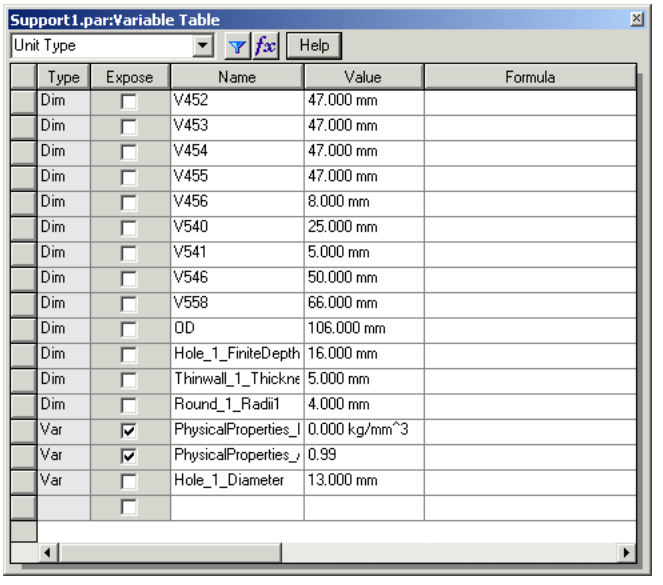
The material between the holes and the outside diameter of the support part looks a little thin, so you will modify the dimension while in the assembly.

On the Tools menu, point to Variables, then click Peer Variables.



In the Assembly window, use QuickPick to select the support part.

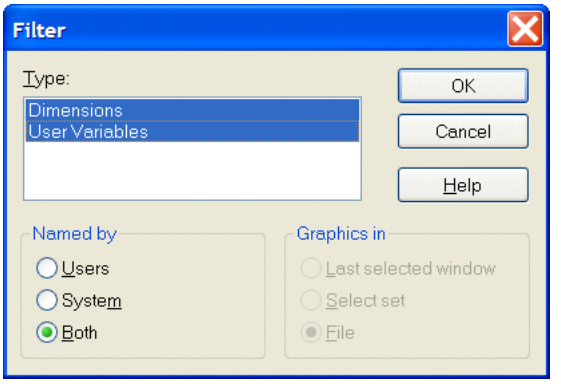
Step 39: The Variable Table for the support part is displayed as shown in the illustration. If your variable table display is different, you need to set the filter options to display the dimensions.





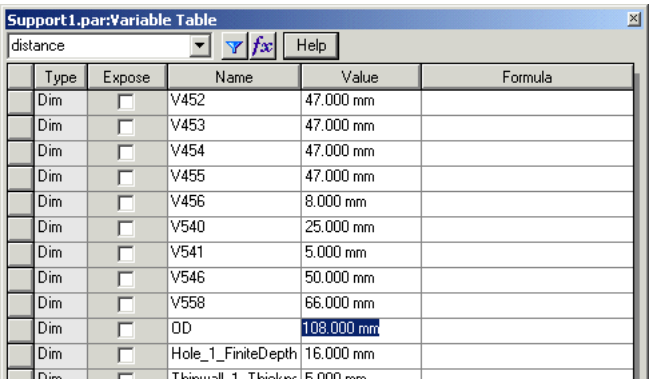
On the Variable Table, click the Filter button.

On the Filter dialog box, in the Type list, select the Dimension and User Variables options, then set the Named By option to Both, then click OK. The dimensions should now be displayed.



In the Variable Table, scroll down until you see the variable you named OD earlier. Note that the other variable names shown in the illustration will probably not match the names you see.

Select the dimensional value for the OD variable as shown, then type 108, and press the ENTER key to increase the outside diameter two millimeters.



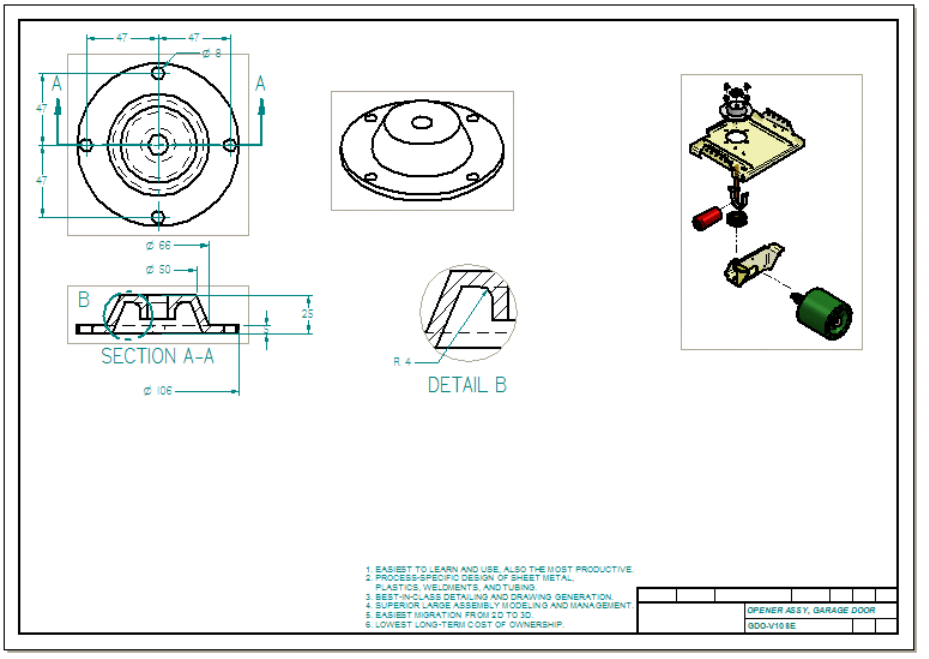
Step 40: On the ribbon bar, click the Close button to dismiss the Variable Table.

Step 41: **Save and Close the Assembly**



On the Main toolbar, click the Save button to save the work you have done so far.

On the File menu, click Close to close the assembly and return to the drawing.



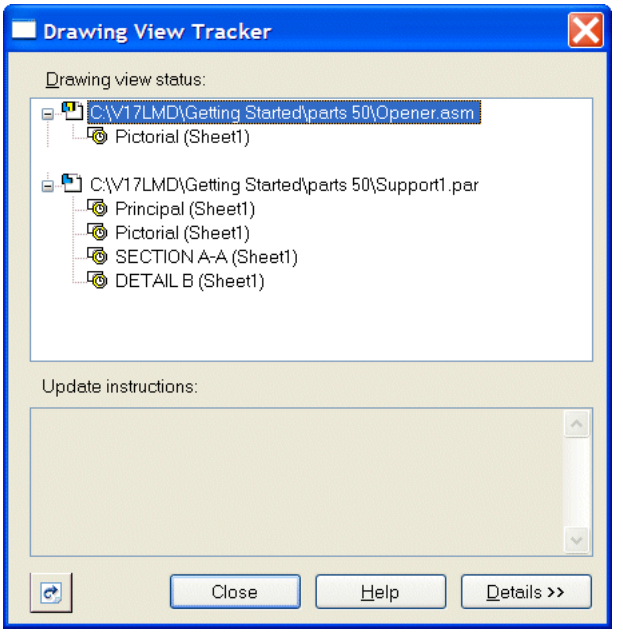


When you return to the Draft environment, a dialog box is displayed to inform you that one or more drawing views are out of date, and to explain that the Drawing View Tracker tool can help you manage this change. Click OK to dismiss the dialog box.

Step 42: Update the Drawing Views

Notice the gray boxes the system has drawn around the drawing views. These boxes notify you that the assembly has changed since the drawing views were created. The design change you made caused the drawing views to no longer be up-to-date with the assembly and the support part. In the next few steps you will evaluate the changes and update the drawing views.

On the Tools menu, click Drawing View Tracker to display the Drawing View Tracker dialog box.





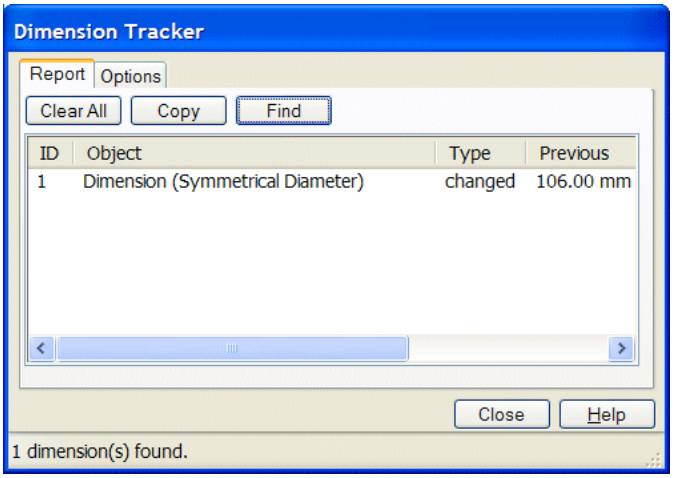
Notice the symbols adjacent to the drawing view entries. These symbols indicate that the drawing views are out-of-date.



On the Drawing View Tracker dialog box, click the Update Views button.

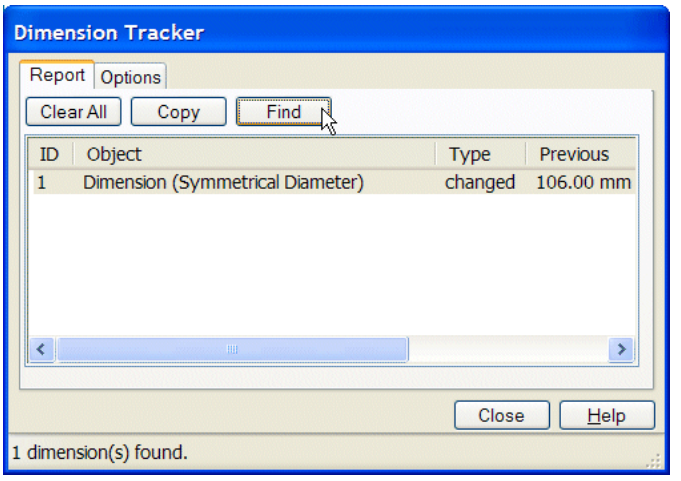
Step 43: On the Drawing View Tracker dialog box, click the Close button.

After Solid Edge processes the changes, the Dimension Tracker dialog box is displayed, indicating that the design change affected one of the drawing dimensions. On complex drawings, Dimension Tracker makes it easy to track dimensional changes and annotations whose positions have changed.

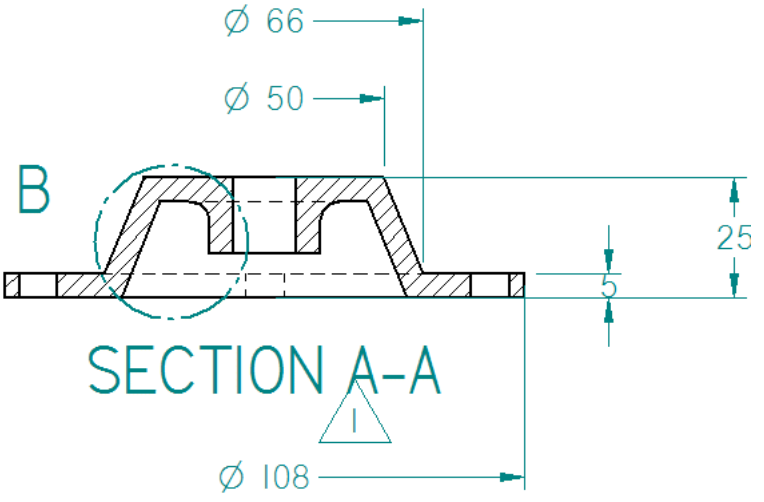


Step 44: **Find the Changed Dimension**

On the Dimension Tracker dialog box, click the entry for the changed dimension, then click the Find button.



Solid Edge zooms in to make it easy to find the changed dimension. A symbol is also added adjacent to the changed dimension. You can define the shape of the symbol as part of your company's standards.



If you choose to remove the change symbols from your drawing, on the Dimension Tracker dialog box, click the Clear All button.

On the Dimension Tracker dialog box, click the Close button.

Step 45: Fit the Drawing Sheet



On the Main toolbar, click the Fit button.

Step 46: Create a Parts List

In the next few steps, you will place a parts list of the assembly on the drawing. You will also set several options that control the display of the parts list.

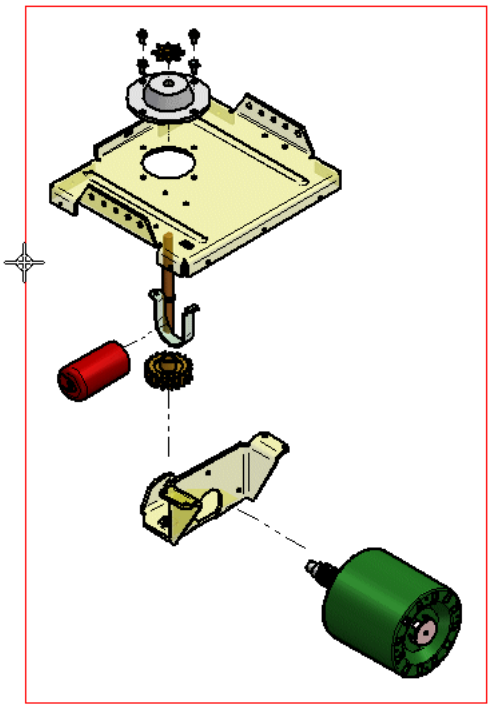


On the Drawing Views toolbar, click the Parts List button.



On the Parts List ribbon bar, ensure that the Auto-Balloon option is set.

Position the cursor over any element in the assembly drawing view and notice that a red box is displayed around the drawing view. Click the left mouse button.



Step 47: Define the Format of the Parts List



On the Parts List ribbon bar, click the Properties button to display the Parts List Properties dialog box.

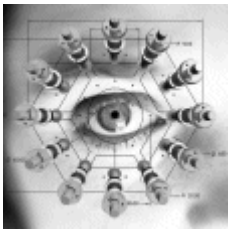
- On the Parts List Properties dialog box, on the General tab, in the Saved Settings list, set the ANSI option.
- On the Columns tab, in the Available Columns list, select the Author option, then click the Add button.
- On the Parts List Properties dialog box, click OK.
- On the Parts List ribbon bar, click the Finish button.

Notice that the parts list is placed in the bottom, left corner of the drawing. The Parts List Properties dialog box allows you to specify the size and location of the parts list, and literally dozens of other properties. You can then save all your parts list properties for quick reuse later, which makes standards definition and enforcement easy.

Item Number	Document Number	Title	Material	Quantity	Author
1	SS101	SUPPORT, SHAFT	steel	1	RIGGS
2	WHB-6X12	BOLT, WASHER HEAD	STEEL	4	PATTON
3	COIL-X1A	COIL	PURCHASED	1	CONERLY
4	GD-S8	GEAR, DRIVE	STEEL	1	RIGGS
5*	TF-101-1	FRAME, TOP	STEEL, GALVANIZED	1	ROBINSON
6	ASM-MTR-1	MOTOR ASSEMBLY		1	WILLIAMS
7	CB-101-1	BRACKET, COIL	STEEL, GALVANIZED	1	SMITH
8	GH-20B	GEAR, HELICAL	STEEL	1	MAHAN
9	SV-101	SHAFT, VERTICAL	STEEL, 4140	1	WATSON
10*	MM-101	MOUNT, MOTOR	STEEL, GALVANIZED	1	ROBINSON

In the Author column, you should see your name listed for the support part.

Notice that the balloons are placed neatly around the part. You can also control the size and shape of the balloons, based on your company's standards.



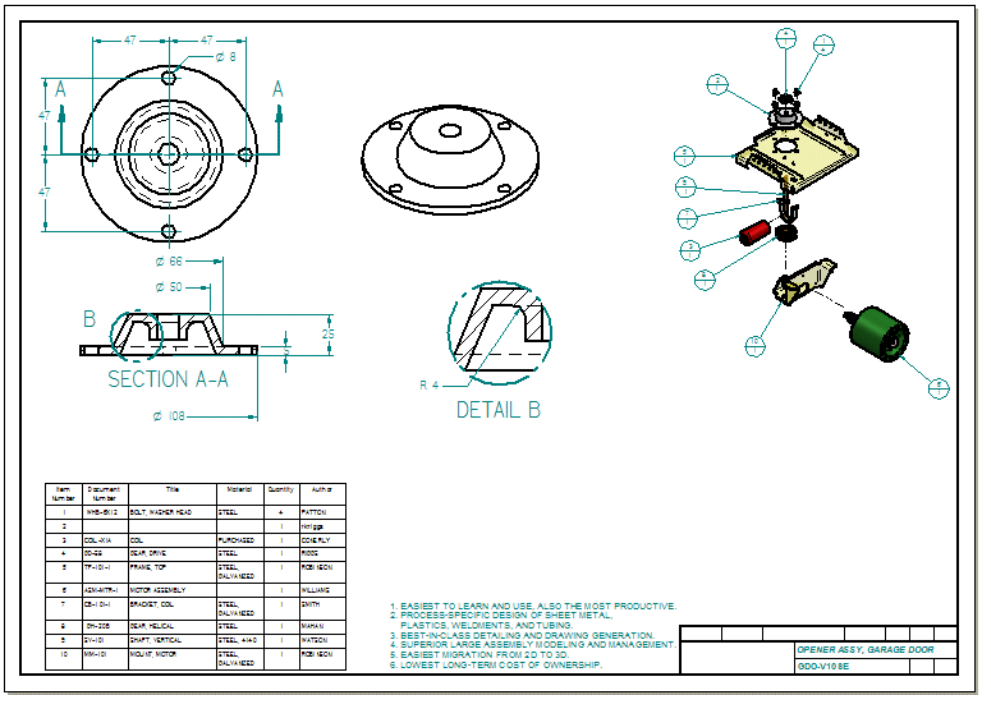
The importance of Bill of Materials management is paramount in a PDM system. The BOM is the culmination of all engineering knowledge that is contained in the design files. Insight manages the metadata along with Solid Edge model-specific data so that non-Solid Edge users can extract BOM information from the managed documents.

Step 48: Print the Drawing

The existing printers connected to your computer or on your network are fully capable of printing Solid Edge drawings.



On the Main toolbar, click the Print button to create a printed copy of the drawing.



Step 49: Save and Close the Drawing



On the Main toolbar, click the Save button to save the work you have done so far.

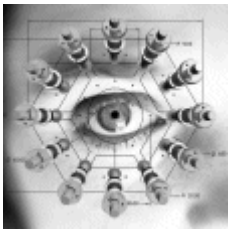
On the File menu, click Close to close the drawing.

Congratulations!

You have created your very first part, assembly, and drawing in Solid Edge. You have experienced firsthand what independent studies and thousands of our customers confirm: that Solid Edge helps designers do more work with less effort than other CAD systems. Because you learn quickly and design faster, you will realize a greater return on investment when you choose Solid Edge.

If you would like to continue learning more about Solid Edge, there are over two dozen tutorials available that cover all aspects of the Solid Edge product. To access these tutorials, on the Help menu, click Tutorials to display the Tutorial menu.

- Part Modeling
- Assembly Modeling
- Drawing Production
- Sheet Metal Modeling
- Weldments
- Variable Table
- Feature Libraries
- Family of Parts
- Family of Assemblies
- Virtual Components
- Inter-part Associativity
- Sensors
- Hole Tables
- 2D To 3D
- Part Patterns
- Tubing
- Wiring
- Surface Modeling
- Virtual Studio+



Ask your Solid Edge Reseller to show you the advantages of Solid Edge Insight. Unlike conventional product data management systems, Solid Edge Insight is not a separate software package with additional commands and long implementation timelines. Instead, Insight is an integral component of Solid Edge, available at no additional cost, requires no user training, and can be implemented in hours instead of months or years.